STAT 305 Handouts

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Table of contents

Preface

A selection of Handouts for STAT 305.

1 Randomness and Probability

Probability comes up in a wide variety of situations. Consider just a few examples.

1. The probability that you roll doubles in a turn of a board game.
2. The probability you win the next Powerball lottery if you purchase a single ticket, 4-8-15-16-42, plus the Powerball number, 23.
3. The probability that a "randomly selected" Cal Poly student is a California resident.
4. The probability that the high temperature in San Luis Obispo next Tuesday is above 80 degrees F.
5. The probability that the Philadelphia Eagles win the next Superbowl.
6. The probability that the Republican candidate wins the 2032 U.S. Presidential Election.
7. The probability that extraterrestrial life currently exists somewhere in the universe.
8. The probability that you ate an apple on April 17, 2014.

Example 1.1. How are the situations above similar, and how are they different? What is one feature that all of the situations have in common? Is the interpretation of "probability" the same in all situations? The goal here is to just think about these questions, and not to compute any probabilities (or to even think about how you would).

- A phenomenon is **random** if there are multiple potential outcomes, and there is **uncertainty** about which outcome will occur.
- Uncertainty is understood in broad terms, and in particular does not only concern future occurrences.
- Many phenomena involve physical randomness, like flipping a coin or drawing powerballs at random from a bin, or in statistical applications of random sampling or random assignment.
- But in many other situations, randomness just vaguely reflects uncertainty.
- Random does *not* mean haphazard. In a random phenomenon, while individual outcomes are uncertain, there is a *regular distribution of outcomes over a large number of (hypothetical) repetitions.*

- Also, random does *not* necessarily mean equally likely. In a random phenomenon, certain outcomes or events might be more or less likely than others.
- The **probability** of an event associated with a random phenomenon is a number in the interval $[0, 1]$ measuring the event's likelihood or degree of uncertainty. A probability can take any value in the continuous scale from 0% to 100%.
- There are two main interpretations of probability.
  - **Long run relative frequency.** The probability of an event can be interpreted as the proportion of times that the event would occur in a very large number of hypothetical repetitions of the random phenomenon.
  - **Subjective probability.** There are many situations where the outcome is uncertain, but it does not make sense to consider the situation as repeatable. In such situations, a subjective (a.k.a., personal) probability describes the degree of likelihood a given individual ascribes to a certain event. Think of subjective probabilities as measuring relative degrees of likelihood rather than long run relative frequencies.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation. In either case, the same basic logical consistency requirements must be satisfied.
- A **simulation** involves an artificial recreation of the random phenomenon, usually using a computer. The probability of an event can be approximated by simulating the random phenomenon a large number of times and determining the proportion of simulated repetitions on which the event occurred out of the total number of repetitions in the simulation.

Example 1.2. One of the oldest documented problems in probability is the following: If three fair six-sided dice are rolled, what is more likely: a sum of 9 or a sum of 10?

1. Explain how you could conduct a *simulation* to investigate this question.

2. In 1 million repetitions of a simulation, a sum of 9 occurred in 115392 repetitions and a sum of 10 occurred in 125026 repetitions. Use the simulation results to approximate the probability that the sum is 9; repeat for a sum of 10.

3. It can be shown that the theoretical probability that the sum is 9 is $25/216 = 0.116$. Write a clearly worded sentence interpreting this probability as a long run relative

frequency.

4. It can be shown that the theoretical probability that the sum is 10 is $27/216 = 0.125$. How many times more likely is a sum of 10 than a sum of 9?

Example 1.3. The weather forecast calls for a 30% chance of rain in your city tomorrow. You ask Donny Don't to interpret the 30% as a long run relative frequency. Donny says: "it will rain in 30% of the city tomorrow". You ask him to elaborate; he says: "Well, there are many different locations in the city. In some of the locations it will rain, in some it won't. It will rain in 30% of the locations, and not in the other 70%. That is, rain will cover 30% of the area of the city, and the other 70% won't have rain." Do you agree? If not, how would you interpret the 30% as a long run relative frequency?

- When interpreting the long run, be careful to define the random phenomenon that is being repeated

Example 1.4. You flip a coin 10 times and it lands on heads 7 times. Is it true that the probability that the coin lands on heads is 7/10=0.7? Explain.

- Distinguish between the short run and the long run

- Observed relative frequencies based on past data (sometimes called "empirical probabilities") are only short run approximations to theoretical probabilities which represent long run relative frequencies

Example 1.5. Your favorite local weatherperson forecasts a 30% chance of rain tomorrow and a 60% chance of rain the next day in your city.

1. Explain how these probabilities are subjective.

2. You ask Donny Don't to interpret these values as relative degrees of likelihood. Donny says: "Well, 30% is not that big, so it's not going to rain that hard tomorrow. Also, 60% is twice is big as 30%, so it's going to rain twice as hard two days from now as it will tomorrow". Do you agree? Explain.

3. Donny says: "Can't we just look at the data from all the days with weather conditions similar to the ones forecast for tomorrow, and see how often it rained on those days to find the probability of rain tomorrow? No subjectivity about that!" How would you respond?

- A **probabilistic forecast** combines observed data and statistical or mathematical models to make predictions.
- Rather than providing a single prediction such as "it will rain tomorrow", probabilistic forecasts provide a range of scenarios and their relative likelihoods.
- Such forecasts are subjective in nature, relying upon the data used and assumptions of the model.
- Changing the data or assumptions can result in different forecasts and probabilities.

Example 1.6. What is your subjective probability that Prof. Ross saw Taylor Swift's Eras Tour live in concert? Consider the following two bets, and suppose you must choose only one.

A) You win \$100 if Professor Ross went to the Eras Tour, and you win nothing otherwise.
B) A box contains 40 green and 60 gold marbles that are otherwise identical. The marbles are thoroughly mixed and one marble is selected at random. You win \$100 if the selected marble is green, and you win nothing otherwise.

1. Which of the above bets would you prefer? Or are you completely indifferent? What does this say about your subjective probability that Prof Ross saw the Eras Tour live?
2. If you preferred bet B to bet A, consider bet C which has a similar setup to B but now there are 20 green and 80 gold marbles. Do you prefer bet A or bet C? What does this say about your subjective probability that Prof Ross saw the Eras Tour live?
3. If you preferred bet A to bet B, consider bet D which has a similar setup to B but now there are 60 green and 40 gold marbles. Do you prefer bet A or bet D? What does this say about your subjective probability that Prof Ross saw the Eras Tour live?
4. Continue to consider different numbers of green and gold marbles. Can you zero in on your subjective probability?

Example 1.7. As of Jan 3, ESPN listed the following probabilities for who will win the 2026 Superbowl.

Team	Probability
Seattle Seahawks	18%
Los Angeles Rams	15%
Denver Broncos	12%
San Francisco 49ers	10%
New England Patriots	10%
Philadelphia Eagles	9%
Other	

According to ESPN (as of Jan 3):

1. Are the above percentages relative frequencies or subjective probabilities? Why?

2. What must be the probability that a team other than the above six teams wins the championship? That is, what value goes in the "Other" row in the table?

3. The Seahawks are how many times more likely than the Eagles to win? The Seahawks are how many times more likely than the Broncos to win?

4. What must be the probability that the 49ers do *not* win the championship? How many times more likely are the 49ers to not win than to win (this ratio is the "odds against" the 49ers winning).

5. How could you construct a circular spinner (like from a kids game) to simulate the champion according to these probabilities? According to this model, what would you expect the results of 10000 repetitions of a simulation of the champion to look like?

Example 1.8. Suppose your subjective probabilities for the 2026 Superbowl satisfy the following conditions.

- The 49ers and Eagles are equally likely to win
- The Rams are 1.5 times more likely than the 49ers to win
- The Seahawks are 2 times more likely than the Rams to win
- The winner is as likely to be among these four teams—49ers, Eagles, Rams, Seahawks—as not.

Construct a table of your subjective probabilities like the one in Example 1.7.

- The previous examples illustrate two interpretations of probability: long run relative frequencies and subjective probabilities.
- We will use these interpretations interchangeably.
- With subjective probabilities it is often helpful to consider what might happen in a simulation.
- It is also useful to consider long run relative frequencies in terms of relative degrees of likelihood.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation.
- A probability takes a value in the sliding scale from 0 to 1 (or 0 to 100%).
- Don't just focus on computation; always remember to properly interpret probabilities.

Example 1.9. Consider a Cal Poly student who frequently has blurry, bloodshot eyes, generally exhibits slow reaction time, always seems to have the munchies, and disappears at 4:20 each day. Which of the following, A or B, has a higher probability? Assume the two probabilities are not equal.

- A: The student has a GPA above 3.0.
- B: The student has a GPA above 3.0 and smokes marijuana regularly.

- Warning! Your psychological judgment of probabilities is often inconsistent with the mathematical logic of probabilities.

Example 1.10. Ron and Leslie agree to the following bet. They'll ask Professor Ross if he saw the Eras Tour live. If he did, Leslie will pay Ron $200; if not, Ron will pay Leslie $100. (Neither has any direct information about whether or not Prof Ross saw the Eras Tour.)

1. Given this setup, which of the following is being judged as more likely: that Prof Ross saw the Eras Tour, or that he did not? Why?

2. What are this bet's "odds"?

3. Ron and Leslie agree that this is a fair bet, and neither would accept worse odds. What is their subjective probability that Professor Ross saw the Eras Tour?

4. Suppose they were to hypothetically repeat this bet many times, say 3000 times. Given the probability from the previous part, how many times would you expect Leslie to win? To lose? What would you expect Leslie's net dollar winnings to be? In what sense is this bet "fair"? (Remember: Leslie's winnings are Ron's losses and vice versa.)

- The **odds** of an event is a ratio involving the probability that the event occurs and the probability that the event does not occur.
- Odds can be expressed as either "in favor" of or "against" the event occurring, depending on the order of the ratio.

2 Working with Probabilities

- It is often helpful to think of probabilities as percentages or proportions.
- Furthermore, when working with multiple percentages, it is also helpful to construct hypothetical **two-way tables** (a.k.a., contingency tables) of counts.
- For the purposes of constructing the table and computing related probabilities, any value can be used for the hypothetical total count.
- When dealing with percentages (or proportions or probabilities) be sure to ask "percent *of what*?" Thinking in fraction terms, be careful to identify the correct reference group which corresponds to the denominator.

Example 2.1. Do American adults (18+) think it's acceptable to curse out loud in public? Assume[1] that

- 62% of American adults age 18-29 think it is acceptable
- 45% of American adults age 30-49 think it is acceptable
- 24% of American adults age 50-64 think it is acceptable
- 11% of American adults age 65+ think it is acceptable

Also assume that among American adults (18+)

- 20% of American adults are age 18-29
- 33% of American adults are age 30-49
- 25% of American adults are age 50-64
- 22% of American adults are age 65+

1. Consider a hypothetical group of 10000 American adults and assume the percentages provided apply to this group. Fill in the counts in each of the cells of the following table.

	18-29	30-49	50-64	65+	Total
Acceptable					
Not acceptable					
Total					10000

[1]The values in this problem are based on a March 12, 2025 article by the Pew Research Center.

2. Randomly select an American adult *from this group of 10000.* Compute the probability that they think cursing out loud in public is acceptable.

3. Randomly select an *American adult.* Compute the probability that they think cursing out loud in public is acceptable. (Hint: did the 10000 matter?)

4. Compute the probability that an American adult who thinks cursing out loud in public is acceptable is age 18-29. (Answer with both an unreduced fraction and a demical/percent.)

5. Compute the probability that an American adult who is age 18-29 thinks cursing out loud in public is acceptable. (Answer with both an unreduced fraction and a demical/percent.)

6. Compute the probability that an American adult is age 18-29 and thinks cursing out loud in public is acceptable. (Answer with both an unreduced fraction and a demical/percent.)

7. Compare the unreduced fractions for the previous three parts. What is the same? What is different?

8. Suppose that we were only told that 35.67% of American adults overall think cursing out loud in public is acceptable, and that we not given the values 62%, 45%, 24%, 11%. Would we be able to complete the two-way table?

- **Warning!** In general, knowing probabilities of individual events alone is not enough to determine probabilities of combinations of them.

Example 2.2. Suppose that 47% of American adults[2] have a pet dog and 25% have a pet cat.

1. Donny Don't says that 72% (which is 47% + 25%) of American adults have a pet dog or a pet cat. Is that necessarily true? If not, is it even possible (in principle anyway) for this to be true? Under what circumstance (however unrealistic) would this be true? Construct a corresponding two-way table.

2. Given only the information provided, what is the smallest possible percentage of American who adults have a pet dog or a pet cat? Under what circumstance (however unrealistic) would this be true? Construct a corresponding two-way table.

3. Donny Don't says that 11.75% (which is 47% $\times$ 25%) of Americans have both a pet dog *and* a pet cat. Explain to Donny why that's not necessarily true. Without further information, what can you say about the percent of American adults who have both a pet dog and a pet cat?

[2]These values are based on the 2018 General Social Survey.

4. Suppose that 14% of American adults have both a pet dog *and* a pet cat. What is the percentage of American adults who have a pet dog *or* a pet cat? Construct a corresponding two-way table. Use your table to show Donny how to correct his error from part 1.

5. What percentage of American adults who have a pet dog also have a pet cat? Is it 25%?

6. What percentage of American adults who do not have a pet dog have a pet cat? Is this the same value as in the previous part?

7. What percentage of American adults who have a pet cat also have a pet dog? Is it 47%?

8. Describe in words the percentage that results from subtracting the answer to the previous part from 100%.

Example 2.3. A woman's chances of giving birth to a child with Down syndrome increase with age. The CDC estimates[3] that a woman in her mid-to-late 30s has a risk of conceiving a child with Down syndrome of about 1 in 250. A nuchal translucency (NT) scan, which involves a blood draw from the mother and an ultrasound, is often performed around the 13th week of pregnancy to test for the presence of Down syndrome (among other things). If the baby has Down syndrome, the probability that the test is positive is about 0.9. However, when the baby does not have Down syndrome, there is still a probability that the test returns a (false) positive of about[4] 0.05. Suppose that the NT test for a pregnant woman in her mid-to-late 30s comes back positive for Down syndrome. What is the probability that the baby actually has Down syndrome?

1. Before proceeding, make a guess for the probability in question.

   0-20% 20-40% 40-60% 60-80% 80-100%

2. Donny Don't says: 0.90 and 0.05 should add up to 1, so there must be a typo in the problem. Do you agree?

3. Construct a hypothetical two-way table of counts to represent the given information.

4. Use the table to find the probability in question: If NT test for a pregnant woman in her mid-to-late 30s is positive, what is the probability that the baby actually has Down syndrome?

5. The probability in the previous part might seem very low to you. Explain why the probability is so low.

[3]Source: http://www.cdc.gov/ncbddd/birthdefects/downsyndrome/data.html

[4]Estimates of these probabilities vary between different sources. The values in the exercise were based on https://www.ncbi.nlm.nih.gov/pubmed/17350315

6. Compare the probability of having Down Syndrome before and after the positive test. How much more likely is a baby who tests positive to have Down Syndrome than a baby for whom no information about the test is available?

- Remember to ask “percentage *of what*”? For example, the percentage of *babies who have Down syndrome* that test positive is a very different quantity than the percentage of *babies who test positive* that have Down syndrome.
- Probabilities are often conditional on information.
- Conditional probabilities (e.g., probability of Down Syndrome *given a positive test*) can be highly influenced by the original unconditional probabilities (e.g. probability of Down Syndrome), sometimes called the **base rates**. Don’t neglect the base rates when evaluating probabilities.
- The example illustrates that when the base rate for a condition is very low and the test for the condition is less than perfect there will be a relatively high probability that a positive test is a *false positive.*

3 Interpreting Probabilities and "Expected" Values

- A probability takes a value in the sliding scale from 0 to 100%.
- Don't just focus on computation; always remember to properly interpret probabilities.

Example 3.1. In each of the following parts, which of the two probabilities, a or b, is larger, or are they equal? You should answer conceptually without attempting any calculations. Explain your reasoning.

1. Randomly select a man.

    a. The probability that a randomly selected man who is greater than six feet tall plays in the NBA.
    b. The probability that a randomly selected man who plays in the NBA is greater than six feet tall.

2. Randomly select a baby girl who was born in 1950.

    a. The probability that a randomly selected baby girl born in 1950 is alive today.
    b. The probability that a randomly selected baby girl born in 1950, who was alive at the end of 2020, is alive today.

- A probability is a measure of the likelihood or degree of uncertainty or plausibility of an event.
- A "conditional" probability revises this measure to reflect any additional information about the outcome of the underlying random phenomenon.
- In a sense, all probabilities are conditional upon some information, even if that information is vague ("well, it has to be one of these possibilities"). Be careful to clearly identify what information is reflected in probabilities
- When interpreting probabilities, consider the conditions under which the probabilities were computed, in the proper direction

Example 3.2. In each of the following parts, which of the two probabilities, a or b, is larger, or are they equal? You should answer conceptually without attempting any calculations. Explain your reasoning.

1. Flip a coin *which is known to be fair* 10 times.

   a. The probability that the results are, in order, HHHHHHHHHH.
   b. The probability that the results are, in order, HHTHTTTHHT.

2. Flip a coin which is known to be fair 10 times.

   a. The probability that all 10 flips land on H.
   b. The probability that exactly 5 flips land on H.

3. In the Powerball lottery there are roughly 300 million possible winning number combinations, all equally likely.

   a. The probability you win the next Powerball lottery if you purchase a single ticket, 4-8-15-16-42, plus the Powerball number, 23
   b. The probability you win the next Powerball lottery if you purchase a single ticket, 1-2-3-4-5, plus the Powerball number, 6.

4. Continuing with the Powerball

   a. The probability that the numbers in the winning number are not in sequence (e.g., 4-8-15-16-42-23)
   b. The probability that the numbers in the winning number are in sequence (e.g., 1-2-3-4-5-6)

5. Continuing with the Powerball

   a. The probability that you win the next Powerball lottery if you purchase a single ticket.
   b. The probability that someone wins the next Powerball lottery. (FYI: especially when the jackpot is large, there are hundreds of millions of tickets sold.)

- When interpreting probabilities, be careful not to confuse "the particular" with "the general".
  - **"The particular:"** A very specific event, surprising or not, often has low probability.
  - **"The general:"** While a very specific event often has low probability, if there are many like events their combined probability can be high.
- Even if an event has extremely small probability, given enough repetitions of the random phenomenon, the probability that the event occurs on *at least one* of the repetitions is often high.
- In general, even though the probability that something very specific happens to you today is often extremely small, the probability that something similar happens to someone some time is often quite high.

- When assessing a numerical probability, always ask "probability of what"? Does the probability represent "the particular" or "the general"? Is it the probability that the event happens in a single occurrence of the random phenomenon, or the probability that the event happens at least once in many occurrences?
- Also distinguish between assumption and observation. For example, if you assume that a coin is fair and the flips are independent, then all possible H/T sequences are equally likely. However, if you observe the coin landing on heads on 20 flips in a row, then that might cast doubt on your assumption that the coin is fair.

Example 3.3. Shuffle a standard deck of 52 playing cards (13 face values in each of 4 suits) and deal two cards without replacement.

1. What is the probability that the first card dealt is a heart?
2. What is the probability that the second card dealt is a heart?
3. What is the probability that the second card dealt is a heart if the first card dealt is a heart?
4. What is the probability that the second card dealt is a heart if the first card dealt is not a heart?
5. Revisit part 2. What is the probability that the second card dealt is a heart? Create a two-way table to answer this question.

- Be careful to distinguish between conditional and unconditional probabilities.
- A conditional probability reflects additional information about the outcome of the random phenomenon.
- In the absence of such information, we must continue to account for all the possibilities.
- When computing probabilities, be sure to only reflect information that is known. Especially when considering a phenomenon that happens in stages, don't assume that when considering what happens second that you know what happened first.

Example 3.4. Within both the colleges of Agriculture and Architecture at Cal Poly, about 49% of admitted students are female, about 84% of admitted students went to high school in CA, and the median GPA of admitted students is about 4.1.

An orientation group of 100 newly admitted Cal Poly students includes 75 students in Agriculture and 25 students in Architecture. A student is randomly selected from this group. The selected student is Maddie, who is female, went to high school in CA, and had a high school GPA of 4.1.

Donny Don't says, "The information about Maddie applies equally well to Agriculture or Architecture and doesn't help us decide which college she's in, so it's just 50/50. Given the information about Maddie, the conditional probability that she is in Agriculture is 0.5." Do you agree? If not, what is the conditional probability that Maddie is in the college of Agriculture given the information about her?

Example 3.5. This is a very simplified example illustrating the basic idea of how insurance works. Every year an insurance company sells many thousands of car insurance policies to drivers within a particular risk class. Each policyholder pays a "premium" of \$1000 at the start of the year, and the insurance company agrees to pay for the cost of all damages that occur during the year. Suppose that each policy incurs damage of either \$0, \$5000, \$20000, or \$50000 with the following probabilities.

Amount of damage (\$)	Profit (\$)	Probability
0	1000	0.910
5000	-4000	0.070
20000	-19000	0.019

Amount of damage ($)	Profit ($)	Probability
50000	-49000	0.001

The insurance company's profit on a policy at the end of the year is the difference between the premium of $1000 and any damage paid out. For example, a policy that incurs no damage results in a profit of $1000; a policy that incurs $5000 in damage results in a profit of -$4000 (that is, a loss of $4000) for the insurance company.

1. Interpret the probabilities 0.91, 0.07, 0.019, and 0.001 as long run relative frequencies in this context.

2. Compute the probability that a policy results in a positive profit for the insurance company.

3. Imagine 100,000 hypothetical policies. How many of these policies would you expect to result in a profit of $1000? -$4000? -$19000? -$49000?

4. What do you expect the total profit for these 100,000 policies to be?

5. What do you expect the average profit per policy for these 100,000 policies to be?

6. Compute the probability that a policy has a profit equal to the value from part 5.

7. Compute the probability that a policy has a profit greater than the value from part 5.

8. Is the value from part 5 the most likely value of profit for a single policy?

9. Is the value from part 5 the profit you would expect for a single policy?

10. Explain in what sense the value from part 5 is "expected".

- The long run average value of a random quantity is called its "expected value".
- Be careful: the term "expected value" is somewhat of a misnomer.
- The expected value is *not* necessarily the value we expect on a single repetition of the random phenomenon, nor the most likely value (or even a possible value).
- Rather, the expected is the value we expect to see on average in the long run over many repetitions.
- A probability can be interpreted as a long run relative frequency; an expected value can be interpreted as a long run average value.

Example 3.6. Continuing Example 3.5. We considered what we would expect for 100000 hypothetical policies, but what about an unspecified large number of policies?

1. Imagine that we have recorded the profit for each of a large number of policies (not necessarily 100000). Explain in words the process by which you would compute the average profit per policy. (In other, more general, words: how do you compute an average of a list of numbers?)

2. Given that the profit of any policy is either 1000, -4000, -19000, or -49000, how could we simplify the calculation of the sum in the previous part? Write a general expression for the average profit per policy in this scenario.

3. What do you think the expression in the previous part converges to in the long run?

4. Explain how the value in the previous part is a “probability-weighted average value”.

5. Compute the expected value of damage (not profit) as a probability-weighted average value.

6. Interpret the value from the previous part as a long run average value in this context.

7. How is the expected value of profit related to the expected value of damage? Does this make sense? Why?

- An expected value can be computed as a “probability-weighted average value”
- But this is just a more compact way of computing an average in the usual way: add up all the values and divide by the number of values.

4 Outcomes, Events, and Random Variables

Probability models can be applied to any situation in which there are multiple potential outcomes and there is uncertainty about which outcome will occur. We'll introduce the basic mathematical objects associated with a probability model: outcomes, events, and random variables.

4.1 Outcomes

- Due to the wide variety of types of random phenomena, an **outcome** can be virtually anything
- In particular, an outcome does *not* have to be a number.
- The **sample space**, denoted Ω (the uppercase Greek letter "omega"), is the set of all possible outcomes of a random phenomenon. An **outcome**, denoted ω (the lowercase Greek letter "omega"), is an element of the sample space: $\omega \in \Omega$.
- Mathematically, the sample space Ω is a *set* containing all possible outcomes, while an individual outcome ω is a *point* or *element* in Ω.
- A random phenomenon is modeled by a *single* sample space, with respect to which all objects (events, random variables) are defined. Whenever possible, a sample space outcome should be defined to provide the maximum amount of information about the outcome of random phenomenon.
- In practice we rarely enumerate the sample space as we'll for some of the examples in this class. Nonetheless, there is always some underlying sample space corresponding to all possible outcomes of the random phenomenon.

Example 4.1. Roll a four-sided die.

1. If we only consider a single roll of the die, what is an appropriate sample space?

2. Now consider two rolls of the die, and record the result of each roll in sequence as an ordered pair. For example, the outcome $(3, 1)$ represents a 3 on the first roll and a 1 on the second; this is not the same outcome as $(1, 3)$. How many possible outcomes are there? Identify the sample space.

3. We might be interested in the sum of the two dice. Explain why it is still advantageous to define the sample space as in the previous part, rather than as $\Omega = \{2, \ldots, 8\}$.

- **Multiplication principle for counting** Suppose that stage 1 of a process can be completed in any one of n_1 ways. Further, suppose that for each way of completing the stage 1, stage 2 can be completed in any one of n_2 ways. Then the two-stage process can be completed in any one of $n_1 \times n_2$ ways.
- This rule extends naturally to a ℓ-stage process, which can then be completed in any one of $n_1 \times n_2 \times n_3 \times \cdots \times n_\ell$ ways.
- It is not important whether there is a "first" or "second" stage. What is important is that there are distinct stages, each with its own number of "choices".

Example 4.2. Regina and Cady plan to meet for lunch. They will definitely arrive between noon and 1, but their exact arrival times are uncertain. Rather than dealing with clock time, it is helpful to represent noon as time 0 and measure time as minutes after noon, including fractions of a minute, so that arrival times take values in the continuous interval [0, 60].

1. If we only consider Regina's arrival time, specify an appropriate sample space.

2. Now consider both arrival times and specify an appropriate definition of an outcome and draw a picture representing the sample space.

4.2 Events

- An event is something that could happen or might be true.
- An *event* is a collection of outcomes that satisfy some criteria.
- Mathematically, an **event** A is a *subset* of the sample space: $A \subseteq \Omega$.
- Events are typically denoted with capital letters near the start of the alphabet, with or without subscripts (e.g. A, B, C, A_1, A_2). Events can be composed from others using basic set operations like unions ($A \cup B$), intersections ($A \cap B$), and complements (A^c).
  - Read A^c as "not A".
  - Read $A \cap B$ as "A and B"
  - Read $A \cup B$ as "A or B". Note that unions ($\cup$, "or") are always inclusive. $A \cup B$ occurs if A occurs but B does not, B occurs but A does not, or both A and B occur.
- A collection of events $A_1, A_2, \ldots$ are **disjoint** (a.k.a. mutually exclusive) if $A_i \cap A_j = \emptyset$ for all $i \neq j$. That is, multiple events are disjoint if none of the events have any outcomes in common.
- If the sample space outcomes are represented by rows in a spreadsheet, then an event is a subset of rows that satisfies some criteria

Example 4.3. The "matching problem" is one well known probability problem. The general setup involves n distincts objects labeled $1, \ldots, n$ which are placed in n distinct boxes labeled $1, \ldots, n$, with exactly one object placed in each box.

1. Consider the matching problem with $n = 3$. Label the objects 1, 2, 3, and the spots 1, 2, 3, with spot 1 the correct spot for object 1, etc. Specify an appropriate definition of an outcome, determine the number of outcomes, and specify the sample space.

2. Using the sample space from the previous part, identify the following events.

   a. B, the event that no objects are put in the correct spot.

b. In words, what does B^c represent?

c. A, the event that all objects are put in the correct spot.

d. C, the event that exactly 2 objects are put in the correct spot.

e. A_3, the event that object 3 is put (correctly) in spot 3.

3. For a general n, how many possible outcomes are there?

4. For a general n let A the event that all objects are put in the correct spot, let B the event that no objects are put in the correct spot, and let A_i be the event that object i is put (correctly) in spot i, $i = 1, \ldots, n$. What is the relationship between A and $A_1, \ldots, A_n$? What is the relationship between B and $A_1, \ldots, A_n$?

Example 4.4. Using the sample space from Example 4.2, identify the following events using pictures.

1. Identify A, the event that Regina arrives after Cady.

2. Identify B, the event that either Regina or Cady arrives before 12:30.

3. Identify C, the event that they arrive within 15 minutes of each other.

4. Identify D, the event that Regina arrives before 12:24.

4.3 Random variables

- Roughly, a *random variable* assigns a number measuring some quantity of interest to each outcome of a random phenomenon.
- Mathematically, a **random variable (RV)** X is a *function* that takes an outcome in the sample space as input and returns a real number as output
- The random variable itself is typically denoted with a capital letter (X); possible values of that random variable are denoted with lower case letters (x).
  - Think of the capital letter X as a label standing in for a formula like "the number of heads in 4 flips of a coin" and
  - x as a dummy variable standing in for a particular value like 3.
- **Discrete random variables** take at most countably many possible values (e.g., $0, 1, 2, ...$). They are often counting variables (e.g., the number of Heads in 10 coin flips).
- **Continuous random variables** can take any real value in some interval (e.g., $[0, 1]$, $[0, \infty)$, $(-\infty, \infty)$.). That is, continuous random variables can take uncountably many

different values. Continuous random variables are often measurement variables (e.g., height, weight, income).

- A function of a random variable is also a random variable: if X is a RV then so is $g(X)$
- Sums and products, etc., of random variables *defined on the same sample space* are random variables. IfX and Y are RVs defined on the same sample space then so are $X+Y$, $X-Y$, XY
- If the sample space outcomes are represented by rows in a spreadsheet, then random variables correspond to columns.
- Expressions like $X=x$ or $\{X=x\}$ represent *events*: for which outcomes is the value of the random variable X equal to the value x

Example 4.5. Roll a four-sided die twice, and record the result of each roll in sequence. Recall the sample space from Example 4.1. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same).

Outcome	X	Y
(1, 1)		
(1, 2)		
(1, 3)		
(1, 4)		
(2, 1)		
(2, 2)		
(2, 3)		
(2, 4)		
(3, 1)		
(3, 2)		
(3, 3)		
(3, 4)		
(4, 1)		
(4, 2)		
(4, 3)		
(4, 4)		

1. Complete the table to identify the values of X and Y for each outcome in the sample space.

2. Identify $\{Y=1\}$.

3. Identify $\{Y = 3\}$.

4. Identify $\{X \leq 4\}$.

5. Identify $\{X \leq 4, Y = 3\}$

Example 4.6. Matching problem. For $n = 3$ objects labeled 1, 2, 3, are placed at random in spots labeled 1, 2, 3, with spot 1 the correct spot for object 1, etc. Let the random variable X count the number of objects that are put in the correct spot. Let I_1 be equal to 1 if object 1 is placed (correctly) in spot 1, and define I_2, I_3 similarly.

Outcome	X	I_1	I_2	I_3
123				
132				
213				
231				
312				
321				

1. Complete the table to identify the value of X, I_1, I_2, I_3 for each outcome in the sample space.
2. What is the relationship between I_3 and event A_3 from Example 4.3?

3. How can you express X in terms of I_1, I_2, I_3?

- The **indicator (a.k.a., Bernoulli) random variable** corresponding to event A is equal to 1 if A occurs and 0 otherwise
- Indicator random variables can be used for incremental counting and are often useful in problems involving "find the expected number of"

Example 4.7. Regina and Cady will definitely arrive between noon and 1, but their exact arrival times are uncertain. Recall the sample space from Example 4.2. Let R be the random variable representing Regina's arrival time (minutes after noon), and Y for Cady.

1. What does the random variable $T = \min(R, Y)$ represent? What are the possible values of T?

2. What does the random variable $W = |R - Y|$ represent? What are the possible values of W?

3. Let N be the number of people (out of 2) who arrive before 12:30. How can you represent N in terms of R and Y. (Hint: use indicators.)

4. Identify each of the random variables above as discrete or continuous.

5. Interpret each of the following in words and draw a picture representing it.

   a. $\{R > Y\}$.

   b. $\{T < 30\}$.

   c. $\{W < 15\}$.

   d. $\{R < 24\}$.

5 Probability Models

- A **probability measure**, typically denoted P, assigns probabilities to *events* to quantify their relative likelihoods according to the assumptions of the model of the random phenomenon.
- The probability of event A, computed according to probability measure $\mathrm{P}(A)$, is denoted $\mathrm{P}(A)$.
- A valid probability measure P must satisfy the following three logical consistency "axioms".
  - For any event A, $0 \le \mathrm{P}(A) \le 1$.
  - If Ω represents the sample space then $\mathrm{P}(\Omega) = 1$.
  - (*Countable additivity.*) If $A_1, A_2, A_3, \ldots$ are disjoint then

$$\mathrm{P}(A_1 \cup A_2 \cup A_3 \cup \cdots) = \mathrm{P}(A_1) + \mathrm{P}(A_2) + \mathrm{P}(A_3) + \cdots$$

- Additional properties of a probability measure follow from the axioms
  - *Complement rule.* For any event A, $\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$.
  - *Subset rule.* If $A \subseteq B$ then $\mathrm{P}(A) \le \mathrm{P}(B)$.
  - *Addition rule for two events.* If A and B are any two events

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

  - *Law of total probability.* If $C_1, C_2, C_3 \ldots$ are disjoint events with $C_1 \cup C_2 \cup C_3 \cup \cdots = \Omega$, then

$$\mathrm{P}(A) = \mathrm{P}(A \cap C_1) + \mathrm{P}(A \cap C_2) + \mathrm{P}(A \cap C_3) + \cdots$$

- A **probability model** (or **probability space**) is the collection of all outcomes, events, and random variables associated with a random phenomenon along with the probabilities of all events of interest under the assumptions of the model.
- The axioms of a probability measure are minimal logical consistent requirements that ensure that probabilities of different events fit together in a valid, coherent way.

Example 5.1. Suppose that the playoff for some sport is down to the final 4 teams, which we'll label teams 1, 2, 3, 4 (in no particular order). We're interested in which team will win the championship. The sample space is $\{1, 2, 3, 4\}$. Table 5.1 lists all possible *events*.

1. Add a description in words for each of the events.

2. Pat assumes that the four teams are equally likely to win the championship. Let P denote Pat's correponding probability measure. Compute $\mathrm{P}(A)$ for each event in Table 5.1.

Table 5.1: All possible events associated with a single roll of a four-sided die.

A	Description	$\mathrm{P}(A)$	$\mathrm{Q}(A)$	$\tilde{\mathrm{P}}(A)$
$\emptyset$				
$\{1\}$				
$\{2\}$				
$\{3\}$				
$\{4\}$				
$\{1,2\}$				
$\{1,3\}$				
$\{1,4\}$				
$\{2,3\}$			0.5	
$\{2,4\}$				
$\{3,4\}$			0.7	
$\{1,2,3\}$			0.6	
$\{1,2,4\}$				
$\{1,3,4\}$				
$\{2,3,4\}$				
$\{1,2,3,4\}$				

Example 5.2. Continuing Example 5.1. Quentin assumes the teams are not equally likely to win the championship. Let Q denote Quentin's probability measure. A few probabilities are provided in Table 5.1; compute $\mathrm{Q}(A)$ for all other events in Table 5.1. What is Quentin's assumed probability that each team wins the championship?

Example 5.3. Continuing Example 5.1. Tilda assumes that

- Team 1 is twice as likely to win the championship as team 4
- Team 2 is three times as likely to win the championship as team 4
- Team 3 is 1.5 times as likely as to win the championship as team 4

Let $\tilde{\mathrm{P}}$ denote Tilda's probability measure. Compute $\tilde{\mathrm{P}}(A)$ for all events in Table 5.1. What is Tilda's assumed probability that each team wins the championship?

- The axioms of a probability measure are minimal logical consistent requirements that ensure that probabilities of different events fit together in a valid, coherent way.
- In particular, the axioms do NOT require outcomes to be equally likely
- A single probability measure corresponds to a particular set of assumptions about the random phenomenon.
- There can be many probability measures defined on a single sample space, each one corresponding to a different probability model for the random phenomenon.
- Probabilities of events can change if the probability measure changes.

Example 5.4. The general meeting problem involves multiple people, but we'll first consider the arrival time of just a single person, who we'll call Han. Suppose that Han arrives "uniformly at random" at a time in $[0, 60]$.

1. Compute the probability that Han arrives before 12:15.

2. Compute the probability that Han arrives after 12:45.

3. Let P denote the corresponding probability measure. Suggest a general formula for $\mathrm{P}([a,b])$, the probability that Han arrives between a and b minutes after noon for $0 \leq a < b \leq 60$.

4. Compute the probability that Han's arrival time, truncated to the nearest minute, is 0 minutes after noon; that is, find the probability that Han arrives between 12:00 and 12:01.

5. Continue to compute the probability that Han's arrival time truncated to the nearest minute is $1, 2, 3, \ldots, 59$, and sketch a plot with arrival time (truncated minutes after noon) on the horizontal axis and probability on the vertical axis. Is the plot what you would expect for arriving "uniformly at random"?

Example 5.5. Assume that the probability that Han arrives between a and b minutes after noon is $(b/60)^2 - (a/60)^2$, for $0 \leq a < b \leq 60$. Let Q denote the corresponding probability measure; notice that $\mathrm{Q}([0,60]) = (60/60)^2 - (0/60)^2 = 1$. Compute the following probabilities and compare your answers to the corresponding parts from Example 5.4.

1. Compute the probability that Han arrives before 12:15.

2. Compute the probability that Han arrives after 12:45.

3. Compute the probability Han's arrival time, truncated to the nearest minute, is 0 minutes after noon; that is, find the probability that Han arrives between 12:00 and 12:01.

4. Compute the probability Han's arrival time, truncated to the nearest minute, is 59 minutes after noon; that is, find the probability that Han arrives between 12:59 and 1:00.

5. Continue to compute the probability that Han's arrival time truncated to the nearest minute is $1, 2, 3, \ldots, 59$, and sketch a plot with arrival time (truncated minutes after noon) on the horizontal axis and probability on the vertical axis. What assumptions about Han's arrival time does this probability measure reflect?

Example 5.6. Continuing with the uniform probability measure $\mathbf{P}$ of Example 5.4.

1. Compute the probability that Han arrives between 12:00 and 12:01, within 1 minute after noon.

2. Compute the probability that Han arrives between 12:00:00 and 12:00:01, within 1 second after noon.

3. Compute the probability that Han arrives between 12:00:00.000 and 12:01:00.001, within 1 millisecond after noon.

4. Compute the probability that Han arrives at the exact time 12:00:00.00000... (with infinite precision).

5. What is the probability that Han arrives "at noon"? Discuss.

Example 5.7. Continuing with the non-uniform probability measure **Q** of Example 5.5.

1. Compute the probability that Han arrives between 12:00 and 12:01, within 1 minute after noon.

2. Compute the probability that Han arrives between 12:00:00 and 12:00:01, within 1 second after noon.

3. Compute the probability that Han arrives at the exact time 12:00:00.00000... (with infinite precision).

4. Compute the probability that Han arrives between 12:59 and 1:00, within 1 minute before 1:00.

5. Compute the probability that Han arrives between 12:59:59 and 1:00:00, within 1 second before 1:00.

6. Compute the probability that Han arrives at the exact time 1:00:00.00000... (with infinite precision).

7. Which is more likely: that Han arrives "at noon" or "at 1:00"? Discuss.

- For a continuous sample space, the probability of any particular outcome is 0.
- Particular outcomes represent "infinite precision" which is not practical in real applications
- For continuous sample spaces it makes more sense to consider "close to" probabilities rather than "equals to" probabilities.
- "Close to" events correspond to *intervals* of reasonable practical precision and these intervals can have non-zero probability.
- Certain outcomes can be more likely than others in the "close to" sense.

Example 5.8. Back to the Regina, Cady meeting problem. Assume that Regina and Cady each arrive at a time uniformly at random between noon and 1:00, independently of each other, so that they arrive "uniformly at random" in the sample space of Example 4.2. Let P denote the corresponding probability measure.

Let R be the random variable representing Regina's arrival time (minutes after noon), and Y for Cady, and let $T = \min(R, Y)$ and $W = |R - Y|$.

Compute and interpret the following.

1. $\mathrm{P}(R > Y)$

2. $\mathrm{P}(T < 30)$

3. $\mathrm{P}(W < 15)$

4. $\mathrm{P}(R < 24)$

5. $\mathrm{P}(W < 1)$

6. $\mathrm{P}(W < 1/60)$

7. $\mathrm{P}(W = 0)$

8. What is the probability that Regina and Cady arrive “at the same time”? Discuss.

6 Distributions of Random Variables (A Brief Introduction)

- The **joint (probability) distribution** of a collection of random variables identifies the possible values that the random variables can take and their relative likelihoods.
- We will see many ways of describing a distribution, depending on how many random variables are involved and their types (discrete or continuous).
- In the context of multiple random variables, the distribution of any one of the random variables is called a **marginal distribution**.

Example 6.1. Roll a fair four-sided die twice. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same).

1. Construct a table and plot displaying the marginal distribution of Y.

2. Describe the distribution of Y in terms of long run relative frequency.

3. Describe the distribution of Y in terms of relative degree of likelihood.

4. Construct a table and plot displaying the joint distribution of X and Y.

5. Construct a table and plot displaying the marginal distribution of X.

- The expected value $\mathrm{E}(X)$ of a discrete random variable X is defined by the probability-weighted average according to the underlying probability measure.
- The expected value of a random variable can be interpreted as the long-run average value of the random variable

Exercise 6.1. Consider the matching problem with $n = 4$: objects labeled 1, 2, 3, 4, are placed at random in spots labeled 1, 2, 3, 4, with spot 1 the correct spot for object 1, etc. Let the random variable X count the number of objects that are put back in the correct spot. Let P denote the probability measure corresponding to the assumption that the objects are equally likely to be placed in any spot, so that the 24 possible placements are equally likely.

1. Find the distribution of X by creating an appropriate table and plot.

2. Compute $\mathrm{E}(X)$.

3. Is the value from part 2 the most likely value of X? Explain.

4. Is the value from part 2 the value that we would "expect" to see for X in a single repetition of the phenomenon? Explain.

5. Explain in what sense the value from part 2 is “expected”.

7 Conditioning

- Conditioning concerns how probabilities of events or distributions of random variables are influenced by information about the occurrence of events or the values of random variables.
- A probability is a measure of the likelihood or degree of uncertainty of an event. A *conditional probability* revises this measure to reflect any "new" information about the outcome of the underlying random phenomenon.

Example 7.1. The probability that a randomly selected American adult believes in human-driven climate change[1] is 0.54.

The probability[2] that a randomly selected American adult is a Democrat is 0.28.

1. Donny Don't says that the probability that a randomly selected American adult both (1) is a Democrat, *and* (2) believes in human-driven climate change is equal to 0.28×0.54. Do you agree?

2. Suppose that the probability that a randomly selected American adult both is a Democrat and believes in human-driven climate change is 0.19. Construct an appropriate two-way table of probabilities.

3. Compute the probability that a randomly selected American adult *who is a Democrat* believes in human-driven climate change.

[1]Probabilities are estimated based on this 2024 survey.
[2]Estimate based on Gallup poll

4. Compute the probability that a randomly selected American adult *who believes in human-driven climate change* is a Democrat.

5. How can the probability in the two previous parts be written in terms of the probabilities provided (0.54, 0.28, 0.19)?

- The **conditional probability of event A given event B**, denoted $\mathrm{P}(A|B)$, is defined as (provided $\mathrm{P}(B) > 0$)
$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$$
- The conditional probability $\mathrm{P}(A|B)$ represents how the likelihood or degree of uncertainty of event A should be updated to reflect information that event B has occurred.
- In general, knowing whether or not event B occurs influences the probability of event A. That is,
$$\text{In general, } \mathrm{P}(A|B) \neq \mathrm{P}(A)$$
- Be careful: order is essential in conditioning. That is,
$$\text{In general, } \mathrm{P}(A|B) \neq \mathrm{P}(B|A)$$
- Within the context of two events, we have joint, conditional, and marginal probabilities.
  - Joint: unconditional probability involving both events, $\mathrm{P}(A \cap B)$.
  - Conditional: conditional probability of one event given the other, $\mathrm{P}(A|B)$, $\mathrm{P}(B|A)$.
  - Marginal: unconditional probability of a single event $\mathrm{P}(A)$, $\mathrm{P}(B)$.
- The relationship $\mathrm{P}(A|B) = \mathrm{P}(A \cap B)/\mathrm{P}(B)$ can be stated generically as
$$\text{conditional} = \frac{\text{joint}}{\text{marginal}}$$
- In many problems conditional probabilities are provided or can be determined directly.

Example 7.2. Suppose that

- 67% of Democrats believe in human-driven climate
- 46% of Independents believe in human-driven climate
- 34% of Republicans believe in human-driven climate

Also suppose that

- 28% of American adults are Democrats
- 42% of American adults are Independents
- 30% of American adults are Republicans

1. Define the event A to represent "believes in human-driven climate change" and D, I, R to correspond to affiliation in each of the parties. If the probability measure P corresponds to selecting an American adult uniformly at random, write all the percentages above as probabilities using proper notation.

2. Construct an appropriate two-way table of probabilities.

3. Now suppose that the randomly selected American believes in human-driven climate change. How does this information change the probability that the selected American belongs to each political party? Answer by computing appropriate probabilities (and using proper notation).

4. How many times more likely is it for an American *adult* to believe in human-driven climate change and be Independent than to:

   a. believe in human-driven climate change and be Democrat

b. believe in human-driven climate change and be Republican

5. How many times more likely is it for an American adult *who believes in human-driven climate change* to be Independent than to be:

   a. Democrat

   b. Republican

6. What do you notice about the answers to the two previous parts?

- The process of conditioning can be thought of as **"slicing and renormalizing".**
  - Extract the "slice" corresponding to the event being conditioned on (and discard the rest). For example, a slice might correspond to a particular row or column of a two-way table.
  - "Renormalize" the values in the slice so that corresponding probabilities add up to 1.
- Slicing determines the *shape*; renormalizing determines the *scale*.
- Slicing determines relative probabilities; renormalizing just makes sure they add up to 1.

Example 7.3. Roll a fair four-sided die twice. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). The table below represents the joint distribution of X and Y.

$x \setminus y$	1	2	3	4
2	1/16	0	0	0
3	0	2/16	0	0
4	0	1/16	2/16	0
5	0	0	2/16	2/16
6	0	0	1/16	2/16
7	0	0	0	2/16
8	0	0	0	1/16

1. Compute and interpret in context $\mathrm{P}(X = 5|Y = 4)$.

2. Construct a table to represent the conditional distribution of X given $Y = 4$ by "slicing and renormalizing".

3. Interpret the the conditional distribution of X given $Y = 4$ as a long run relative frequency distribution.

4. Compute $\mathrm{E}(X|Y = 4)$.

5. Interpret the value from the previous part as a long run average value in context.

6. Construct a table to represent the conditional distribution of X given $Y = 3$, and

compute $\mathrm{E}(X|Y = 3)$

7. Construct a table to represent the conditional distribution of X given $Y = 2$, and compute $\mathrm{E}(X|Y = 2)$

8. Construct a table to represent the conditional distribution of X given $Y = 1$, and compute $\mathrm{E}(X|Y = 1)$

8 Probability Rules

8.1 Multiplication rule

- Rearranging the definition of conditional probability we get the **Multiplication rule:** the probability that two events both occur is

$$\begin{aligned} P(A \cap B) &= P(A|B)P(B) \\ &= P(B|A)P(A) \end{aligned}$$

- The multiplication rule says that you should think "multiply" when you see "and".
- However, be careful about *what* you are multiplying: to find a joint probability you need an unconditional and an appropriate conditional probability.
- You can condition either on A or on B, provided you have the appropriate marginal probability; often, conditioning one way is easier than the other.
- Be careful: the multiplication rule does *not* say that $P(A \cap B)$ is the same as $P(A)P(B)$.

- The multiplication rule is useful in situations where conditional probabilities are easier to obtain directly than joint probabilities.

Example 8.1. A standard deck of playing cards has 52 cards, 13 cards (2 through 10, jack, king, queen, ace) in each of 4 suits (hearts, diamonds, clubs, spades). Shuffle a deck and deals cards one at a time without replacement.

1. Find the probability that the first card dealt is a heart.

2. If the first card dealt is a heart, determine the conditional probability that the second card is a heart.

3. Find the probability that the first two cards dealt are hearts.

4. Find the probability that the first two cards dealt are hearts and the third card dealt is a diamond.

- The multiplication rule extends naturally to more than two events (though the notation gets messy). For three events, we have

$$\mathrm{P}(A_1 \cap A_2 \cap A_3) = \mathrm{P}(A_1)\mathrm{P}(A_2|A_1)\mathrm{P}(A_3|A_1 \cap A_2)$$

- And in general,

$$\mathrm{P}(A_1 \cap A_2 \cap A_3 \cap A_4 \cap \cdots) = \mathrm{P}(A_1)\mathrm{P}(A_2|A_1)\mathrm{P}(A_3|A_1 \cap A_2)\mathrm{P}(A_4|A_1 \cap A_2 \cap A_3)\cdots$$

- The multiplication rule is useful for computing probabilities of events that can be broken down into component "stages" where conditional probabilities at each stage are readily available. At each stage, condition on the information about all previous stages.

Example 8.2. The birthday problem concerns the probability that at least two people in a group of n people have the same birthday[1]. Ignore multiple births and February 29 and assume that the other 365 days are all equally likely[2].

1. If $n = 30$, what do you think the probability that at least two people share a birthday is: 0-20%, 20-40%, 40-60%, 60-80%, 80-100%? How large do you think n needs to be in order for the probability that at least two people share a birthday to be larger than 0.5? Just make guesses before proceeding to calculations.

[1] You should really click on this birthday problem link.

[2] Which isn't quite true. However, a non-uniform distribution of birthdays only increases the probability that at least two people have the same birthday. To see that, think of an extreme case like if everyone were born in September.

2. Now consider $n = 3$ people, labeled 1, 2, and 3. What is the probability that persons 1 and 2 have different birthdays?

3. What is the probability that persons 1, 2, and 3 all have different birthdays *given* that persons 1 and 2 have different birthdays?

4. What is the probability that persons 1, 2, and 3 all have different birthdays?

5. When $n = 3$. What is the probability that at least two people share a birthday?

6. For $n = 30$, find the probability that at least two people have the same birthday.

7. Write a clearly worded sentence interpreting the probability in the previous part as a long run relative frequency.

8. When $n = 100$ the probability is about 0.9999997. If you are in a group of 100 people and no one shares your birthday, should you be surprised? Discuss.

8.2 Law of total probability

Example 8.3. There exist multiple penguin species throughout Antarctica, including the *Adelie* (A), *Chinstrap* (C), and *Gentoo* (G). Suppose that among Antarctic penguins[3] of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Suppose that 83.4% of Adelie penguins are below average weight, 89.7% of Chinstrap penguins are below average weight, and 4.9% of Gentoo penguins are below average weight (that is, below 4200g).

Randomly select a penguin. Compute the probability that it is below average weight.

- **Law of total probability.** If $C_1, \dots, C_k$ are disjoint with $C_1 \cup \cdots \cup C_k = \Omega$, then

$$\begin{aligned} \mathrm{P}(A) &= \sum_{i=1}^{k} \mathrm{P}(A \cap C_i) \\ &= \sum_{i=1}^{k} \mathrm{P}(A|C_i)\mathrm{P}(C_i) \end{aligned}$$

- The events $C_1, \dots, C_k$, which represent the "cases", form a *partition* of the sample space; each outcome $\omega \in \Omega$ lies in exactly one of the C_i.
- The law of total probability says that we can interpret the unconditional probability $\mathrm{P}(A)$ as a probability-weighted average of the case-by-case conditional probabilities $\mathrm{P}(A|C_i)$ where the weights $\mathrm{P}(C_i)$ represent the probability of encountering each case.
- Conditioning and using the law of probability is an effective strategy in solving many problems, even when the problem doesn't seem to involve conditioning.
- For example, when a problem involves iterations or steps it is often useful to *condition on the result of the first step.*

Example 8.4. You and your friend are playing the "lookaway challenge".

The game consists of possibly multiple rounds. In the first round, you point in one of four directions: up, down, left or right. At the exact same time, your friend also looks in one of those four directions. If your friend looks in the same direction you're pointing, you win! Otherwise, you switch roles and the game continues to the next round — now your friend points in a direction and you try to look away. As long as no one wins, you keep switching

[3] Throughout "penguin" refers to an Antarctic penguin of one of these three species. This example is based on the famous Palmer penguins data set.

off who points and who looks. The game ends, and the current "pointer" wins, whenever the "looker" looks in the same direction as the pointer.

Suppose that each player is equally likely to point/look in each of the four directions, independently from round to round. What is the probability that you win the game?

1. Why might you expect the probability to not be equal to 0.5?

2. If you start as the pointer, what is the probability that you win in the first round?

3. If p denotes the probability that the player who starts as the pointer wins the game, what is the probability that the player who starts as the looker wins the game? (Note: p is the probability that the person who starts as pointer wins the whole game, not just the first round.)

4. Let A be the event that the person who starts as the pointer wins the game, and B be the event that the person who starts as the pointer wins in the first round. What is $\mathrm{P}(A|B)$?

5. Find a simple expression for $\mathrm{P}(A|B^c)$ in terms of p. The key is to consider this question: if the player who starts as the pointer does not win in the first round, how does the game behave from that point forward?

6. Condition on the result of the first round and set up an equation to solve for p.

7. Interpret the probability from the previous part.

8.3 Bayes Rule

- **Bayes' rule for events** specifies how a prior probability $P(H)$ of event H is updated in response to the evidence E to obtain the posterior probability $P(H|E)$.

$$\mathrm{P}(H|E) = \frac{\mathrm{P}(E|H)\mathrm{P}(H)}{\mathrm{P}(E)}$$

- Event H represents a particular hypothesis (or model or case)
- Event E represents observed evidence (or data or information)
- $\mathrm{P}(H)$ is the unconditional or **prior probability** of H (prior to observing evidence E)
- $\mathrm{P}(H|E)$ is the conditional or **posterior probability** of H after observing evidence E.
- $\mathrm{P}(E|H)$ is the **likelihood** of evidence E given hypothesis (or model or case) H
- Bayes rule is often used when there are multiple hypotheses or cases. Suppose $H_1, \dots, H_k$ is a series of distinct hypotheses which together account for all possibilities, and E is any event (evidence).
- Combining Bayes' rule with the law of total probability,

$$\begin{aligned} P(H_j|E) &= \frac{P(E|H_j)P(H_j)}{P(E)} \\ &= \frac{P(E|H_j)P(H_j)}{\sum_{i=1}^k P(E|H_i)P(H_i)} \end{aligned}$$

Example 8.5. Continuing Example 8.3. Suppose that among Antarctic penguins of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Suppose that 83.4% of Adelie penguins are below average weight, 89.7% of Chinstrap penguins are below average weight, and 4.9% of Gentoo penguins are below average weight (that is, below 4200g).

Randomly select a penguin and suppose it is below average weight. We are interested in classifying the species of the penguin.

1. Frame this problem in the context of Bayes rule: Identify the hypotheses, prior probabilities, evidence, likelihood, and posterior probabilities.

2. Prior to observing the penguin's weight, how many times more likely is it to be Adelie than to be Gentoo?

3. How many times more likely is it for a penguin to be below average weight given that it is Adelie than given that it is Gentoo?

4. Compute the posterior probability of each species given that the penguin is below average weight. Given that the penguin is below average weight, how many times more likely is it to be Adelie than to be Gentoo?

5. How are the ratios in the three previous parts related?

- Bayes rule is often used when there are multiple hypotheses or cases. Suppose $H_1, \dots, H_k$ is a series of distinct hypotheses which together account for all possibilities, and E is any event (evidence).

$$
\begin{aligned}
P(H_j|E) &= \frac{P(E|H_j)P(H_j)}{P(E)} \\
&= \frac{P(E|H_j)P(H_j)}{\sum_{i=1}^{k} P(E|H_i)P(H_i)}
\end{aligned}
$$

$$
P(H_j|E) \propto P(E|H_j)P(H_j)
$$

- The symbol $\propto$ is read "is proportional to". The relative *ratios* of the posterior probabilities of different hypotheses are determined by the product of the prior probabilities and the likelihoods, $P(E|H_j)P(H_j)$.
- The marginal probability of the evidence, $P(E)$, in the denominator simply normalizes the numerators to ensure that the updated conditional probabilities given the evidence sum to 1 over all the distinct hypotheses.
- **In short, Bayes' rule says that posterior is proportional to the product of prior and likelihood**

$$
\textbf{posterior} \propto \textbf{prior} \times \textbf{likelihood}
$$

Example 8.6. Continuing Example 8.5. Suppose that among Antarctic penguins of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Suppose that 83.4% of Adelie penguins are below average weight, 89.7% of Chinstrap penguins are below average weight, and 4.9% of Gentoo penguins are below average weight (that is, below 4200g).

Randomly select a penguin. We are interested in classifying the species of the penguin.

1. Before you know the weight of the penguin (or any other information) what specify would you predict it to be? Why? What is the probability that your classification is correct?

2. If the penguin is below average weight, what species would you classify it as? Why?

3. Now suppose that the penguin is not below average weight. Use a Bayes table to compute the posterior probability of each species given that the penguin is not below average weight. If the penguin is not below average weight, what species would you classify it as? Why?

4. Suppose that you classify any randomly selected penguin based on whether or not it is below average weight as in the two previous parts. What is the posterior probability that you classify the penguin correctly? How does this compare to your probability of being correct before you observed the penguin's weight?

- Bayesian analysis is often an iterative process.
- Posterior probabilities are updated after observing some information or data. These probabilities can then be used as prior probabilities before observing new data.
- Posterior probabilities can be sequentially updated as new data becomes available, with the posterior probabilities after the previous stage serving as the prior probabilities for the next stage.
- The final posterior probabilities only depend upon the cumulative data. It doesn't matter if we sequentially update the posterior after each new piece of data or only once after all the data is available; the final posterior probabilities will be the same either way. Also, the final posterior probabilities are not impacted by the order in which the data are observed.

Example 8.7. Suppose that you are presented with six boxes, labeled 0, 1, 2, ..., 5, each containing five marbles. Box 0 contains 0 green and 5 gold marbles, box 1 contains 1 green and 4 gold, and so on with box i containing i green and $5-i$ gold. One of the boxes is chosen uniformly at random (perhaps by rolling a fair six-sided die), and then you will randomly select marbles from that box, without replacement. Imagine the boxes appear identical and you can't see inside; all you observe is the color of the marbles you select. Based on the colors of the marbles selected, you will update the probabilities of which box had been chosen.

1. Suppose that a single marble is selected and it is green. Which box do you think is the most likely to have been chosen? Make a guess for the posterior probabilities for each box. Then construct a Bayes table to compute the posterior probabilities. How do they compare to the prior probabilities?

2. Now suppose a second marble is selected from the same box, without replacement, and its color is gold. Which box do you think is the most likely to have been chosen given these two marbles? Make a guess for the posterior probabilities for each box. Then construct a Bayes table to compute the posterior probabilities, *using the posterior probabilities from the previous part after the selection of the green marble as the new prior probabilities before seeing the gold marble.*

3. Now construct a Bayes table corresponding to the original prior probabilities (1/6 each) and the combined evidence that the first ball selected was green and the second was gold. How do the posterior probabilities compare to the previous part?

9 Independence

Example 9.1. Do Americans think it is important for federal judges to be impartial in how they decide court cases? Consider the following data from the Pew Research Center. (Indepedendents are classified as either lean Democrat or lean Republican.)

	Democrat/lean Dem (D)	Republican/lean Rep (D^c)	Total
Important (I)	1674	1531	3205
Not important (I^c)	146	133	279
Total	1820	1664	3484

Suppose a person is randomly selected from this sample. Consider the events

$$I = \{\text{person thinks it is important for judges to be impartial}\}$$
$$D = \{\text{person is a Democrat/leans Dem}\}$$

1. Compute and interpret $\mathrm{P}(I)$.

2. Compute and interpret $\mathrm{P}(I|D)$.

3. Compute and interpret $\mathrm{P}(I|D^c)$.

4. What do you notice about $\mathrm{P}(I)$, $\mathrm{P}(I|D)$, and $\mathrm{P}(I|D^c)$?

5. Compute and interpret $\mathrm{P}(D)$.

6. Compute and interpret $\mathrm{P}(D|I)$.

7. Compute and interpret $\mathrm{P}(D|I^c)$.

8. What do you notice about $\mathrm{P}(D)$, $\mathrm{P}(D|I)$, and $\mathrm{P}(D|I^c)$?

9. Compute and interpret $\mathrm{P}(D \cap I)$.

10. What is the relationship between $\mathrm{P}(D \cap I)$ and $\mathrm{P}(D)$ and $\mathrm{P}(I)$?

11. When randomly selecting a person from this particular group, would you say that events

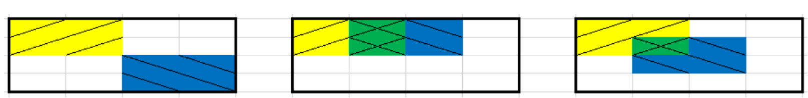
D and I are independent? Why?

- Events A and B are **independent** if the knowing whether or not one occurs does not change the probability of the other.
- For events A and B (with $0 < \mathrm{P}(A) < 1$ and $0 < \mathrm{P}(B) < 1$) the following are equivalent. That is, if one is true then they all are true; if one is false, then they all are false.

$$
\begin{aligned}
A \text{ and } B &\text{ are independent} \\
\mathrm{P}(A \cap B) &= \mathrm{P}(A)\mathrm{P}(B) \\
\mathrm{P}(A^c \cap B) &= \mathrm{P}(A^c)\mathrm{P}(B) \\
\mathrm{P}(A \cap B^c) &= \mathrm{P}(A)\mathrm{P}(B^c) \\
\mathrm{P}(A^c \cap B^c) &= \mathrm{P}(A^c)\mathrm{P}(B^c) \\
\mathrm{P}(A|B) &= \mathrm{P}(A) \\
\mathrm{P}(A|B) &= \mathrm{P}(A|B^c) \\
\mathrm{P}(B|A) &= \mathrm{P}(B) \\
\mathrm{P}(B|A) &= \mathrm{P}(B|A^c)
\end{aligned}
$$

Example 9.2. Each of the three Venn diagrams below represents a sample space with 16 equally likely outcomes. Let A be the yellow / event, B the blue \ event, and their intersection $A \cap B$ the green $\times$ event. Suppose that areas represent probabilities, so that for example $\mathrm{P}(A) = 4/16$.

In which of the scenarios are events A and B independent?



- Do not confuse "disjoint" with "independent".

- Disjoint means two events do not "overlap". Independence means two events *"overlap in just the right way"*.
- You can pretty much forget "disjoint" exists; you will naturally apply the addition rule for disjoint events correctly without even thinking about it.
- Independence is much more important and useful, but also requires more care.

Example 9.3. Roll two fair six-sided dice, one green and one gold. There are 36 total possible outcomes (roll on green, roll on gold), all equally likely. Consider the event $E = \{\text{the green die lands on } 1\}$. Answer the following questions by computing and comparing appropriate probabilities.

1. Consider $A = \{\text{the gold die lands on } 6\}$. Are A and E independent?

2. Consider $B = \{\text{the sum of the dice is } 2\}$. Are B and E independent?

3. Consider $C = \{\text{the sum of the dice is } 7\}$. Are C and E independent?

- Independence concerns whether or not the occurrence of one event affects the *probability* of the other.
- Given two events it is not always obvious whether or not they are independent.

- Independence depends on the underlying probability measure. Events that are independent under one probability measure might not be independent under another.
- Independence is often assumed. Whether or not independence is a valid assumption depends on the underlying random phenomenon.

Example 9.4. As in the birthday problem, assume that birthdays are distributed equally over 365 days. Randomly select 3 people.

- A_{12} is the event that the first and second persons selected have the same birthday

- A_{13} is the event that the first and third persons selected have the same birthday
- A_{23} is the event that the second and third persons selected have the same birthday

1. Are the two events A_{12} and A_{13} independent?

2. Are the two events A_{12} and A_{23} independent?

3. Are the two events A_{13} and A_{23} independent?

4. Are the three events A_{12}, A_{13}, and $A23$ independent?

- A collection of events $A_1, A_2, A_3, ...$ are **independent** if:
  - any pair of events $A_i, A_j, (i \neq j)$ satisfies $\mathrm{P}(A_i \cap A_j) = \mathrm{P}(A_i)\mathrm{P}(A_j)$,
  - and any triple of events A_i, A_j, A_k (distinct i, j, k) satisfies $\mathrm{P}(A_i \cap A_j \cap A_k) = \mathrm{P}(A_i)\mathrm{P}(A_j)\mathrm{P}(A_k)$,
  - and any quadruple of events satisfies $\mathrm{P}(A_i \cap A_j \cap A_k \cap A_m) = \mathrm{P}(A_i)\mathrm{P}(A_j)\mathrm{P}(A_k)\mathrm{P}(A_m)$,
  - and so on.
- Intuitively, a collection of events is independent if knowing whether or not any combination of the events in the collection occur does not change the probability of any other event in the collection.

Example 9.5. A certain system consists of four identical components. Suppose that the probability that any particular component fails is 0.1, and failures of the components occur independently of each other. Find the probability that the system fails if:

1. The components are connected in *parallel*: the system fails only if *all* of the components fail.

2. The components are connected in *series*: the system fails whenever *at least one* of the components fails.

3. Donny Don't says the answer to the previous part is $0.1+0.1+0.1+0.1=0.4$. Explain the error in Donny's reasoning.

- When events are independent, the multiplication rule simplifies greatly.

$$\mathrm{P}(A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_n) = \mathrm{P}(A_1)\mathrm{P}(A_2)\mathrm{P}(A_3)\cdots\mathrm{P}(A_n) \quad \text{if } A_1, A_2, A_3, \ldots, A_n \text{ are independent}$$

- When a problem involves independence, you will want to take advantage of it. Work with "and" events whenever possible in order to use the multiplication rule.
- For example, for problems involving "at least one" (an "or" event) take the complement to obtain "none" (an "and" event).

10 Introduction to Simulation

A probability model of a random phenomenon consists of a sample space of possible outcomes, associated events and random variables, and a probability measure which encapsulates the model assumptions and determines probabilities of events and distributions of random variables. We will study strategies for computing probabilities, distributions, expected values, and more, but in many situations explicit computation is difficult. Simulation provides a powerful tool for working with probability models and solving complex problems.

- **Simulation** involves using a probability model to artificially recreate a random phenomenon, many times, usually using a computer.
- Given a probability model, we can simulate outcomes, occurrences of events, and values of random variables, according to the specifications of the probability measure which reflects the model assumptions.
- In general, a simulation involves the following steps.

  1. **Set up.** Define a probability space, and related random variables and events.
  2. **Simulate.** Simulate — according to the assumptions reflected in the probability measure — outcomes, occurrences of events, and values of random variables.
  3. **Summarize.** Summarize simulation output in plots and summary statistics (relative frequencies, averages, standard deviations, correlations, etc) to describe and approximate probabilities, distributions, and related characteristics.
  4. **Sensitivity analysis.** Investigate how results respond to changes in the assumptions or parameters of the simulation.

- Many random phenomena can be concretely represented in terms of "box models" and "spinners".

Example 10.1. Consider rolling a four-sided die.

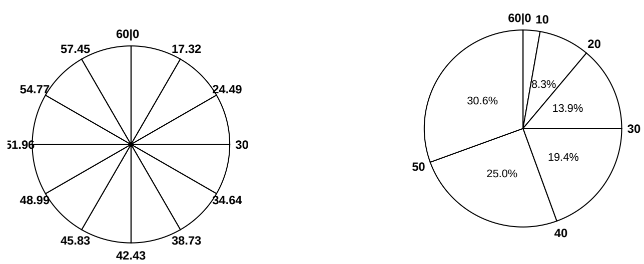
1. Set up a box model for simulating a roll of a fair four-sided die. Then set up a spinner. Suppose you spin this spinner many times. Sketch a plot of the simulated values.

2. Set up a box model for simulating a roll of a die that lands on 1, 2, 3, 4, with respective probability 0.1, 0.2, 0.3, 0.4. Then set up a spinner. Suppose you spin this spinner many times. Sketch a plot of the simulated values.

Example 10.2. Consider the meeting problem.

1. Suppose Regina arrives uniformly at random in the continuous time interval between noon and 1:00 (that is, between 0 and 60 minutes after noon). Construct a circular spinner to simulate Regina's arrival time. In particular, what values go at the "3 o'clock", "6 o'clock", and "9 o'clock" points on the spinner's axis? (Here "3 o'clock" refers to the direction on the "clock" (spinner), one quarter of the way around, and not arrival time.) Suppose you spin this spinner many times. Sketch a histogram of the simulated arrival times.

2. Suppose you use the following spinner to simulate Han's arrival time. Suppose you spin this spinner many times. Sketch a histogram of the simulated arrival times. Roughly what does such a model assume about Han's arrival time?



(a) Divided into equal area sectors

(b) Spinner axis marked in increments of 10 minutes

Figure 10.1: The same spinner is displayed on both sides, with different features highlighted on the left and right. Only selected rounded values are displayed, but in the idealized model the spinner is infinitely precise so that any value between 0 and 60 is possible. Notice that the values on the axis are *not* evenly spaced.

1. Suppose that Cady has a probability of 0.25 of arriving in each of the intervals $[0, 23]$, $[23, 30]$, $[30, 37]$, $[37, 60]$. Also assume that Cady has a probability of (roughly):

   - 0.025 of arriving in the interval $[0, 10]$
   - 0.135 of arriving in the interval $[10, 20]$
   - 0.135 of arriving in the interval $[40, 50]$
   - 0.025 of arriving in the interval $[50, 60]$

   Sketch a spinner for simulating Cady's arrival time. Suppose you spin this spinner many times. Sketch a histogram of the simulated arrival times. Roughly what does such a model assume about Cady's arrival time?

2. Now suppose Regina, Han, and Cady arrive independently of each other, according to the assumptions in the previous parts. Let T (minutes after noon) be the time of the first arrival, and let W be the amount of time (minutes) the first to arrive waits for the last to arrive. Explain how you can use simulation to approximate the joint and marginal distributions of T and W.

Example 10.3. Suppose that

- 65% of American adults age 18-29 use Snapchat
- 24% of American adults age 30-49 use Snapchat
- 12% of American adults age 50-64 use Snapchat
- 2% of American adults age 65+ use Snapchat

Also suppose that

- 20% of American adults are age 18-29
- 33% of American adults are age 30-49
- 25% of American adults are age 50-64
- 22% of American adults are age 65+

Consider simulating a randomly selected U.S. adult and determining the person's age group and whether or not they use Snapchat.

1. How could you perform one repetition of the simulation using spinners based solely on the percentages provided above, without first computing any other probabilities? (Hint: you'll need a few spinners, but you might not spin them all in a single repetition.)

2. How could you perform one repetitionof the simulation using just a single spinner? (Hint: You'll need to compute some probabilities first.)

11 Approximating Probabilities

- A probability can be interepreted as a theoretical long run relative frequency.
- The probability of event A can be approximated by simulating, according to the assumptions corresponding to the probability measure P, the random phenomenon a large number of times and computing the relative frequency of A.

$$\mathrm{P}(A) \approx \frac{\text{number of repetitions on which } A \text{ occurs}}{\text{number of repetitions}}, \quad \text{for a large number of P-repetitions}$$

- In practice, many repetitions of a simulation are performed on a computer to approximate what happens in the "long run". However, we often start by carrying out a few repetitions, (1) by hand to help make the process more concrete, or (2) by computer to check that our code is working correctly
- A probability can be approximated by a relative frequency from a large number of simulated repetitions, but there is some simulation margin of error, due to natural variability in the simulation.
- The margin of error when approximating a single probability based on a simulated relative frequency is roughly on the order $1/\sqrt{N}$, where N is the *number of independently simulated values used to calculate the relative frequency.*
  - Use a margin of error of $2/\sqrt{N}$ when approximating many probabilities based on a single simulation.
- Be careful when conditioning, especially when conditioning on values of continuous random variables.
- The larger N the more precise the estimate, but there is cost to simulating and storing more repetitions in terms of computational time and memory. More importantly, beware a false sense of precision: A precise estimate of a probability under the assumptions of the model is not necessarily a comparably precise estimate of the true probability.

Example 11.1. Roll a fair four-sided die twice and let X be the sum and Y the larger of the two rolls.

1. Explain how you would use simulation to approximate
   a. $\mathrm{P}(X = 6)$

b. $\mathrm{P}(X = 6, Y = 4)$

2. In 10000 repetitions of the simulation, $X = 6$ on 1915 repetitions. Approximate $\mathrm{P}(X = 6)$, including a margin of error.

3. Explain how you would use simulation to approximate the distribution of X. What would the margin of error be?

Example 11.2. Recall the three person meeting problem in Example 10.2; let W be the time the first person waits for the last person.

1. Explain in full detail how you could conduct an appropriate simulation using spinners and use the results to approximate the probability $\mathrm{P}(W < 15)$.

2. In 10000 simulated repetitions, $W < 15$ in 1821. Approximate and interpret $\mathrm{P}(W < 15)$; include a margin of error.

Example 11.3. Consider simulating a randomly selected U.S. adult and determining whether or not the person has a college degree and whether or not they can fix a problem with a car's engine. Let C be the event that the selected adult has a college degree, E be the event that the selected adult can fix an engine, and P correspond to randomly selecting an American adult. Suppose that[1] $\mathrm{P}(C) = 0.40$, $\mathrm{P}(E) = 0.28$, and $\mathrm{P}(C \cap E) = 0.07$.

1. In 10000 repetitions of an appropriate simulation

   - C occurs on 3989 repetitions
   - E occurs on 2837 repetitions
   - Both C and E occur on 680 repetitions.

   Use the simulation results to approximate $\mathrm{P}(E|C)$.

2. What is the margin of error for your estimate in the previous part?

3. What is another method for performing the simulation and estimating $\mathrm{P}(E|C)$ that has a margin of error of 0.01? What is the disadvantage of this method?

Example 11.4. Continuing Example 11.1, roll two fair four-sided dice and let X be the sum and Y the larger of the rolls.

1. For each of the following, describe how to approximate the probability with a margin of error of 0.01.

   a. $\mathrm{P}(X = 6|Y = 4)$.

[1]Values are based on this Pew Research survey.

b. $\mathrm{P}(Y = 4|X = 6)$.

2. Explain how to approximate the conditional distribution of X given $Y = 4$.

Example 11.5. Continuing Example 11.5. We want to approximate the conditional probability that $W < 15$ given that the first person arrives at 12:10. Explain in detail how you would conduct a simulation to approximate this probability. (Be careful!)

12 Expected Values: Approximation and Some Properties

- The expected value of a random variable X is its *probability-weighted average value* $\mathrm{E}(X)$
- $\mathrm{E}(X)$ can be interpreted as the *long run average value* of X
- We can approximate $\mathrm{E}(X)$ by simulating many values of the random variable and computing the average (mean) in the usual way: add up the simulated values of X and divide by the number of simulated values.

Example 12.1. Let X be the number of vehicles crashes involving fatalities that occur in San Luis Obispo County in a randomly selected week. Suppose that X has distribution[1]

x	$\mathrm{P}(X = x)$
0	0.589
1	0.312
2	0.083
3	0.014
4	0.002

1. Intepret the probabilities as long run relative frequencies in this context

2. Sketch a spinner for simulating values of X.

[1]Source: the California Highway Patrol Statewide Integrated Traffic Records System (CHP-SWITRS) public dataset available through the Berkeley Transportation Incident Mapping System (TIMS).

3. Describe in detail how you could use simulation to approximate $\mathrm{E}(X)$.

4. Compute $\mathrm{E}(X)$ and compare to the simulated approximation.

5. Interpret $\mathrm{E}(X)$ in context.

6. Suppose you want to approximate $\mathrm{E}(X^2)$. (We'll see why soon.) Donny Don't says: "Just square the value from part 3". Is that true? If not, how would you use simulation to approximate $\mathrm{E}(X^2)$?

7. Compute $\mathrm{E}(X^2)$. Is it equal to $\mathrm{E}(X)^2$?

- In general the order of transforming and averaging is not interchangeable.
- Whether in the short run or the long run, in general

$$\begin{aligned}\text{Average of } g(X) &\neq g(\text{Average of } X)\\ \text{Average of } g(X,Y) &\neq g(\text{Average of } X, \text{Average of } Y)\end{aligned}$$

- In terms of expected values (long run averages), in general

$$\begin{aligned}\mathrm{E}(g(X)) &\neq g(\mathrm{E}(X))\\ \mathrm{E}(g(X,Y)) &\neq g(\mathrm{E}(X),\mathrm{E}(Y))\end{aligned}$$

Example 12.2. Continuing Example 12.1. Suppose that the number of fatal crashes in any one week period follows the distribution in Example 12.1, independently from week to to week. Now consider a two week period, say this week and next week. Let X be the number of fatal crashes in the first week and Y in the second week, so that $X+Y$ is the total number of fatal crashes in the two week period.

1. Are X and Y the same random variable?

2. Do X and Y have the same distribution? Explain.

3. What is $\mathrm{E}(Y)$. Why?

4. Describe how you would conduct a simulation to approximate the distribution of $X+Y$.

5. Describe how you would use simulation to approximate $\mathrm{E}(X+Y)$.

6. Use simulation to approximate $\mathrm{E}(X)$, $\mathrm{E}(Y)$, and $\mathrm{E}(X+Y)$. How are they related?

7. Compute $\mathrm{P}(X+Y=0)$.

8. Compute $\mathrm{P}(X+Y=1)$.

9. The following table represents the joint distribution of X and Y. Explain how this table was constructed.

x, y	0	1	2	3	4
0	0.3469	0.1838	0.0489	0.0082	0.0012
1	0.1838	0.0973	0.0259	0.0044	0.0006
2	0.0489	0.0259	0.0069	0.0012	0.0002
3	0.0082	0.0044	0.0012	0.0002	0.0000
4	0.0012	0.0006	0.0002	0.0000	0.0000

10. Compute the distribution of $X+Y$.

11. Compute $\mathrm{E}(X+Y)$. Interpret the value in context.

12. How is $\mathrm{E}(X+Y)$ related to $\mathrm{E}(X)$ and $\mathrm{E}(Y)$?

- In general the order of transforming and averaging is not interchangeable.
- However, the order is interchangeable for *linear* transformations.
- **Linearity of averages.**: If X and Y are random variables and a and b are non-random constants, whether in the short run or the long run,

$$\begin{aligned}
\text{Average of } (mX+b) &= m(\text{Average of } X) + b \\
\text{Average of } (X+Y) &= \text{Average of } X + \text{Average of } Y
\end{aligned}$$

- **Linearity of expected value.**: If X and Y are random variables and m and b are non-random constants,

$$\begin{aligned}
\mathrm{E}(mX+b) &= m\mathrm{E}(X) + b \\
\mathrm{E}(X+Y) &= \mathrm{E}(X) + \mathrm{E}(Y)
\end{aligned}$$

- The expected value of the sum of X and Y is the sum of the expected value of X and the expected value of Y *regardless of the relationship between X and Y.*

Example 12.3. Continuing Example 12.2.

1. Explain how we could use simulation to approximate the conditional distribution of $X+Y$ given $X=1$.

2. Explain how we could use simulation to approximate $\mathrm{E}(X+Y|X=1)$.

3. Compute the conditional distribution of $X+Y$ given $X=1$.

4. Compute $\mathrm{E}(X+Y|X=1)$

13 Variance and Standard Deviation

The values of a random variable vary. The distribution of a random variable describes its pattern of variability. The expected value of a random variable summarizes the distribution in just a single number, the long run average value. But the expected value does not tell us much about the degree of variability of the random variable. Do the values of the random variable tend to be close to the expected value, or are they spread out? Variance and standard deviation are numbers that address these questions.

Example 13.1. Let X be a random variable with a Uniform(0, 60) distribution (like we have seen in the meeting problem).

1. Sketch a histogram of many simulated values of X.

2. Explain how you could use simulation to approximate $\mathrm{E}(X)$. What do you think the value of $\mathrm{E}(X)$ is for this X?

3. Explain how we could use simulation to approximate $\mathrm{E}(|X - \mathrm{E}(X)|)$. Why might we consider this quantity as a measure of variability? What do you think the value of $\mathrm{E}(|X - \mathrm{E}(X)|)$ is for this X?

4. The variance of X is defined as $\mathrm{Var}(X) = \mathrm{E}(|X - \mathrm{E}(X)|^2)$. Explain how you could use simulation to approximate $\mathrm{E}(|X - \mathrm{E}(X)|^2)$. Why might we consider variance as a measure of variability?

5. The standard deviation of X is defined as $\text{SD}(X) = \sqrt{\text{Var}(X)}$. Explain why we take the square root. Hint: consider measurement units, e.g., if X is measured in minutes what are the measurement units of variance and standard deviation? Why might we consider standard deviation as a measure of variability?

6. Is $\text{SD}(X)$ equal to $\text{E}(|X - \text{E}(X)|)$? Why or why not?

7. Explain how we can use simulation to approximate $\text{E}(X^2)$. Is it equal to $\text{E}(X)^2$?

8. Use simulation to approximate $\text{E}(X^2) - \text{E}(X)^2$. What do you notice?

- The **variance** of a random variable X is

$$\begin{aligned}\text{Var}(X) &= \text{E}\left((X - \text{E}(X))^2\right) \\ &= \text{E}\left(X^2\right) - \left(\text{E}(X)\right)^2\end{aligned}$$

- The **standard deviation** of a random variable is

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

- Variance is the long run average squared deviation from the mean.
- Standard deviation measures, roughly, the long run average distance from the mean. The measurement units of the standard deviation are the same as for the random variable itself.

- The definition $\mathrm{E}((X - \mathrm{E}(X))^2)$ represents the concept of variance. However, variance is usually computed using the following equivalent but slightly simpler formula.

$$\mathrm{Var}(X) = \mathrm{E}\left(X^2\right) - \left(\mathrm{E}\left(X\right)\right)^2$$

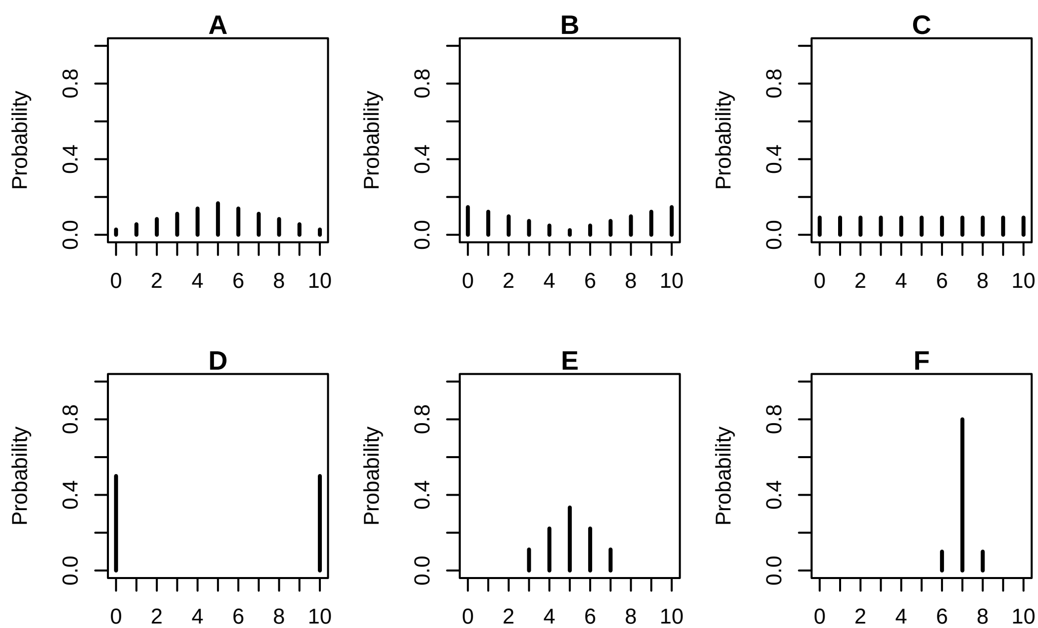
- That is, variance is the expected value of the square of X minus the square of the expected value of X.

- In some cases, we have the expected value and variance and we want to compute $\mathrm{E}(X^2)$. Rearranging the above formula yields

$$\mathrm{E}\left(X^2\right) = \mathrm{Var}(X) + \left(\mathrm{E}\left(X\right)\right)^2$$

- Variance has many nice theoretical properties (which we'll see later). Whenever you need to compute a standard deviation, first find the variance and then take the square root at the end.

Example 13.2. The plots below summarize hypothetical distributions of quiz scores in six classes. All plots are on the same scale. Each quiz score is a whole number between 0 and 10 inclusive.



1. Donny Dont says that C represents the smallest SD, since there is no variability in the heights of the bars. Do you agree that C represents "no variability? Explain.

2. What is the smallest *possible* value the SD of quiz scores could be? What would need to be true about the distribution for this to happen? (This scenario might not be represented by one the plots.)

3. Without doing any calculations, arrange the classes in order based on their SDs from smallest to largest.

4. In one of the classes, the SD of quiz scores is 5. Which one? Why?

5. Is the SD in F greater than, less than, or equal to 1? Why?

6. Provide a ballpark estimate of SD in each case.

Example 13.3. This is a simplified example of *batch testing*[1] for a disease. In batch testing,

[1]Thanks to Allan Rossman for this example. His blog has more investigations into batch testing, including what happens as n and p change. Cal Poly implemented versions of batch testing for COVID through saliva samples and wastewater testing.

specimens (like blood or saliva samples) from a group of people are pooled together into one batch, which then undergoes one test. If none of the people has the disease, then the batch test result will be negative, and no further tests are required. But if at least one person in the batch has the disease, then the batch test result will be positive, and then each person in the batch must be tested individually. (We're assuming no false positives and no false negatives for this example.)

Suppose that $n = 8$ people to be tested are split into two groups of 4 people each. Within each group of 4 people, specimens are combined into a batch to be tested. If anyone in the batch has the disease, then the batch test will be positive, and those 4 people will need to be tested individually. Assume that each person has probability $p = 0.1$ of having the disease, independently from person to person. (The assumption of independence might be unreasonable if the people are in close contact with one another.)

Let X represent the total number of tests that are conducted.

1. What are the possible values of X?

2. Find the distribution of X.

3. Compute and interpret $\mathrm{E}(X)$. In what sense is batch testing betting than just testing all 8 individuals initially?

4. Compute $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$.

Example 13.4. SAT scores have a mean of 1050 and a standard deviation of 200. ACT scores have a mean of 21 and a standard deviation of 5.5. The shape of the distributions of scores is similar for the two tests.

Darius's score on the SAT is 1500. Alfred's score on the ACT is 31. We'll investigate who scored relatively better on their test.

1. Compute the deviation from the mean for Darius's SAT score. How large is this value relative to the standard deviation for SAT scores (which measures, roughly, the average deviation from the mean)?

2. Compute the deviation from the mean for Alfred's ACT score. How large is this value relative to the standard deviation for ACT scores?

3. Who scored relatively better on their test? Explain.

4. Could we answer the previous question if we did not know that the shapes were similar?

- Standard deviation provides a "ruler" by which we can judge a particular realized value of a random variable relative to the distribution of values.
- If X is a random variable with expected value $\mathrm{E}(X)$ and standard deviation $\mathrm{SD}(X)$, then the **standardized random variable** is

$$Z = \frac{X - \mathrm{E}(X)}{\mathrm{SD}(X)}$$

- However, keep in mind that comparing standardized values is most appropriate for distributions that have *similar shapes*.
- Standardization is most natural for random variables that follow a Normal distribution.

- A Normal distribution has two parameters: its mean (typically denoted μ) and its standard deviation (typically denoted σ).
- For Normal distributions the probability that a value is within (or beyond) a given number of standard deviations from the mean follows a specific pattern called the "empirical rule". For example:
  - 50% of values are within 0.67 standard deviations of the mean
  - 68% of values are within 1 standard deviation of the mean
  - 95% of values are within 2 standard deviations of the mean
  - 99.7% of values are within 3 standard deviations of the mean
- We will study Normal distributions in much more detail later

14 Covariance and Correlation

- Quantities like expected value and variance summarize characteristics of the *marginal* distribution of a single random variable.
- When there are multiple random variables their *joint* distribution is of interest.
- Covariance and correlation summarize a characteristic of the joint distribution of two random variables, namely, the degree to which they "co-deviate from the their respective means".

14.1 Covariance

- The **covariance** between two random variables X and Y is

$$\begin{aligned}
\mathrm{Cov}(X, Y) &= \mathrm{E}\left[(X - \mathrm{E}[X])\,(Y - \mathrm{E}[Y])\right] \\
&= \mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)
\end{aligned}$$

- Covariance is the long run average of the products of the paired deviations from the mean
- Covariance is the expected value of the product minus the product of expected values.
  - Remember: in general $\mathrm{E}(XY)$ is not necessarily equal to $\mathrm{E}(X)\mathrm{E}(Y)$

Example 14.1. In the meeting problem, assume the two arrival times X and Y each have a Uniform(0, 60) distribution, independently. Let $T = \min(X, Y)$ be the time of the first arrival and let $W = |X - Y|$ be the waiting time of the first person for the second to arrive.

1. Describe in detail how you would use simulation to approximate $\mathrm{Cov}(T, W)$.

2. Is $\mathrm{Cov}(T, W)$ positive, negative, or zero? Explain why.

- $\mathrm{Cov}(X, Y) > 0$ (positive association): above average values of X tend to be associated with above average values of Y
- $\mathrm{Cov}(X, Y) < 0$ (negative association): above average values of X tend to be associated with below average values of Y
- $\mathrm{Cov}(X, Y) = 0$ indicates that the random variables are **uncorrelated**: there is no overall positive or negative association.
  - X and Y are uncorrelated if and only if $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(X)$
  - But be careful: if X and Y are uncorrelated there can still be a relationship between X and Y; there is just no overall positive or negative association.

Example 14.2. Roll a fair four-sided die twice. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). The table below represents the joint distribution of X and Y.

In previous exercises we computed $\mathrm{E}(X) = 5$ and $\mathrm{E}(Y) = 3.125$

$x \setminus y$	1	2	3	4
2	1/16	0	0	0
3	0	2/16	0	0
4	0	1/16	2/16	0
5	0	0	2/16	2/16
6	0	0	1/16	2/16
7	0	0	0	2/16
8	0	0	0	1/16

1. Compute $\mathrm{E}(XY)$.

2. Compute $\mathrm{Cov}(X, Y)$.

Example 14.3. Consider the probability space corresponding to two rolls of a fair four-sided die. Let

- X be the sum of the two rolls
- Y the larger of the two rolls
- W the number of rolls equal to 4
- Z the number of rolls equal to 1
- $V = W + Z$

Without doing any calculations, determine if the covariance between each of the following pairs of variables is positive, negative, or zero. Explain your reasoning conceptually.

1. $\mathrm{Cov}(X, W)$

2. $\mathrm{Cov}(X, Z)$

3. $\mathrm{Cov}(X, V)$

4. $\mathrm{Cov}(W, Z)$

5. $\mathrm{Cov}(X, V) = 0$. Are X and V independent?

- $\mathrm{Cov}(X, Y) = 0$ indicates that the random variables are **uncorrelated**: there is no overall positive or negative association.
  - X and Y are uncorrelated if and only if $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$

- But be careful: if X and Y are uncorrelated there can still be a relationship between X and Y; there is just no overall positive or negative association.
- If X and Y are independent then $\mathrm{Cov}(X, Y) = 0$
- But the converse is not true; there are many examples of uncorrelated random variables that are not independent.

Example 14.4. The covariance between height in inches and weight in pounds for football players is 96.

1. What are the measurement units of the covariance?

2. Suppose height were measured in feet instead of inches. Would the shape of the joint distribution (say, represented by a scatterplot) change? Would the strength of the association between height and weight change? Would the value of covariance change?

- The numerical value of the covariance depends on the measurement units of both variables, so interpreting it can be difficult.
- Covariance is a measure of joint association between two random variables that has many nice theoretical properties, but the *correlation (coefficient)* is often a more practical measure.

14.2 Correlation

- The **correlation (coefficient)** between random variables X and Y is

$$\begin{aligned}\mathrm{Corr}(X, Y) &= \mathrm{Cov}\left(\frac{X - \mathrm{E}(X)}{\mathrm{SD}(X)}, \frac{Y - \mathrm{E}(Y)}{\mathrm{SD}(Y)}\right) \\ &= \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X)\mathrm{SD}(Y)}\end{aligned}$$

- The correlation for two random variables is the covariance between the corresponding standardized random variables. Therefore, correlation is a *standardized* measure of the association between two random variables.

- A correlation coefficient has no units and is measured on a universal scale. Regardless of the original measurement units of the random variables X and Y

$$-1 \le \text{Corr}(X,Y) \le 1$$

- $\text{Corr}(X,Y) = 1$ if and only if $Y = aX + b$ for some $a > 0$
- $\text{Corr}(X,Y) = -1$ if and only if $Y = aX + b$ for some $a < 0$
- Therefore, correlation is a standardized measure of the strength of the *linear* association between two random variables.
- The closer the correlation is to 1 or -1, the closer the joint distribution of (X,Y) pairs hugs a straight line, with positive or negative slope.
- Because correlation is computed between standardized random variables, correlation is not affected by a linear rescaling of either variable (e.g., a change in measurement units from minutes to seconds)
- Covariance is the correlation times the product of the standard deviations.

$$\text{Cov}(X,Y) = \text{Corr}(X,Y)\text{SD}(X)\text{SD}(Y)$$

Example 14.5. The covariance between height in inches and weight in pounds for football players is 96. Heights have mean 74 and SD 3 inches. Weights have mean 250 pounds and SD 45 pounds. Find the correlation between weight and height.

15 Equally Likely Outcomes and Counting Rules

- For a sample space Ω with finitely many possible outcomes, assuming **equally likely outcomes** corresponds to a probabiliy measure P which satisfies

$$\mathrm{P}(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega} \qquad \text{when outcomes are equally likely}$$

- Computing probabilities in the equally likely case reduces to just counting outcomes.
- But remember:
  - In most situations, outcomes are not equally likely
  - Even if the sample space outcomes are equally likely, the possible values of related random variables are usually not.
- However, rules for counting outcomes are useful even in situations where the outcomes are not equally likely

Example 15.1. Suppose that to send an internet packet from the east coast of the US to the west coast, a packet must go through a major east-coast city (Boston, New York, Washington DC, or Atlanta), then a major midwest city (Chicago, St. Louis, or New Orleans), and then a major west-coast city (San Francisco or Los Angeles). How many possible routes are there?

- **Multiplication principle for counting.** Suppose that stage 1 of a process can be completed in any one of n_1 ways. Further, suppose that for each way of completing the stage 1, stage 2 can be completed in any one of n_2 ways. Then the two-stage process can be completed in any one of $n_1 \times n_2$ ways.
- This rule extends naturally to a ℓ-stage process, which can then be completed in any one of $n_1 \times n_2 \times n_3 \times \cdots \times n_\ell$ ways.

- In the multiplication principle it is not important whether there is a "first" or "second" stage. What is important is that there are distinct stages, each with its own number of "choices".

Example 15.2. Suppose the board of directors of a corporation has identified 5 candidates — Ariana, Beyonce, Cardi, Drake, Elvis — for three distinct executive positions: chief executive officer (CEO), chief financial officer (CFO), and chief operating officer (COO). In the interest of fairness, the board assigns 3 of the 5 candidates to the positions completely at random. No individual can hold more than one of the positions.

When calculating probabilities below, consider the sample space of all possible executive teams.

1. How many executive teams are possible?

2. What is the probability that Ariana is CEO, Beyonce is CFO, and Cardi is COO?

3. What is the probability that Ariana is CEO and Beyonce is CFO?

4. What is the probability that Ariana is CEO?

- **Number of "ordered" arrangements.** The number of "ordered" arrangements of k items, selected *without* replacement from a set of n distinct items is

$$n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!}$$

- Recall the **factorial** notation: $m! = m(m-1)(m-2)\cdots(3)(2)(1)$. For example, $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$. By definition, $0!=1$.

Example 15.3. Your boss is forming a committee of 3 people for a new project team, and 5 people — Ariana, Beyonce, Cardi, Drake, Elvis— have volunteered to be on the committee.

In the interest of fairness, 3 of the 5 people will be selected uniformly at random to form the committee.

1. How many possible committees consist of Ariana, Beyonce, Cardi? How many executive teams from the previous example consisted of Ariana, Beyonce, Cardi? How is this example different from the executive team example?

2. How many different possible committees of 3 people can be formed from the 5 volunteers?

- The following is the relationship between "ordered" and "unordered" counting.

$$
\begin{aligned}
&\text{(number of "ordered" selections of } k \text{ from } n) \\
= \quad &\text{(number of "unordered" selections of } k \text{ from } n) \\
&\quad \times \text{(number of ways of arranging the } k \text{ items in order)}\,.
\end{aligned}
$$

- "Ordered" and "unordered" are somewhat misnomers. It is not important whether there is a "first", "second", "third", etc. What is important is whether or not there are *distinct stages*, each with its own number of "choices".
  - In Example 15.2, it doesn't matter if we pick the CEO first and the CFO second or the CFO first and the CEO second; what does matter is that choosing the CEO is a distinct stage from choosing the CFO.
- **Number of permutations.** The number of ways of arranging k items in order is

$$k \times (k-1) \times (k-2) \times \cdots \times 3 \times 2 \times 1 = k!$$

- **Number of combinations.** The number of ways to choose k items *without* replacement from a group of n distinct items where *"order" does not matter*, denoted $\binom{n}{k}$, is

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!} = \frac{n!}{k!(n-k)!}$$

- The quantity on the right is just a compact way of representing the quantity in the middle. But since factorials can be very large, it's best to use the quantity in the middle to compute. In R: `choose(n, k)`. In Python: `math.comb(n, k)`
- The symbol $\binom{n}{k}$ is by definition equal to the quantity in the middle above. It is read as "n choose k" and is referred to as a **binomial coefficient**.

Example 15.4. Continuous Example 15.3. Your boss is forming a committee of 3 people for a new project team, and 5 people — Ariana, Beyonce, Cardi, Drake, Elvis— have volunteered to be on the committee. In the interest of fairness, 3 of the 5 people will be selected uniformly at random to form the committee.

1. Find the probability that the committee consists of Ariana, Beyonce, and Cardi.

2. Find the probability that Ariana and Beyonce are on the committee.

3. Find the probability that Ariana is on the committee.

Example 15.5. A standard deck of 52 cards contains 4 suits (hearts, diamonds, spades, clubs) each consisting of 13 different face values (2 through 10, jack, queen, king, ace). The deck is well shuffled and a hand of 5 cards is dealt (without replacement as is usual in practice).

1. How many possible hands are there?

2. What is the probability the hand contains exactly 4 hearts?

3. What is the probability the hand contains 3 aces and 2 kings?

4. What is the probability the hand is a full house (3 cards of one face value and 2 of another)?

Example 15.6. *Capture-recapture* sampling is a technique often used to estimate the size of a population. Suppose you want to estimate N, the number of monarch butterflies in Pismo Beach. Assume that N is a fixed but unknown number[1]; the population size doesn't change over time. You first capture a sample of N_1 butterflies, selected randomly without replacement, and tag them and release them. At a later date, you then capture a second sample of n butterflies, selected randomly from the N butterflies *without* replacement. Let X be the number of butterflies in the second sample that have tags (because they were also caught in the first sample). (Assume that the tagging has no effect on behavior, so that selection in the first sample is independent of selection in the second sample.)

In practice, N is unknown and the goal is to estimate it based on what we observe for X; we'll see later how to do this later. For now, we'll start with a simpler, but unrealistic, example where N is known. Assume there are $N = 52$ butterflies, $N_1 = 13$ are tagged in the first sample, and $n = 5$ is the size of the second sample.

1. What are the possible values of X?

2. Describe in detail how you could use simulation to approximate the distribution of X.

3. Find $\mathrm{P}(X = 0)$ in two ways. Interpret the value in context.

4. Find the probability that in the second sample the first butterfly selected is tagged but the rest are not.

[1]Another approach—a *Bayesian* approach—is to treat the unknown number N as a *random variable* with a distribution that quantifies its uncertainty.

5. Find the probability that in the second sample the first four butterflies selected are not tagged but the fifth is.

6. Find $P(X = 1)$ in two ways. Interpret the value in context.

7. Find $P(X = 2)$ in two ways. Interpret the value in context.

8. Suggest a formula for determining the distribution of X.

9. Suggest a simple shortcut formula for $E(X)$.

10. Use the distribution of X to compute $E(X)$. Does it match with the shortcut formula? Intepret the value in context.

- Consider a population of size $N = N_1 + N_0$ (e.g., a box with N tickets)
  - Each individual in the population can be classified as "success" (1) or "failure" (0)
  - N_1 individuals in the population are "success" (1)
  - N_0 individuals in the population are "failure" (0)
- Randomly select a sample of n individuals from a population with N_1 successes and N_0 failures *without replacement* and let X be the number of individuals in the selected sample that are successes. The distribution of X is defined to be the **Hypergeometric(N_1, N_0, n) distribution**.
  - If successes are labeled as 1 and failures as 0, the random variable X which counts the number of successes in the sample is equal to the sum of the 1/0 labels of the individuals in the sample.
- A discrete random variable X has a **Hypergeometric distribution** with with parameters n, N_0, N_1, all nonnegative integers—with $N = N_0 + N_1$ and $p = N_1/N$—if its distribution satisfies[2]

$$\mathrm{P}(X = x) = \frac{\binom{N_1}{x}\binom{N_0}{n-x}}{\binom{N_0+N_1}{n}}, \quad x = 0, 1, \dots, n;\ x \le N_1;\ x \ge n - N_0$$

- If X has a Hypergeometric(n, N_1, N_0) distribution

$$\mathrm{E}(X) = np$$
$$\mathrm{Var}(X) = np(1-p)\left(\frac{N-n}{N-1}\right)$$

[2] We must have $X \le N_1$ since there can't be more successes in the sample than there are in the population. Similarly, we must have $X \ge n - N_0$, that is $n - X \le N_0$, since there can't be more failures in the sample than there are in the population. Often the population sizes N_0 and N_1 are large relative to the sample size n in which case X simply takes values $0, 1, \dots, n$.

16 Discrete Random Variables: Probability Mass Functions

- Roughly, a random variable assigns a number measuring some quantity of interest to each outcome of a random phenomenon.
- The **(probability) distribution** of a random variable specifies the possible values of the random variable and a way of determining corresponding probabilities.
- We will see many ways of describing a distribution, depending on how many random variables are involved and their types (discrete or continuous)
- A **discrete** random variable can take on only countably many isolated points on a number line. These are often counting type variables.
  - Note that "countably many" includes the case of countably infinite, such as $\{0, 1, 2, ...\}$.
- The **probability mass function (pmf)** of a *discrete* RV X, defined on a probability space with probability measure P, is a function $p_X : \mathbb{R} \mapsto [0, 1]$ which specifies each possible value of the RV and the probability that the RV takes that particular value: $p_X(x) = \mathrm{P}(X = x)$ for each possible value of x.
- Certain common distributions have special names and properties. We often specify a distribution by name along with values of relevant parameters.

Example 16.1. Roll a fair four-sided die twice and let X be the sum and Y the larger of the two rolls.

1. Verify that the pmf of X is

$$p_X(x) = \frac{4 - |x - 5|}{16}, \qquad x = 2, 3, \ldots, 8$$

2. Is the following the pmf of Y? Discuss.

$$p_Y(u) = \frac{2u-1}{16}$$

- In a sequence of **Bernoulli(p) trials**
  - There are only two possible outcomes, "success" (1) or not (0, "failure"), on each trial.
  - The unconditional/marginal probability of success is the same on every trial, and equal to p.
  - The trials are independent.

Example 16.2. Recall the "lookaway challenge of Example 8.4.

The game consists of possibly multiple rounds. In the first round, you point in one of four directions: up, down, left or right. At the exact same time, your friend also looks in one of those four directions. If your friend looks in the same direction you're pointing, you win! Otherwise, you switch roles and the game continues to the next round — now your friend points in a direction and you try to look away. As long as no one wins, you keep switching off who points and who looks. The game ends, and the current "pointer" wins, whenever the "looker" looks in the same direction as the pointer.

Suppose that each player is equally likely to point/look in each of the four directions, independently from round to round.

Let X be the number of rounds until the game ends.

1. Can the rounds be considered Bernoulli trials?

2. What are the possible values that X can take? Is X discrete or continuous?

3. Compute and interpret $\mathrm{P}(X = 1)$.

4. Compute and interpret $\mathrm{P}(X = 2)$.

5. Compute and interpret $\mathrm{P}(X = 3)$.

6. Find the probability mass function of X.

7. Construct a table, plot, and spinner representing the distribution of X.

8. How can you use the distribution of X to compute the probability that the player who starts as the pointer wins the game? (In Example 8.4 we computed this to be $4/7$ using the law of total probability.)

9. Compute and interpret $\mathrm{P}(X > 3)$. Can you think of a way to do this without summing several terms?

10. Compute and interpret $\mathrm{P}(X > 5|X > 2)$. Compare to the previous part. What do you notice? Does this make sense?

11. What seems like a reasonable shortcut formula for $\mathrm{E}(X)$ in terms of p? Consider the case $p = 0.25$ first and then general p.

12. Compute $\mathrm{E}(X)$ using the pmf of X. Did the shortcut work?

13. Interpret $\mathrm{E}(X)$ in context.

14. Compute $\mathrm{Var}(X)$.

15. Would $\mathrm{Var}(X)$ be bigger or smaller if $p = 0.9$? If $p = 0.1$?

- Perform Bernoulli(p) trials until a success occurs and then stop. Let X count the total number of trials, including the single success. The distribution of X is defined to be the

Geometric(p) distribution.

- The Geometric(p) probability mass function follows from the fact that if X has a Geometric distribution, then $X = x$ if and only if
  - the first $x - 1$ trials are failures, and
  - the xth (last) trial results in success.
- A discrete random variable X has a Geometric(p) distribution if and only if its probability mass function is

$$p_X(x) = p(1-p)^{x-1}, \qquad x = 1, 2, 3, \ldots$$

- If X has a Geometric(p) distribution then

$$\begin{aligned} \mathrm{P}(X > x) &= (1-p)^x, \qquad x = 1, 2, 3, \ldots \\ \mathrm{E}(X) &= \frac{1}{p} \\ \mathrm{Var}(X) &= \frac{1-p}{p^2} \end{aligned}$$

Example 16.3. Continuing Example 16.2.

1. Specify how you could simulate a value of X using the "simulate from the probability space" method.

2. Specify how you could simulate a value of X using the "simulate from the distribution" method.

3. Explain how you could use simulation to approximate the distribution of X and its expected value.

- There are two basic methods for simulating a value x of a random variable X.

- **Simulate from the probability space.** Simulate an outcome ω from the underlying probability space and set $x = X(\omega)$.
- **Simulate from the distribution.** Simulate a value x directly from the distribution of X.

- The "simulate from the distribution" method corresponds to constructing a spinner representing the marginal distribution of X and spinning it once to generate x. This method does require that the distribution of X is known. However, as we will see in many examples, *it is common to specify the distribution of a random variable directly without defining the underlying probability space.*

Example 16.4. Donny Dont is thoroughly confused about the distinction between a random variable and its distribution. Help him understand by by providing a simple concrete example of *two different random variables* X and Y that have the *same distribution.* Can you think of X and Y for which $\mathrm{P}(X = Y) = 0$?

- Do not confuse a random variable with its distribution!
- A random variable measures a numerical quantity which depends on the outcome of a random phenomenon
- The distribution of a random variable specifies the long run pattern of variation of values of the random variable over many repetitions of the underlying random phenomenon.

17 Binomial Distributions

Example 17.1. Biathlon is a winter sport that combines cross-country skiing and rifle shooting. In Olympic sprint biathlon, each competitor competes in a ski race (10km for men, 7.5km for women) during which they also shoot at a total of 10 targets. Suppose that Campbell has a probability of 0.87 of successfully hitting any single target, independently.

Let X be the number of targets, out of 10, that Campbell successfully hits in their next sprint race.

1. Can the shots at the targets be considered Bernoulli trials?

2. What are the possible values of X?

3. Describe in detail how you could use simulation to approximate the distribution of X.

4. Compute and interpret $\mathrm{P}(X = 10)$.

5. Compute the probability that the Campbell makes their first 9 shots and misses the last.

6. Explain why the previous part is not $\mathrm{P}(X = 9)$.

7. How many outcomes (S/F sequences of length 10) satisfy the event $\{X = 9\}$? Answer without listing all the possibilities.

8. Compute and interpret $\mathrm{P}(X = 9)$.

9. Compute and interpret $\mathrm{P}(X = 8)$.

10. Suggest a general formula for the probability mass function of X. Then use the pmf to create a table and spinner representing the distribution of X.

11. Suggest a simple shortcut formula for $\mathrm{E}(X)$.

12. Compute and interpret $\mathrm{E}(X)$. Did the shortcut formula work?

13. Compute $\text{Var}(X)$.

- Consider a *fixed number*, n, of Bernoulli(p) trials and let X count the number of successes. The distribution of X is defined to be the **Binomial(n, p) distribution**.
- A Binomial distribution is specified by two parameters:
  - n (an integer): the fixed number of Bernoulli trials, or the sample size
  - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
- A discrete random variable X has a **Binomial distribution** with parameters n, a nonnegative integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad\qquad x = 0, 1, 2, \dots, n$$

  - In the Binomial(n, p) situation, there are $\binom{n}{x}$ S/F sequences that result in exactly x successes—and thus $n - x$ failures—in n trials,
  - and each of these sequences has probability $p^x(1-p)^{n-x}$
- If X has a Binomial(n, p) distribution then

$$\begin{aligned} \text{E}(X) &= np \\ \text{Var}(X) &= np(1-p) \end{aligned}$$

Example 17.2. In each of the following situations determine whether or not X has a Binomial distribution. If so, specify n and p. If not, explain why not.

1. Roll a die 20 times; X is the number of times the die lands on an even number.

2. Roll a die 20 times; X is the number of times the die lands on 6.

3. Roll a die until it lands on 6; X is the total number of rolls.

4. Roll a die until it lands on 6 three times; X is the total number of rolls.

5. Roll a die 20 times; X is the sum of the numbers rolled.

6. Shuffle a standard deck of 52 cards (13 hearts, 39 other cards) and deal 5 *without replacement*; X is the number of hearts dealt. (Hint: be careful about why.)

7. Roll a fair six-sided die 10 times and a fair four-sided die 10 times; X is the number of 3s rolled (out of 20).

Example 17.3. Xavier bets on red on each of 3 spins of a roulette wheel. After the third spin, Xavier leaves and Zander arrives and bets on black on the next two spins. Let X be the number of bets that Xavier wins and let Z be the number that Zander wins.

1. Identify the distribution of X and the distribution of Z.

2. Are X and Z independent? Explain.

3. Compute $\mathrm{P}(X = 2, Z = 1)$.

4. What does $X+Z$ represent in this context? Identify the distribution of $X+Z$ without doing any calculations.

5. Compute $\mathrm{P}(X+Z=3)$.

6. Suppose that Yolanda bets on black on all 5 spins of the wheel. Does $X+Y$ have a Binomial distribution? Explain.

7. Suppose instead that Zavier bet on the number 7 in his two bets. Does $X+Z$ have a Binomial distribution in this case? Explain.

- If X and Y are independent, X has a Binomial(n_X, p) distribution, and Y has a Binomial(n_Y, p) distribution then $X+Y$ has a Binomial(n_X+n_Y, p)
- A Binomial($1, p$) distribution is also known as a **Bernoulli(p)** distribution, taking a value of 1 with probability p and 0 with probability $1-p$. Any *indicator random variable* has a Bernoulli distribution.
- If $X_1, X_2, \dots, X_n$ are independent each with a Bernoulli(p) distribution, then $X_1+\cdots+X_n$ has a Binomial(n, p) distribution.

18 Negative Binomial Distributions

Example 18.1. Suppose that 86% of Cal Poly students are CA residents. Randomly select Cal Poly students one at a time, independently[1], until you select 5 CA residents, then stop. Let X be the total number of students selected.

1. Does X have a Binomial distribution? A Geometric distribution? Why or why not?

2. What are the possible values of X? Is X discrete or continuous?

3. Describe in words how you could use simulation to approximate the distribution of X.

4. Compute and interpret $\mathrm{P}(X = 5)$.

5. Compute the probability that the first student is not a CA resident but the next 5 are.

[1]Technically this means *with replacement*, but since the population of Cal Poly students is much larger than the sample we're selecting then independence is a reasonable assumption even if selecting *without* replacement.

6. Why is $\mathrm{P}(X = 6)$ different from the probability in the previous part?

7. How many outcomes satisfy the event $\{X = 6\}$? Answer without listing all the possibilities. Hint: careful, it's not $\binom{6}{5}$; why not?

8. Compute and interpret $\mathrm{P}(X = 6)$

9. Compute the probability that the first two students are not CA residents but the next 5 are.

10. Why is $\mathrm{P}(X = 7)$ different from the probability in the previous part?

11. How many outcomes satisfy the event $\{X = 7\}$? Answer without listing all the possibilities. Hint: careful, it's not $\binom{7}{5}$; why not?

12. Compute and interpret $\mathrm{P}(X = 7)$

13. Suggest a formula for the probability mass function of X. Then use the pmf to create a table representing the distribution of X.

14. What seems like a reasonable shortcut formula for $\mathrm{E}(X)$?

15. Use the theoretical pmf to compute $\mathrm{E}(X)$. Did the shortcut formula work?

16. Interpret $\mathrm{E}(X)$ for this example.

17. Compute $\mathrm{Var}(X)$.

18. Would the variance be larger or smaller if we kept selecting until we obtain 10 CA residents instead of 5?

- Perform Bernoulli(p) trials until a fixed number, r, of successes and then stop. Let X count the total number of trials, including the r successes. The distribution of X is defined to be the **Negative Binomial(r, p) distribution**.
- A Negative Binomial distribution is specified by two parameters:
  - r (an integer): the fixed number of successes
  - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
- A discrete random variable X has a **Negative Binomial distribution** with parameters r, a positive integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \qquad x = r, r+1, r+2, ...$$

  - In the NegativeBinomial(r, p) situation, there are $\binom{x-1}{x-r} = \binom{x-1}{r-1}$ S/F sequences that result in exactly $x - r$ failures—and thus $r - 1$ succeses—in the first $x - 1$ trials and success on trial x
  - and each of these sequences has probability $p^r (1-p)^{x-r}$
- If X has a NegativeBinomial(r, p) distribution

$$\mathrm{E}(X) = \frac{r}{p}$$

$$\mathrm{Var}(X) = \frac{r(1-p)}{p^2}$$

Example 18.2. What is another name for a NegativeBinomial(1, p) distribution?

Example 18.3. Xavier bets on red on roulette until he wins 3 bets and then he leaves. After Xavier leaves, Zander arrives and bets on black until he wins 2 bets and then he leaves. Let X be the number of bets that Xavier makes and let Z be the number of bets that Zander makes.

1. Identify the distribution of X and the distribution of Z.

2. Are X and Z independent? Explain.

3. What does $X + Z$ represent in this context? Identify the distribution of $X + Z$ without doing any calculations.

4. Suppose that Yolanda starts making bets on the same game as Xavier, and she bets on black until she wins 5 bets. Does $X + Y$ have a Negative Binomial distribution? Explain.

5. Suppose instead that Zavier bets on the number 7 until he wins 5 bets. Does $X + Z$ have a Negative Binomial distribution in this case? Explain.

- If X and Y are independent, X has a NegativeBinomial(r_X, p) distribution, and Y has a NegativeBinomial(r_Y, p) distribution then $X + Y$ has a NegativeBinomial($r_X + r_Y$, p)
- A NegativeBinomial(1, p) distribution is a Geometric(p) distribution.
- If $X_1, X_2, \ldots, X_r$ are independent each with a Geometric(p) distribution, then $X_1 + \cdots + X_r$ has a NegativeBinomial(r, p) distribution.

In a Negative Binomial situation, the number of successes is fixed so what's random is the number of failures and hence the number of trials. Unfortunately, there is not a standard definition of Negative Binomial in terms of what is counted: failures or trials. It's a good idea to always check the documentation of a software package to see what definition is used.

- Perform Bernoulli(p) trials until a fixed number, r, of successes and then stop. Let X count the total number of *failures* (excluding the r successes). The distribution of X is defined to be the **Pascal(r, p) distribution**.

  - The possible values of a NegativeBinomial(r, p) distribution are $r, r+1, r+2, \ldots$
  - The possible values of a Pascal(r, p) distribution are $0, 1, 2, \ldots$

- If X has a NegativeBinomial(r, p) distribution then $(X - r)$ has a Pascal(r, p) distribution.
- Similarly, if Y has a Pascal(r, p) distribution then $(Y + r)$ has a NegativeBinomial(r, p) distribution.
- Some software packages call "Negative Binomial" what we are calling "Pascal".

19 Poisson Distributions

- Binomial and Negative Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of "relatively rare" events that occur over a certain interval of time/in a certain location

Example 19.1. Recall the matching problem with a general n: there are n objects that are shuffled and placed uniformly at random in n spots with one object per spot. Note: n is a fixed and known number, but we are considering different values of it.

We have seen that $\mathrm{E}(X) = 1$ for any n and that unless n is small (e.g., $n \leq 6$) the approximate distribution of X is the Poisson(1) distribution.

1. Starting with $x = 0$, describe the distribution of X in terms of relative likelihoods:
   - 1 is [blank] times as likely as 0
   - 2 is [blank] times as likely as 1
   - 3 is [blank] times as likely as 2
   - 4 is [blank] times as likely as 3
   - 5 is [blank] times as likely as 4
   - 6 is [blank] times as likely as 5
   - In general, x is [blank] times as likely as $x - 1$, for $x = 1, 2, 3, ...$

2. Specify the pmf of X.

3. Use the pmf to compute $\mathrm{E}(X)$.

Table 19.1: Poisson(1) distribution, the approximate distribution of X, the number of matches in the matching problem for general n and uniformly random placement of objects in spots.

x	P(X=x)
0	0.3679
1	0.3679
2	0.1839
3	0.0613
4	0.0153
5	0.0031
6	0.0005
7	0.0001

4. Use the pmf to compute $\mathrm{Var}(X)$.

- A random variable X has a **Poisson distribution** with parameter $\mu > 0$ if the possible values are $0, 1, 2, ...$ and the pmf follows the

$$\text{Poisson}(\mu) \text{ pattern:} \quad x \text{ is } \frac{\mu}{x} \text{ as likely as } x-1 \qquad (\text{for } x = 1, 2, 3, ...)$$

- As a function of $x = 0, 1, 2, ...$, a Poisson(μ) pmf increases for $x < \mu$ and then decreases for $x > \mu$
  - If μ is a whole number, then the Poisson pmf assigns the same probability to the values μ and $\mu - 1$.
  - If $\mu < 1$ then $x = 0$ has the highest probability and the pmf decreases from there.
- A discrete random variable X has a Poisson distribution with parameter $\mu > 0$ if and only if its probability mass function p_X satisfies

$$p_X(x) = \frac{e^{-\mu}\mu^x}{x!}, \quad x = 0, 1, 2, ...$$

  - The Poisson pattern implies that the shape of a Poisson pmf as a function of x is given by $\mu^x/x!$.

$$p(x) = p(0)\frac{\mu^x}{x!}, \qquad x = 0, 1, 2, ...$$

– The constant $p(0) = e^{-\mu}$ simply renormalizes the pmf so that the probabilities sum[1] to 1.

- If X has a Poisson(μ) distribution then

$$\begin{aligned} \mathrm{E}(X) &= \mu \\ \mathrm{Var}(X) &= \mu \end{aligned}$$

Example 19.2. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Assume that X has a Poisson(2.4) distribution.

1. Interpret the parameter 2.4 in context.

2. Compute $\mathrm{P}(X = 2.4)$.

3. Identify the most likely value of X.

4. Specify the pmf of X.

5. Compute $\mathrm{P}(X = 0)$ and interpret the value in context.

[1] Recall the series expansion $e^{\mu} = \sum_{x=0}^{\infty} \frac{\mu^x}{x!}$

6. Construct a table and spinner corresponding to the distribution of X.

Example 19.3. Suppose that the number of fatal crashes in SLO County in a week follows, approximately, a Poisson(0.53) distribution, independently from week to week. Let X be the number of fatal crashes in a period of 4 consecutive weeks (close enough to a "month").

1. What is $\mathrm{E}(X)$? You should be able to compute it without knowing the distribution of X.

2. Compute $\mathrm{P}(X = 0)$.

3. Compute $\mathrm{P}(X = 1)$.

4. How could you use simulation to approximate the distribution of X?

5. Use simulation to approximate the distribution of X. Does the approximate distribution follow (approximately) the Poisson pattern?

6. Use a Poisson pmf to compute $\mathrm{P}(X = 5)$.

- **Poisson aggregation.** If X and Y are independent, X has a Poisson(μ_X) distribution, and Y has a Poisson(μ_Y) distribution, then $X + Y$ has a Poisson($\mu_X + \mu_Y$) distribution.
- If component counts are independent and each has a Poisson distribution, then the total count also has a Poisson distribution.

20 Poisson Approximation

- Poisson models are often used to model the distribution of random variables that count the number of "relatively rare" events that occur over a certain interval of time/in a certain location

Example 20.1. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Assume that X has a Poisson(2.4) distribution.

1. In what ways is this like the Binomial situation?

2. In what ways is this NOT like the Binomial situation?

3. In what sense are home runs "relatively rare"?

- Binomial and Negative Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of "relatively rare" events that occur over a certain interval of time/in a certain location

Example 20.2. Recall the birthday problem: in a group of n people what is the probability that at least two have the same birthday? (Ignore multiple births and February 29 and assume that the other 365 days are all equally likely.) We will investigate this problem using Poisson approximation. Imagine that we have a trial for each possible *pair* of people in the group, and let "success" indicate that the pair shares a birthday. Consider both a general n and $n = 30$.

1. How many trials are there?

2. Do the trials have the same probability of success? If so, what is it?

3. Are any *two* trials independent? To answer this questions, suppose that three people in the group are Abe, Bill, and Charles and consider any *two* of the trials that involve these three people.

4. Are any *three* trials independent? Consider the three trials that involve Abe, Bill, and Charles.

5. Let X be the number of *pairs* that share a birthday. Does X have a Binomial distribution?

6. If X has an approximate Poisson distribution, what would the parameter have to be? Compare this Poisson distribution with a simulation of X; does it seem like a reasonable approximation?

7. Approximate the probability that at least two people share the same birthday. Compare to the theoretical value.

Poisson paradigm. Let $A_1, A_2, \dots, A_n$ be a collection of n events. Suppose event i occurs with marginal probability $p_i = \mathrm{P}(A_i)$. Let $N = \mathrm{I}_{A_i} + \mathrm{I}_{A_2} + \cdots + \mathrm{I}_{A_n}$ be the random variable which counts the number of the events in the collection which occur. Suppose

- n is "large",
- $p_1, \dots, p_n$ are "comparably small", and
- the events $A_1, \dots, A_n$ are "not too dependent",

Then N has an approximate Poisson distribution with parameter $\mathrm{E}(N) = \sum_{i=1}^{n} p_i$.

Example 20.3. Use Poisson approximation to approximate the probability that at least *three* people in a group of n people share a birthday. How large does n need to be for the probability to be greater than 0.5?

21 Continuous Random Variables

- **The probability that a continuous random variable X equals any particular value is 0.** That is, if X is continuous then $\mathrm{P}(X = x) = 0$ for all x.

- For continuous random variables, it doesn't really make sense to talk about the probability that the random value is *equal to* a particular value.
- Even though any specific value of a continuous random variable has probability 0, *intervals* can have positive probability. In particular, the probability that a continuous random variable is "close to" a specific value can be positive.
- For continuous random variables, finding frequencies of individual values and impulse plots (like we used for discrete RVs) don't make any sense since each simulated value will only occur once in the simulation.
- Instead, we summarize simulated values of continuous random variables in a *histogram*. A **histogram** groups the observed values into "close-to bins" and plots frequencies or *densities* for each bin.
- Typically, in a histogram *areas* of bars represent relative frequencies; in which case the axis which represents the height of the bars is called "density".
- The continuous analog of a probability mass function (pmf) is a **probability density function (pdf)**. However, while pmfs and pdfs play analogous roles, they are different in one fundamental way; namely, a pmf outputs probabilities directly, while a pdf does not.
- A probability density function only provides the density height; the *area under the density curve determines probabilities*.
- For a continuous random variable X with pdf f_X, the probability that X takes a value in the interval $[a, b]$ is the *area under the pdf over the region* $[a, b]$.

Example 21.1. Maggie and Seamus are babies who have just turned one. At their one-year visits to their pediatrician:

- Maggie is 76cm tall and in the 75th percentile of height for females.
- Seamus is 72cm tall and in the 10th percentile of height for males.

Explain what these percentiles mean.

Table 21.1: Percentiles for Example 21.2.

Percentile	Value (minutes)
10th	12.6
20th	26.8
30th	42.8
40th	61.3
50th	83.2
60th	110.0
70th	144.5
80th	193.1
90th	276.3

- A distribution is basically just a collection of *percentiles*.
- Roughly, the value x is the pth percentile of a distribution of a random variable X if p percent of values of the variable are less than or equal to x: $\mathrm{P}(X \le x) = p$.
- The *cumulative distribution function (cdf)* of a random variable fills in the blank for any given x: x is the (blank) percentile. That is, for an input x, the cdf outputs $\mathrm{P}(X \le x)$.
- The **cumulative distribution function (cdf)** (of a random variable X defined on a probability space with probability measure P) is the function, $F_X : \mathbb{R} \mapsto [0,1]$, defined by $F_X(x) = \mathrm{P}(X \le x)$.
- A cdf is defined for all real numbers x regardless of whether x is a possible value of X.
- A cdf defines a distribution by defining all of its percentiles. Given all the percentiles we can determine the probability that the random variable lies in any interval.

Example 21.2. In a certain region, times (minutes) between occurrences of earthquakes (of any magnitude) have a distribution with percentiles displayed in Table 21.1. Suppose an earthquake just occurred and let X be the amount of time until the next earthquake; we'll assume Table 21.1 represents (partially) the distribution of X.

1. Let F be the cdf of X. Evaluate and interpret $F(12.6)$ and $F(26.8)$.

2. Construct a spinner corresponding to this distribution.

3. Sketch a histogram of this distribution with *unequal* bin widths.

4. Sketch the pdf of X.

5. Sketch a histogram of this distribution with equal bin widths.

6. Let $Q(0.1)$ represent the value of x for which $F(x) = 0.1$. Evaluate and interpret $Q(0.1)$.

- The *quantile function* (essentially the inverse cdf) fills in the following blank for a given $p \in [0, 1]$: the $100p$th percentile is (blank).
- For example, evaluating the quantile function at $p = 0.25$ outputs the 25th percentile.
- For a *continuous* random variable with cdf F, the **quantile function** $Q : [0, 1] \mapsto \mathbb{R}$ is the inverse of the cdf, $Q(p) = F^{-1}(p)$.
- A quantile function basically gives you the same information as a cdf, just in the other direction. Both a quantile function and a cdf define a distribution by specifying all the percentiles.

Example 21.3. Recall Example 5.5. Let X be the random variable representing Han's arrival time and assume that the cdf of X satisfies

$$F(x) = (x/60)^2, \qquad 0 \le x \le 60$$

1. Compute and interpret $\mathrm{P}(X \le 30)$.

2. Compute and interpret $\mathrm{P}(X \leq 42.42)$.

3. Compute and interpret $\mathrm{P}(X \leq 51.96)$.

4. What do the previous parts tell you about the percentiles of X?

5. Use the answers to the previous parts to start to construct a spinner for simulating values of X.

6. How could you fill in more of the spinner? For example, where should you place the values 10, 20, 30, 40, 50 on the spinner axis?

7. Suppose you simulate many values of X. Sketch a histogram of X.

8. Sketch the pdf of X.

9. Compute and interpret the 10th percentile.

10. Compute and interpret the 90th percentile.

11. Find an expression for the pth percentile, where $0 < p < 1$, e.g., $p = 0.1$ corresponds to the 10th percentile.

12. Use the answers to the previous parts to start to construct a spinner for simulating values of X. How could you fill in more of the spinner?

- The quantile function can be used to create a spinner for a distribution. Basically, the values on the outside boundary of the spinner are scaled based on the quantile function (which is determined by the cdf). Intervals corresponding to regions of higher density ("more likely") values are stretched out on the spinner boundary; intervals corresponding regions of lower density ("less likely" values) are shrunk.
- **Universality of the Uniform (or "one spinner to rule them all").** Let F be a cdf and Q its corresponding quantile function. Let U have a Uniform(0, 1) distribution and define the random variable $X = Q(U)$. Then the cdf of X is F.
- Universality of the uniform might look complicated but all it basically says is that you can construct a spinner by putting the 25th percentile 25% of the way around, the 75th percentile 75% of the way around, etc.

- Actually, universality of the uniform says we don't have to create a new spinner. We can just spin the Uniform(0, 1) spinner—which would tell us what percentile that we want, e.g., 0.7 for the 70th percentile—and transform each resulting value by plugging it into the quantile function to obtain the percentile in the measurement units of the the variable.

22 Probability Density Functions of Continuous Random Variables

- The continuous analog of a probability mass function (pmf) is a *probability density function (pdf).*
- However, while pmfs and pdfs play analogous roles, they are different in one fundamental way; namely, a pmf outputs probabilities directly, while a pdf does not.
- For a continuous random variable X with pdf f_X, the probability that X takes a value in the interval $[a, b]$ is the *area under the pdf over the region* $[a, b]$.
- The **probability density function (pdf)** (a.k.a. density) of a *continuous* RV X is the function f_X which satisfies

$$\mathrm{P}(a \le X \le b) = \int_a^b f_X(x)dx, \qquad \text{for any } -\infty \le a \le b \le \infty \tag{22.1}$$

- The axioms of probability imply that a valid pdf must satisfy

$$f_X(x) \ge 0 \qquad \text{for all } x, \tag{22.2}$$

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \tag{22.3}$$

- A pdf is 0 outside the range of possible values. Given a specific pdf, the generic bounds like $(-\infty, \infty)$ in integrals should be replaced by the range of possible values.

Example 22.1. Recall Example 21.3. Based on earlier examples, it seems reasonable to assume that the pdf f_X is linear as a function of possible x values, with an intercept of 0 and positive slope; that is,

$$f_X(x) = \begin{cases} cx, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Compute the value of c. Note: this involves more than just reading off a value from a plot. What is the key principle that will determine the value of c?

2. Use the pdf (and geometry) to compute $\mathrm{P}(X < 15)$.

3. Use the pdf (and geometry) to compute $\mathrm{P}(X > 45)$.

4. Use the pdf (and geometry) to find an expression for the cdf of X. (Compare to the cdf from Example 21.3.)

- If X is a continuous random variable with pdf f_X, its cdf F_X satisfies

$$F_X(x) = \int_{-\infty}^{x} f_X(u)du$$

- Roughly, the cdf is the integral of the cdf

Example 22.2. Continuing Example 22.1. Han's arrival time X (minutes after noon) has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

The corresponding cdf satisfies $F_X(x) = (x/60)^2, 0 < x < 60$.

1. Compute and interpret $\mathrm{P}(X = 15)$ and $\mathrm{P}(X = 45)$.

2. Compute the probability that X, truncated to the nearest minute, is 15 minutes after noon; that is, compute $\mathrm{P}(15 \leq X < 16)$. How does this probability compare to $f(15)$?

3. Compute the probability that X, truncated to the nearest minute, is 45 minutes after noon; that is, compute $\mathrm{P}(45 \le X < 46)$. How does this probability compare to $f(45)$?

4. Compute and interpret the ratio of the probabilities from the two previous parts. How does this ratio compare to $f(45)/f(15)$?

- The probability that a continuous random variable X equals any particular value is 0. That is, if X is continuous then $\mathrm{P}(X = x) = 0$ for all x.
- In practical applications involving continuous random variables, "equal to" really means "close to", and "close to" probabilities correspond to intervals which can have positive probability.
- The density $f_X(x)$ at value x is *not* a probability. Rather the height of the pdf $f_X(x)$ is related to $\mathrm{P}(X$ is close to $x)$ in the following sense.
- Let ϵ be a small number in the measurement units of X. Then

$$\mathrm{P}(x - 0.5\epsilon < X < x + 0.5\epsilon) \approx f_X(x)\epsilon$$

  For example, $\epsilon = 0.01$ represents "close to" as rounding to two decimal places.
- In other words,

$$f_x(x) \approx \frac{\mathrm{P}(x - 0.5\epsilon < X < x + 0.5\epsilon)}{\epsilon} = \frac{F_X(x + 0.5\epsilon) - F_X(x - 0.5\epsilon)}{\epsilon}$$

- Taking the limit as $\epsilon \to 0$, we see that the pdf is the derivative of the cdf

$$f_X(x) = \frac{d}{dx} F_X(x)$$

- For any two possible values x_1 and x_2

$$\frac{\mathrm{P}(X \text{ is close to } x_1)}{\mathrm{P}(X \text{ is close to } x_2)} \approx \frac{f_X(x_1)}{f_X(x_2)}$$

  regardless of how "close to" is defined (as long as it's reasonably "small")

Example 22.3. Continuing Example 21.2. Let X be the times (minutes) until the next earthquake occurs (of any magnitude in a certain region). Assume that the pdf of X is

$$f_X(x) = (1/120)e^{-x/120}, \qquad x \geq 0$$

1. Sketch the pdf of X. What does this tell you about waiting times?

2. Without doing any integration, approximate the probability that X rounded to the nearest second is 30 minutes.

3. Without doing any integration determine how much more likely that X rounded to the nearest second is to be 30 than 60 minutes.

4. Compute and interpret $\mathrm{P}(X < 30)$.

5. Compute and interpret $\mathrm{P}(X > 360)$.

6. Find the cdf of X.

7. Find the 25th percentile of X.

8. Find the quantile function of X.

9. Sketch a spinner for simulating values of X.

23 Expected Values for Continuous Random Variables

Example 23.1. Continuing Example 22.1. Han's arrival time X (minutes after noon) has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X)$.

2. Explain how you could approximate $\mathrm{E}(X)$ as a sum (as if X were a discrete random variable).

3. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?

4. Compute and interpret $\mathrm{E}(X)$.

5. Compute and interpret $\mathrm{P}(X \leq \mathrm{E}(X))$.

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $\mathrm{E}(X)$ representing the probability-weighted average value of X.
- The expected value of a continuous random variable with pdf f_X is defined as

$$\mathrm{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Replace the generic bounds $(-\infty, \infty)$ with the possible values of the random variable
- Note well that $\mathrm{E}(X)$ represents *a single number*.
- The expected value is the "balance point" (center of gravity) of a distribution.
- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $\mathrm{E}(\cdot)$ as
  - Simulate lots of values of what's inside $(\cdot)$
  - Compute the average. This is a "usual" average, regardless of whether the variable is discrete or continuous; just sum all the simulated values and divide by the number of simulated values.

Example 23.2. Continuing Example 22.3. Let X be the times (minutes) until the next earthquake occurs (of any magnitude in a certain region). Assume that the pdf of X is

$$f_X(x) = (1/120)e^{-x/120}, \qquad x \geq 0$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X)$.

2. Donny Dont says $\mathrm{E}(X) = \int_0^{\infty}(1/120)e^{-x/120}dx = 1$. Do you agree?

3. Compute and interpret $\mathrm{E}(X)$.

4. Compute and interpret $\mathrm{P}(X \le \mathrm{E}(X))$.

Example 23.3. Continuing Example 23.1. Han's arrival time X (minutes after noon) has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X^2)$.

2. Explain how you could approximate $\mathrm{E}(X^2)$ as a sum (like a discrete random variable).

3. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?

4. Compute $\mathrm{E}(X^2)$.

5. Compute $\mathrm{Var}(X)$.

6. Compute $\mathrm{SD}(X)$.

- The **"law of the unconscious statistician" (LOTUS)** says that the expected value of a transformed random variable can be found without finding the distribution of the transformed random variable, simply by applying the probability weights of the original random variable to the transformed values.

$$\text{Discrete } X \text{ with pmf } p_X: \qquad \mathrm{E}[g(X)] = \sum_x g(x) p_X(x)$$

$$\text{Continuous } X \text{ with pdf } f_X: \qquad \mathrm{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

- LOTUS says we don't have to first find the distribution of $Y = g(X)$ to find $\mathrm{E}[g(X)]$; rather, we just simply apply the transformation g to each possible value x of X and then apply the corresponding weight for x to $g(x)$.
- Whether in the short run or the long run, in general

$$\text{Average of } g(X) \neq g(\text{Average of } X)$$

- In terms of expected values, in general

$$\mathrm{E}(g(X)) \neq g(\mathrm{E}(X))$$

  The left side $\mathrm{E}(g(X))$ represents first transforming the X values and then averaging the transformed values. The right side $g(\mathrm{E}(X))$ represents first averaging the X values and then plugging the average (a single number) into the transformation formula.

Example 23.4. Continuing Example 23.2. Let X be the times (minutes) until the next earthquake occurs (of any magnitude in a certain region). Assume that the pdf of X is

$$f_X(x) = (1/120)e^{-x/120}, \qquad x \geq 0$$

1. Explain how you could use simulation to approximate $\mathrm{E}(X^2)$.

2. Donny Dont says: "I can just use LOTUS and replace x with x^2, so $\mathrm{E}(X^2)$ is $\int_{-\infty}^{\infty}(1/120)x^2e^{-x^2/120}dx$". Do you agree?

3. Compute $\mathrm{E}(X^2)$.

4. Compute $\mathrm{Var}(X)$.

5. Compute $\mathrm{SD}(X)$.

24 Exponential Distributions

Exponential distributions are often used to model the *waiting times* between events in a random process that occurs continuously over time.

Example 24.1. NASA tracks data on fireball events, exceptionally bright meteors that are spectacular enough to to be seen over a very wide area[1]. Let X be the time, measured in months (assume 30 days), between any two fireballs and suppose X has pdf

$$f_X(x) = 2.5e^{-2.5x},\ x \geq 0$$

We say that X has an Exponential distribution with rate 2.5 per month.

1. Sketch the pdf of X and verify that f_X is a valid pdf.

2. Without doing any integration, approximate the probability that the time until the next fireball *rounded to the nearest day* is 6 days.

3. Compute the probability that the time until the next fireball is less than 6 days.

4. Compute and interpret $\mathrm{P}(X > 2)$.

[1]Thanks for former STAT 305 student Martin Hsu for this example and data.

5. Find the cdf of X.

6. Find the median time between fireballs.

7. Suggest a shortcut formula for $\mathrm{E}(X)$. Then compute and interpret $\mathrm{E}(X)$. How does the mean compare to the median? Why?

8. Compute and interpret $\mathrm{P}(X \leq \mathrm{E}(X))$.

9. Compute $\mathrm{E}(X^2)$.

10. Find $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$.

11. Suppose Y is the time between fireballs measured in *days*. How does all of the above change?

- A continuous random variable X has an **Exponential distribution** with *rate* parameter $\lambda > 0$ if its pdf is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \ge 0, \\ 0, & \text{otherwise} \end{cases}$$

- If X has an Exponential(λ) distribution then

$$\begin{aligned} \text{cdf:} \quad F_X(x) &= 1 - e^{-\lambda x}, \quad x \ge 0 \\ \mathrm{P}(X > x) &= e^{-\lambda x}, \quad x \ge 0 \\ \text{quantile:} \quad Q(p) &= -(1/\lambda)\ln(1-p), \quad 0 < p < 1 \\ \mathrm{E}(X) &= \frac{1}{\lambda} \\ \mathrm{Var}(X) &= \frac{1}{\lambda^2} \\ \mathrm{SD}(X) &= \frac{1}{\lambda} \end{aligned}$$

- Percentiles for any Exponential distribution follow a particular pattern in terms of multiples of the mean:
  - For any $m > 0$, $\mathrm{P}(X \le m\mathrm{E}(X)) = 1 - e^{-m}$.
  - The pth percentile is $-\ln(1-p)$ times $\mathrm{E}(X)$.
  - See Table 24.1.
- Exponential distributions are often used to model the *waiting time* in a random process until some event occurs.
  - λ is the average *rate* at which events occur over time (e.g., 2 per hour)
  - $1/\lambda$ is the mean time between events (e.g., 1/2 hour)
- If X has an Exponential(λ) distribution and $a > 0$ is a constant, then aX has an Exponential(λ/a) distribution.
  - If X is measured in hours with rate $\lambda = 2$ per hour and mean 1/2 hour
  - Then $60X$ is measured in minutes with rate 2/60 per minute and mean $60(1/2) = 30$ minutes.
- If X has an Exponential(1) distribution and $\lambda > 0$ is a constant then X/λ has an Exponential(λ) distribution.
- If U has a Uniform(0, 1) distribution then $-\ln(1-U)$ has Exponential(1) distribution, and $-(1/\lambda)\ln(1-U)$ has an Exponential(λ) distribution.

Table 24.1: Percentiles for an Exponential distribution in terms of multiples of the mean

Percentile	Relative to mean
10%	0.11 times the mean
25%	0.29 times the mean
39.3%	0.50 times the mean
50%	0.69 times the mean
63.2%	1.00 times the mean
75%	1.39 times the mean
77.7%	1.50 times the mean
86.5%	2.00 times the mean
90%	2.30 times the mean
91.8%	2.50 times the mean
95%	3.00 times the mean
97%	3.50 times the mean
98.2%	4.00 times the mean
98.9%	4.50 times the mean
99.3%	5.00 times the mean

Example 24.2. Continuing Example 24.1. Let X be the waiting time (months) until the next fireball and assume X has an Exponential(2.5) distribution.

1. Find the conditional probability that the waiting time from now until the next fireball is greater than 3 months given that no fireballs occur in the next month. Be sure to write a valid probability statement involving X before computing.

2. Compare to $\mathrm{P}(X > 2)$. What do you notice?

- **Memoryless property.** If W has an Exponential(λ) distribution then for any $w, h > 0$

$$\mathrm{P}(W > w + h \,|\, W > w) = \mathrm{P}(W > h)$$

- Given that we have already waited w units of time, the conditional probability that we wait at least an additional h units of time is the same as the unconditional probability that we wait at least h units of time.

- In this sense, the future waiting time does not "remember" how long we have already waited.
- A continuous random variable W has the memoryless property if and only if W has an Exponential distribution. That is, Exponential distributions are the *only* continuous[2] distributions with the memoryless property.
- If W has an Exponential(λ) distribution then the conditional distribution of the *excess* waiting time, $W - w$, given $\{W > w\}$ is Exponential(λ).

Example 24.3. Xiomara and Rogelio each leave work at noon from different locations to meet the other for lunch. The amount of time, X, that it takes Xiomara to arrive is a random variable with an Exponential distribution with mean 10 minutes. The amount of time, Y, that it takes Rogelio to arrive is a random variable with an Exponential distribution with mean 20 minutes. Assume that X and Y are independent. Let W be the time, in minutes after noon, at which the first person arrives.

1. What is the relationship between W and X, Y?

2. Compute and interpret $\mathrm{P}(W > 40)$.

3. Find $\mathrm{P}(W > w)$ and identify by name the distribution of W.

4. Compute and interpret $\mathrm{E}(W)$. Is it equal to $\min(\mathrm{E}(X), \mathrm{E}(Y))$?

5. Is $\mathrm{P}(Y > X)$ greater than or less than 0.5? Make an educated guess for $\mathrm{P}(Y > X)$. Then run a simulation to approximate the probability.

[2] Geometric distributions are the only discrete distributions with the discrete analog of the memoryless property.

6. Use simulation to approximate the conditional distribution of W given $\{Y > X\}$ and the conditional distribution of W given $\{Y < X\}$. What do you notice?

- **Exponential race** (a.k.a., competing risks.) Let $W_1, W_2, \dots, W_n$ be independent random variables. Suppose W_i has an Exponential distribution with rate parameter λ_i. Let $W = \min(W_1, \dots, W_n)$ and let I be the discrete random variable which takes value i when $W = W_i$, for $i = 1, \dots, n$. Then
  - W has an Exponential distribution with rate $\lambda = \lambda_1 + \cdots + \lambda_n$
  - $\mathrm{P}(I = i) = \mathrm{P}(W = W_i) = \frac{\lambda_i}{\lambda_1 + \cdots + \lambda_n}, i = 1, \dots, n$
  - W and I are independent
- Imagine there are n contestants in a race, labeled $1, \dots, n$, racing independently, and W_i is the time it takes for the ith contestant to finish the race. Then $W = \min(W_1, \dots, W_n)$ is the winning time of the race, and W has an Exponential distribution with rate parameter equal to sum of the individual contestant rate parameters.
- The discrete random variable I is the label of which contestant is the winner. The probability that a particular contestant is the winner is the contestant's rate divided by the total rate. That is, the probability that contestant i is the winner is proportional to the contestant's rate λ_i.
- Furthermore, W and I are independent. Information about the winning time does not influence the probability that a particular contestant won. Information about which contestant won does not influence the distribution of the winning time.

Example 24.4. Database queries to the Cal Poly data warehouse occur randomly throughout the day. During regular business hours, queries arrive at rate 0.8 per second on average, so that the average number of queries that arrive during any t second time interval is $0.8t$. Suppose that the *number* of queries that arrive during any t second time interval follows a marginal Poisson distribution with mean $0.8t$.

We are interested in the distribution of T, the *time* (seconds) until the next query arrives.

1. Interpret the event $\{T > 2\}$. How can you express this as an equivalent event involving the number of queries?

2. Compute $\mathrm{P}(T > 2)$.

3. Compute $\mathrm{P}(T > t)$ as a function of $t > 0$.

4. Identify by name the distribution of T; be sure to specify the values of any relevant parameters.

5. Compute and interpret $\mathrm{E}(T)$.

- Events occur continuously over time according to a **Poisson process** with *rate* $\lambda > 0$ (per unit time) if
  - The number of events that occur in any time interval of length t has a Poisson(λt) distribution.
    * In particular, the distribution does not depend on the starting time or ending time of the interval, just its length.
  - The numbers of events that occur in disjoint intervals of time are independent.
- A random process is a Poisson process with rate λ if and only if
  - it is a counting process, that is, it is a process which counts the number of occurrences of some event over time.

  - the *interarrival times* (times between events) $W_0, W_1, \ldots$ are independent Exponential(λ) random variables

- A random process is a Poisson process with rate λ if and only if
  - It is a counting process
  - The numbers of events that occur in disjoint intervals of time are independent
  - For small h,

$$\begin{aligned}
\text{P(0 events in time interval of length } h) &\approx 1 - \lambda h \\
\text{P(1 events in time interval of length } h) &\approx \lambda h \\
\text{P(at least 2 events in time interval of length } h) &\approx 0
\end{aligned}$$

    * Roughly, in short intervals of time, at most one event will occur, and the probability that an event does occur is proportional to the length of the time interval.
    * That is, if time is chopped up into many short intervals of equal length h, a Poisson Process behaves like a Bernoulli process: each short interval is a trial, an event either occurs (success) or not, the trials (intervals) are independent, the probability of success is the same for each trial (λh), and the process counts the number of successes

Example 24.5. Suppose that elapsed times (hours) between successive fireballs are independent, each having an Exponential(2.5) distribution. Let T be the time elapsed from now until the third fireball occurs.

1. Compute E(T).

2. Compute SD(T).

3. Does T have an Exponential distribution? Explain.

4. Use simulation to approximate the distribution of T.

- A continuous random variable X has a **Gamma distribution** with *shape* parameter α, a positive integer[3], and *rate* parameter[4] $\lambda > 0$ if its pdf is

$$\begin{aligned} f_X(x) &= \frac{\lambda^\alpha}{(\alpha-1)!} x^{\alpha-1} e^{-\lambda x}, & x \geq 0, \\ &\propto x^{\alpha-1} e^{-\lambda x}, & x \geq 0 \end{aligned}$$

  If X has a Gamma(α, λ) distribution then

$$\begin{aligned} \mathrm{E}(X) &= \frac{\alpha}{\lambda} \\ \mathrm{Var}(X) &= \frac{\alpha}{\lambda^2} \\ \mathrm{SD}(X) &= \frac{\sqrt{\alpha}}{\lambda} \end{aligned}$$

- If $W_1, \ldots, W_n$ are independent and each W_i has an Exponential(λ) distribution then $(W_1 + \cdots + W_n)$ has a Gamma(n,λ) distribution.
- Each W_i represents the waiting time *between* two occurrences of some event, so $W_1 + \cdots + W_n$ represents the total waiting time until a total of n occurrences.
- Exponential distributions are continuous analogs of Geometric distributions, and Gamma distributions are continuous analogs of Negative Binomial distributions.
- For a positive integer d, the Gamma($d/2, 1/2$) distribution is also known as the **chi-square distribution with d degrees of freedom**.

[3] There is a more general expression of the pdf which replaces $(\alpha-1)!$ with the *Gamma function* $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u} du$, that can be used to define a Gamma pdf for any $\alpha > 0$. When α is a positive integer, $\Gamma(\alpha) = (\alpha-1)!$.

[4] Like Exponential distributions, Gamma distributions are sometimes parametrized directly by their mean $1/\lambda$, instead of the rate parameter λ. The mean $1/\lambda$ is called the scale parameter.

25 Normal Distributions

Normal distributions are probably the most important distributions in probability and statistics. Any Normal distribution follows the "empirical rule" which specify the percentiles that give a Normal distribution its particular bell shape. The key to working with Normal distributions is to work with standardized values, that is, standard deviations from the mean

- A continuous random variable X has a **Normal** (a.k.a., Gaussian) distribution with mean $\mu \in (-\infty, \infty)$ and standard deviation $\sigma > 0$ if its pdf is

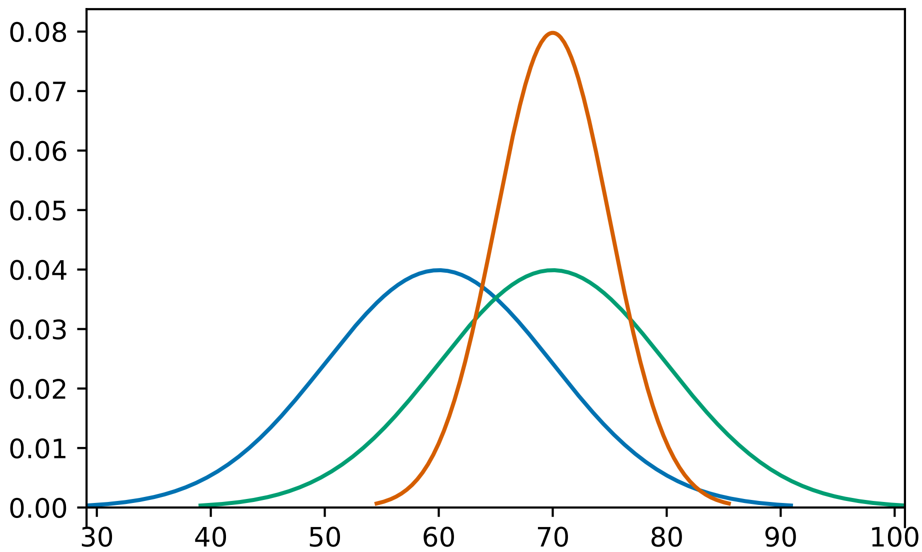
$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \quad -\infty < x < \infty,$$

- If X has a Normal(μ, σ) distribution then

$$\begin{aligned} \mathrm{E}(X) &= \mu \\ \mathrm{SD}(X) &= \sigma \end{aligned}$$

- A Normal density is a particular "bell-shaped" curve which is symmetric about its mean μ. The mean μ is a *location* parameter: μ indicates where the center and peak of the distribution is.
- The standard deviation σ is a *scale* parameter: σ indicates the distance from the mean to where the concavity of the density changes. That is, there are inflection points at $\mu \pm \sigma$.

Example 25.1. The pdfs in the plot below represent the distribution of hypothetical test scores in three classes. The test scores in each class follow a Normal distribution. Identify the mean and standard deviation for each class.



- The key to working with Normal distributions is to work in terms of standard deviations from the mean (that is, standardized values).
  - If X has a Normal(μ, σ) distribution then $Z = \frac{X-\mu}{\sigma}$ has a Normal(0, 1) distribution
  - If Z has a Normal(0, 1) distribution then $X = \mu + \sigma Z$ has a Normal(μ, σ) distribution
- Any Normal distribution follows an "empirical rule" which relates percentiles to standard deviation from the mean and defines the particular bell shape.
- A continuous random variable Z has a **Standard Normal** (a.k.a. Normal(0, 1)) distribution if its pdf is

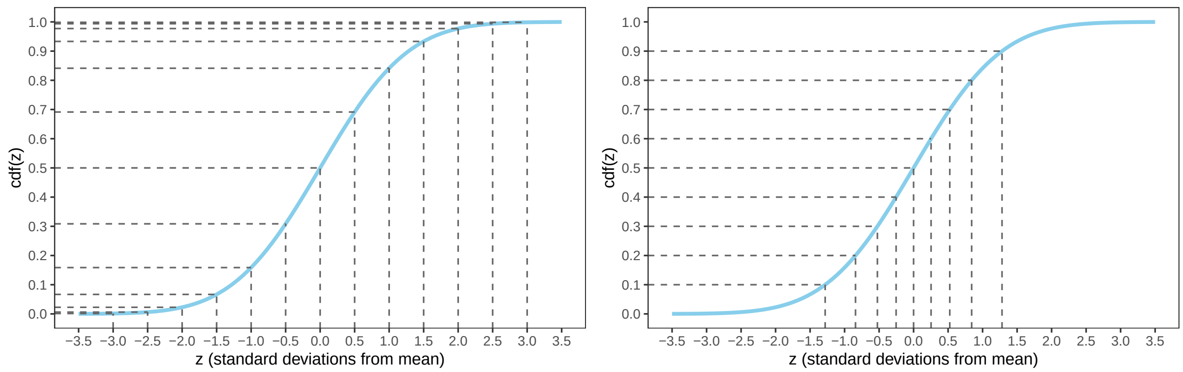
$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty,$$

- The cdf is a Normal(0, 1) distribution is

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}, \quad -\infty < z < \infty,$$

  but integrals of this type do not have a closed form and must be evaluated with software

- Figure 25.1 defines a Normal distribution by specifying all of its percentiles.
- The empirical rule is often described in terms of "within [blank] standard deviations of the mean" as in the following:
  - 38% of values are within 0.5 standard deviations of the mean



(a) Percentiles highlighted in 0.5 increments of standard deviations from the mean

(b) Deciles highligted in terms of standard deviations from the mean

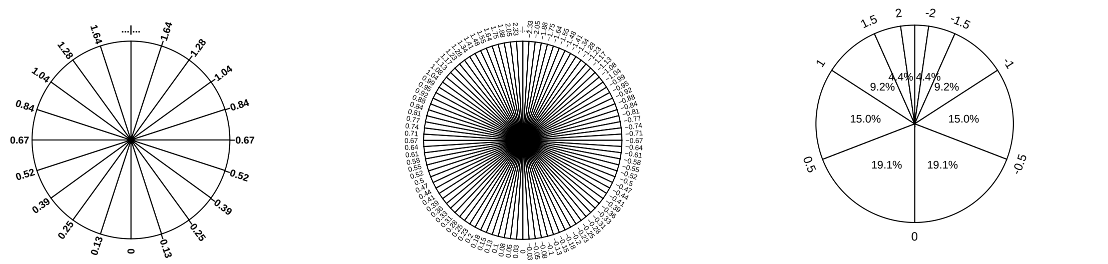
Figure 25.1: The Normal(0, 1) cdf. Provides the cdf for any Normal distribution if the variable is measured in *standard deviations from the mean*. It's the same cdf in both figures just with different values highlighted.

Table 25.1: Select percentiles for a Normal distribution in terms of standard deviations from the mean

(a) In 0.5 increments of standard deviations from the mean

(b) Deciles and quantiles as standard deviations from the mean

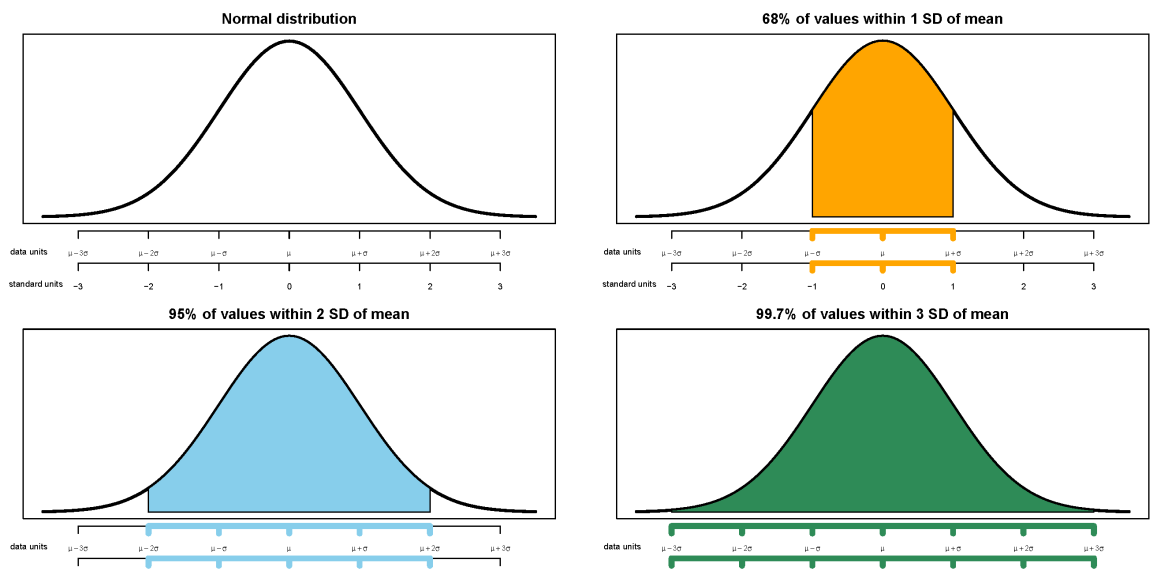
Percentile	Standard deviations from the mean	Percentile	Standard deviations from the mean
0.02%	-3.5	1%	-2.33
0.1%	-3.0	5%	-1.64
0.6%	-2.5	10%	-1.28
2.3%	-2.0	20%	-0.84
6.7%	-1.5	25%	-0.67
15.9%	-1.0	30%	-0.52
30.9%	-0.5	40%	-0.25
50%	0.0	50%	0.00
69.1%	0.5	60%	0.25
84.1%	1.0	70%	0.52
93.3%	1.5	75%	0.67
97.7%	2.0	80%	0.84
99.4%	2.5	90%	1.28
99.9%	3.0	95%	1.64
99.98%	3.5	99%	2.33



(a) Each sector represents an interval with 5% probability (b) Each sector represents an interval with 1% probability (c) Values labeled in increments of 0.5 standard deviations from the mean

Figure 25.2: A continuous *Normal(0, 1)* spinner. The same spinner is displayed in all figures, with different features highlighted. This spinner can be used to simulate values from any Normal distribution by simulating values in terms of standard deviations from the mean.

- 50% of values are withing 0.67 standard deviations of the mean
- 68% of values are within 1 standard deviation of the mean
- 87% of values are within 1.5 standard deviations of the mean
- 95% of values are within 2 standard deviations of the mean
- 99% of values are within 2.6 standard deviations of the mean
- 99.7% of values are within 3 standard deviations of the mean
- 99.99% of values are within 4 standard deviations of the mean



Example 25.2. The wrapper of a package of candy lists a weight of 47.9 grams. Naturally, the

weights of individual packages vary somewhat. Suppose package weights have an approximate Normal distribution with a mean of 49.8 grams and a standard deviation of 1.3 grams.

1. Sketch the distribution of package weights. Carefully label the variable axis. It is helpful to draw two axes: one in the measurement units of the variable, and one in standardized units.

2. Why wouldn't the company print the mean weight of 49.8 grams as the weight on the package?

3. Estimate the probability that a package weighs less than the printed weight of 47.9 grams.

4. Estimate the probability that a package weighs between 47.9 and 53.0 grams.

5. Suppose that the company only wants 1% of packages to be underweight. Find the weight that must be printed on the packages.

6. Find the 99th percentile of package weights. How can you use the work you did in the previous part?

Example 25.3. A bank uses an applicant's score on some criteria to decide whether or not to approve their loan application. Based on past history, the bank has determined that

- Scores for those who repay the loan follow a Normal distribution with mean 60 and SD 8
- Scores for those who do NOT repay the loan follow a Normal distribution with mean 40 and SD 12

When someone applies for a loan the bank does not know whether the applicant will eventually repay the loan. How can the bank use the applicant's score to decide whether or not to approve the loan?

1. Draw sketches of these two normal curves on the same axis.

2. Suggest a general form of a decision rule, based on an applicant's score, for deciding whether or not to approve the applicant's loan.

3. Describe the two kinds of classification errors that could be made in this situation.

4. Determine the probability that we incorrectly reject the application of someone who would repay. Shade in the corresponding region under the Normal curve, and interpret this probability.

5. Determine the probability that we incorrectly approve the application of someone who would NOT repay. Shade in the corresponding region under the Normal curve, and

interpret this probability.

6. In which direction — smaller or larger — would you need to change the decision rule's cutoff value in order to decrease the probability that an applicant who would repay the loan is incorrectly rejected?

7. Would the probability of the other kind of error — incorrectly approving a loan for an applicant who would not repay it — increase or decrease with this new cutoff value?

8. Determine the cutoff value needed to decrease the probability that an applicant who would repay the loan is incorrectly rejected to 0.05.

9. Determine the other error probability with this new cut-off rule.

10. Now repeat the two previous parts with the goal of decreasing to 0.05 the probability of incorrectly approving an applicant who would not repay the loan.

11. If you consider the two kinds of errors to be equally serious, how might you decide which of the three decision rules considered so far is the best?

12. Which error do you think the bank would consider more serious? In which direction would that move the threshold?

- Making decisions/predictions based on data often involves trade-offs.
- Decreasing the probability of one kind of error often comes at the expense of increasing the probability of another kind of error.

26 Expected Values of Linear Combinations of Random Variables

A **linear rescaling** of a random variable X is of the form $aX+b$ where a and b are non-random constants. A **linear combination** of two random variables X and Y is of the form $aX+bY$ where a and b are non-random constants.

Properties of expected value

- Expected value and linear rescaling: If X is a random variable and a, b are non-random constants then
$$\mathrm{E}(aX+b) = a\mathrm{E}(X) + b$$
- Linearity of expected value: For *any* two random variables X and Y,
$$\mathrm{E}(X+Y) = \mathrm{E}(X) + \mathrm{E}(Y)$$
- That is, the expected value of the sum is the sum of expected values, regardless of how the random variables are related.
- Therefore, you only need to know the marginal distributions of X and Y to find the expected value of their sum. (But keep in mind that the *distribution* of $X+Y$ will depend on the joint distribution of X and Y.)
- Combining properties of linear rescaling with linearity of expected value yields the expected value of a linear combination, which extends naturally to more than two random variables.
$$\mathrm{E}(aX+bY+c) = a\mathrm{E}(X) + b\mathrm{E}(Y) + c$$

Properties of variance

- Variance and linear rescaling: If X is a random variable and a, b are non-random constants then
$$\mathrm{SD}(aX+b) = |a|\mathrm{SD}(X)$$
$$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$$
- Variance of sums and differences of random variables
$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$$
$$\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\mathrm{Cov}(X,Y)$$

- The variance of the sum (or difference) is the sum of the variances if and only if X and Y are uncorrelated.

$$\begin{aligned}
\mathrm{Var}(X+Y) &= \mathrm{Var}(X) + \mathrm{Var}(Y) \qquad \text{if } X, Y \text{ are uncorrelated} \\
\mathrm{Var}(X-Y) &= \mathrm{Var}(X) + \mathrm{Var}(Y) \qquad \text{if } X, Y \text{ are uncorrelated}
\end{aligned}$$

- If a, b, c are non-random constants and X and Y are random variables then

$$\mathrm{Var}(aX + bY + c) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\mathrm{Cov}(X, Y)$$

Example 26.1. Recall the SAT score Application assignment.

Let M be the Math score and R be the Reading (and Writing) score of a randomly selected SAT taker. We'll assume (M, R) pairs follow a *Bivariate Normal distribution*.

- Math scores (M) have mean 500 and standard deviation 140
- Reading scores (R) have mean 520 and standard deviation 110

For each of the following scenarios, compute

- $\mathrm{E}(M+R)$
- $\mathrm{Var}(M+R)$
- $\mathrm{SD}(M+R)$

1. $\mathrm{Corr}(M, R) = 0$

2. $\mathrm{Corr}(M, R) = 0.6$

3. $\mathrm{Corr}(M, R) = -0.9$

4. In which of these scenarios is $\mathrm{SD}(M+R)$ largest? Smallest? Can you explain why intuitively?

Example 26.2. Recall the SAT score Application assignment.

Let M be the Math score and R be the Reading (and Writing) score of a randomly selected SAT taker. We'll assume (M, R) pairs follow a *Bivariate Normal distribution*.

- Math scores (M) have mean 500 and standard deviation 140
- Reading scores (R) have mean 520 and standard deviation 110

For each of the following scenarios, compute

- $\mathrm{E}(M-R)$
- $\mathrm{Var}(M-R)$
- $\mathrm{SD}(M-R)$

1. $\mathrm{Corr}(M, R) = 0$

2. $\mathrm{Corr}(M, R) = 0.6$

3. $\mathrm{Corr}(M, R) = -0.9$

4. In which of these scenarios is $\mathrm{SD}(M-R)$ largest? Smallest? Can you explain why intuitively?

- Two continuous random variables X and Y have a **Bivariate Normal** distribution with parameters μ_X, μ_Y, $\sigma_X > 0$, $\sigma_Y > 0$, and $-1 < \rho < 1$ if the joint pdf is

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

- It can be shown that if the pair (X,Y) has a BivariateNormal(μ_X, μ_Y, σ_X, σ_Y, ρ) distribution

$$\begin{aligned}
\mathrm{E}(X) &= \mu_X \\
\mathrm{E}(Y) &= \mu_Y \\
\mathrm{SD}(X) &= \sigma_X \\
\mathrm{SD}(Y) &= \sigma_Y \\
\mathrm{Corr}(X,Y) &= \rho
\end{aligned}$$

- A Bivariate Normal Density has elliptical contours. For each height $c > 0$ the set $\{(x,y) : f_{X,Y}(x,y) = c\}$ is an ellipse. The density decreases as (x,y) moves away from (μ_X, μ_Y), most steeply along the minor axis of the ellipse, and least steeply along the major of the ellipse.
- A scatterplot of (x,y) pairs generated from a Bivariate Normal distribution will have a rough linear association and the cloud of points will resemble an ellipse.
- If X and Y have a Bivariate Normal distribution, then the marginal distributions are also Normal: X has a Normal(μ_X, σ_X) distribution and Y has a Normal(μ_Y, σ_Y).
- If X and Y have a Bivariate Normal distribution and $\mathrm{Corr}(X,Y) = 0$ then X and Y are independent. (Remember, in general it is possible to have situations where the correlation is 0 but the random variables are not independent.)
- **X and Y have a Bivariate Normal distribution if and only if every linear combination of X and Y has a Normal distribution.** That is, X and Y have a Bivariate Normal distribution if and only if $aX + bY$ has a Normal distribution for all a, b.

Example 26.3. Let M be the Math score and R be the Reading (and Writing) score of a randomly selected SAT taker. We'll assume (M,R) pairs follow a *Bivariate Normal distribution.*

- Math scores (M) have mean 500 and standard deviation 140
- Reading scores (R) have mean 520 and standard deviation 110
- The correlation is 0.6

1. Compute the probability that a student has a total score above 1400.

2. Compute the probability that a student has a higher Math than Reading score.

Properties of covariance

$$
\begin{aligned}
\mathrm{Cov}(X, X) &= \mathrm{Var}(X) \\
\mathrm{Cov}(X, Y) &= \mathrm{Cov}(Y, X) \\
\mathrm{Cov}(X, c) &= 0 \\
\mathrm{Cov}(aX + b, cY + d) &= ac\mathrm{Cov}(X, Y) \\
\mathrm{Cov}(X + Y,\ U + V) &= \mathrm{Cov}(X, U) + \mathrm{Cov}(X, V) + \mathrm{Cov}(Y, U) + \mathrm{Cov}(Y, V)
\end{aligned}
$$

Example 26.4. Let X be the number of two-point field goals a basketball player makes in a game, Y the number of three point field goals made, and Z the number of free throws made (worth one point each). Assume X, Y, Z have standard deviations of 2.5, 3.7, 1.8, respectively, and $\mathrm{Corr}(X, Y) = 0.1$, $\mathrm{Corr}(X, Z) = 0.3$, $\mathrm{Corr}(Y, Z) = -0.5$.

1. Find the standard deviation of the number of fields goals in a game (not including free throws)

2. Find the standard deviation of total points scored on fields goals in a game (not including free throws)

3. Find the standard deviation of total points scored in a game.

27 Statistical Estimation (From Probability to Statistics)

Example 27.1. In each of the following scenarios, one of (i) and (ii) is a *probability* question and one is a *statistics* question. Classify each question as "probability" or "statistics". Then discuss some general features of the probability questions, and some general features of the statistics questions

1. U.S. Satisfaction

    - Assume that 27% of Americans are satisfied with the way things are going in the U.S. If a random sample of 1000 Americans is selected, what are the chances that fewer than 250 Americans in the sample are satisfied with the way things are going in the U.S?
    - In a random sample of 1000 Americans, 250 Americans are satisfied with the way things are going in the U.S. What percent of all Americans are satisfied with the way things are going in the U.S?

2. Great white shark length

    - Assume that lengths of female great white sharks follow a Normal distribution with mean 16 ft and standard deviation 3 ft. If a random sample of 40 female great white sharks is selected, what are the chances that the average length in the sample is between 16.5 and 17.5 ft?
    - In a sample of 40 female great white sharks, the sample mean length is 17.0 ft. What is the average length of female great white sharks?

3. In a clinical trial, 200 subjects are randomly assigned to receive either an experimental vaccine or no treatment (e.g., a placebo), and then each subject is tested to see if they developed immunity.

    - If the vaccine is in reality not effective, what are the chances that the proportion of subjects who develop immunity is at least twice as large for those who received the vaccine than for those who did not?

- Among the 100 subjects who received the vaccine, 20 developed immunity; among the 100 subjects who did not receive the vaccine, 10 developed immunity. Is there evidence that the vaccine is effective?

4. Snap streaks

   - Assume that lengths of streaks on Snapchat follow an Exponential distribution, with mean 75 for teen users and mean 40 for adult users. A random sample of 200 Snapchat users is selected, 100 teens and 100 adults. What are the chances that the sample mean streak length for teens is with within 20 of the sample mean streak length for adults?
   - In a random sample of 100 teen Snapchat users, the sample mean streak length is 80. In a random sample of 100 adult Snapchat users, the sample mean streak length is 35. In general, how much longer do streak lengths for teen Snapchat users tend to be than for adult Snapchat users?

- **Probability:** Assume a model for a random (uncertain) process and evaluate probabilities of potential outcomes or events — what might the data look like?
  - In probability, a model is assumed and parameters are assumed to be known, but the data is unknown.
  - In probability, we typically call the model a "distribution" and the data "random variables".
- **Statistical inference:** Observe sample data and make conclusions about the process that generated the data
  - In statistics, the data is observed but parameters are unknown.
  - In statistics, we typically call the model the "population" and the data the "sample".
- *Statistical inference* involves using sample data to make conclusions about the population.
  - **Point estimation:** Estimate an unknown population parameter with a single number based on sample data

- **Interval estimation:** Estimate an unknown population parameter with a plausible range of values based on sample data
- **Hypothesis testing:** Assess the evidence that the sample data provides regarding a particular claim about unknown population parameters.

- The *population distribution* is the distribution of individual values of the variable of interest. A *parameter*, generically denoted θ, is a number that summarizes the population distribution.
- The **population distribution**, denoted p_θ or $p_\theta(x)$, is a probability model for the distribution of values of a variable[1] X over all *individuals* in the population.
  - If the variable X is discrete, then p_θ represents a probability mass function (pmf): $p_\theta(x) = \mathrm{P}(X = x)$
  - If the variable X is continuous, then p_θ represents a probability density function (pdf): $p_\theta(x) \approx \frac{1}{\epsilon}\mathrm{P}(x - \epsilon/2 < X < x + \epsilon/2)$
- A **parameter**, generically denoted θ, is a characteristic of the population distribution, often computed as an expected value of some function of X.
  - In some cases θ represents a vector of multiple parameters.
  - For example, if p_θ represents a Normal distribution, then θ might represent the pair $\theta \equiv (\mu, \sigma)$ consisting of the population mean and the population standard deviation.
  - The **population mean**, typically denoted μ, is the average of the values of the variable over all individuals in the population, $\mu = \mathrm{E}(X)$
  - The **population standard deviation**, typically denoted σ, is the standard deviation of all the individual values in the population, $\sigma = \mathrm{SD}(X)$
- A parameter is a *fixed number* but its value is *unknown*[2].
- Many statistical studies involve *random sampling* from a population.
- A **(simple) random sample** of size n is a collection of random variables $X_1, \ldots, X_n$ that are *independent* and *identically distributed* (i.i.d.)

[1]Recall: By definition $\mathrm{Var}(Y) = \mathrm{E}[(Y - \mathrm{E}(Y))^2]$ but remember the useful computational formula $\mathrm{Var}(Y) = \mathrm{E}(Y^2) - (\mathrm{E}(Y))^2$. Apply this to $Y = \hat{\theta} - \theta$, and remember that $\hat{\theta}$ is random but θ is not.

[2]There are two main philosophical approaches to statistical inference that differ mainly in whether they emphasize "fixed number" or "unknown". In our brief exploration of estimation, we will follow the *frequentist* approach, which treats a parameter as a *fixed number*. The *Bayesian* approach treats an *unknown* parameter— in other words, an *uncertain* parameter—as a *random variable* with a probability distribution that describes the degree of uncertainty. Both approaches are valid, each with advantages and disadvantages. The frequentist approach has been historically more prevalent, but the Bayesian approach has gained in popularity in the last 30 or so years.

– Identically distributed — each individual X_i has the same marginal distribution — the population distribution — since they are sampled from the same population:

$$X_i \sim p_\theta \quad \text{for all } i = 1, \dots, n$$

– Independent — the values are sampled independently, so the joint distribution[3] is the product of marginal distributions:

$$p_\theta(x_1, \dots, x_n) = p_\theta(x_1)p_\theta(x_2)\cdots p_\theta(x_n) \quad \text{for all } x_1, \dots, x_n$$

- A **statistic** is a characteristic of the *sample* which can be computed from the data. More precisely, a statistic is a function of $X_1, \dots, X_n$, but not of θ.
  - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
  - The probability distribution of a statistic is called a **sampling distribution**.
  - The *distribution* of a statistic will depend on θ, but the *value* of a statistic will not.
  - The **sample mean**, typically denoted $\bar{X}$, is the average of the values of the variable over all individuals in the population, $\bar{X} = (X_1 + \cdots + X_n)/n$
- A statistic $\hat{\theta} \equiv T(X_1, \dots, X_n)$ used to estimate a parameter θ is called a **(point) estimator**.
  - Naturally, we hope that there is some correspondence between our estimator $\hat{\theta}$ and the parameter it is estimating θ — for example, estimating the population mean with the sample mean — but the definition does not require this.
  - When the numerical value of a statistic is calculated from observed sample data, its value is called a *(point) estimate* of the parameter.
  - The estimator $\hat{\theta}$ is a RV whose value changes from sample to sample; an estimate is the observed value computed based on the results of a single sample.

Example 27.2. Jerry has just opened a car dealership in a new location. He assumes that the number of cars he'll sell in any randomly selected day has a Poisson(μ) distribution. But he doesn't know the value of μ, so he plans to use data from n days of business to estimate μ. Let $X_1, \dots, X_n$ represent the number of cars sold for a sample of n days.

[3]Abuse of notation in the expression: p_θ on the left represents the joint distribution of $(X_1, \dots, X_n)$, a function with n inputs; p_θ on the right represents that marginal distribution of each X_i (since i.d.), a function with a single input.

1. What plays the role of the population distribution? What is the parameter? How might you interpret the parameter in this context?

2. Suggest one estimator of μ. (Hint: what does μ represent for a Poisson distribution?)

3. Suppose he collects data on $n = 3$ days and observes $x_1 = 3$ cars sold on day 1, $x_2 = 0$ cars sold on day 2, and $x_3 = 2$ cars sold on day 3. Compute the value of your estimator from the previous part for this sample. That is, what is your estimate of μ based on this sample?

4. Suggest some other estimators of μ. (Hint: what else does μ represent for a Poisson distribution?)

5. For a Poisson(μ) distribution μ is both the population mean and the population variance. So one reasonable estimator of μ is the sample mean $\bar{X}$. Another reasonable estimator is the sample variance. But there are two commonly used formulas for the sample variance, each of which provides an estimator of μ

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$
$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

   Compute the value of each of these estimators for the (3, 0, 2) sample. That is, what is your estimate of μ based on this sample if you use each of these estimators?

6. At his previous dealeship at a different location, Jerry sold 2.3 cars per day on average. Consider the estimator of μ: $\frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)$. Explain in words what this estimator represents. In particular, what happens as n increases? Then compute the value of this estimator for the (3, 0, 2) sample.

7. According to the Poisson(μ) distribution, the probability of selling 0 cars in a day is $p_0 = e^{-\mu}$. Rearranging, $\mu = -\log p_0$. Therefore, if $\hat{p}_0$ is the sample proportion of days with 0 cars sold, then another estimator of μ is $-\log \hat{p}_0$. Compute the value of this estimator for the (3, 0, 2) sample.

8. We have seen a few different, seemingly reasonable estimators of μ, and each produced a different estimate of μ based on the same (3, 0, 2) sample data. Which of these numbers is the best *estimate* of μ? That is, which of these estimates is closest to μ?

9. How might you investigate which of the *estimators* corresponds to the best estimation *procedure*?

- Because the true value of the parameter θ is unknown, we never know if the sample data provides a reasonable estimate of θ
- Rather, we evaluate the estimation *procedure* for potential values of θ: How does the estimation procedure perform over many possible samples given a range of potential values of θ?
- Some questions of interest when estimating a parameter θ

- How do we find an estimator of θ?
- What are properties of “good” estimators?
- How can we tell if one estimator is “better” than another?
- Can we find a “best” estimator of θ? (Short answer: no)
- How do we determine the margin of error for an estimator?

28 Maximum Likelihood Estimation

Example 28.1. Suppose we are willing to assume a Poisson(μ) model for the number of cars sold in a day at a particular car dealership. But we don't know the value of μ, so we'll collect data and use it to estimate μ. Suppose first that we observe just a single day; let X be the number of cars sold on this day. (We'll look at a sample of n days soon.)

1. Does X take values on a discrete or continuous scale? What about μ?

2. *First some probability questions.* Assume $\mu = 2.3$. Find the pmf of X and sketch a plot of it; what is this a function of? Find and interpret the probability that in a single day there are 3 cars sold.

3. *Now back to some statistics questions.* Now remember that in reality μ is unknown. Suppose that $x = 3$ cars are sold in a single day. Now write the pmf plugging in everything we know; what is this a function of? Is this function a pmf?

4. Compute and intepret the probability that in a single day there are 3 cars sold if $\mu = 2.3$.

5. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 1$.

6. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 5$.

7. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 3.5$.

8. Suppose that $x = 3$ cars are sold in a single day. Based just on this day, which value — 1, 2.3, 3.5, or 5 — would you choose as your estimate for μ? Why?

9. We obviously have more choices for our estimate of μ than just 1, 2.3, 3.5, or 5. Describe in principle the process you would follow to find the estimate of μ based on a single day with 3 cars sold.

10. Suppose that we observe $x = 3$ cars sold in a single day. Plot the likelihood function; be sure to label your axes. Based on your plot, what would be your estimate of μ?

11. Suppose that we observe $x = 3$ cars sold in a single day. Plot the log[1] of the likelihood function. Compare the plots of the likelihood function and the log-likelihood; what do you notice about where the maximum value occurs?

12. Now consider a general value of x cars solds in a single day. Carefully write the likelihood function, and the log-likelihood function. What are these functions of? How would you use them to determine your estimate of μ given x cars sold in a single day?

- Consider a single observation X from a population p_θ with parameter θ: Let $X \sim p_\theta$. The **likelihood function** given $X = x$ is defined as

$$L(\theta) \equiv L_x(\theta) = p_\theta(x)$$

- That is, the likelihood function is the probability (or density for continuous data X) of observing the given sample data x viewed as a *function of the unknown parameter(s)* θ. The observed value of x (data) is treated as a fixed constant.
- The *maximum likelihood estimate* of θ is the value of θ for which the observed data has the highest likelihood of occurring.
- The **maximum likelihood estimator (MLE)** of θ is the RV $\hat{\theta}$ which maximizes the likelihood, i.e. the value $\hat{\theta}$ which satisfies

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta.$$

- Remember that an estimator $\hat{\theta}$ is a function of X, the observed data.
- It is often convenient to consider the **log-likelihood function** of θ

$$\ell(\theta) \equiv \ell_x(\theta) = \log L_x(\theta)$$

- Because log is an increasing function it follows that $\hat{\theta}$ is the MLE of θ if and only if

$$\ell(\hat{\theta}) \geq \ell(\theta) \text{ for all } \theta.$$

Example 28.2. Suppose we are willing to assume a Poisson(μ) model for the number of cars sold in a day at a particular car dealership. But we don't know the value of μ, so we'll collect

[1]"log" always means natural log, ln.

data and use it to estimate μ. But now we estimate μ based on a sample of n days. Let $X_1, \ldots, X_n$ be the number of cars sold in a random sample of n days. For example, suppose $n = 3$ and we observe $x_1 = 3$, $x_2 = 0$, $x_3 = 2$.

1. *First some probability questions.* Assume $\mu = 2.3$ and $n = 3$. Find the joint pmf of (X_1, X_2, X_3); what is this a function of? Find and interpret the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$.

2. *Now back to some statistics questions.* Now remember that in reality μ is unknown. Suppose that the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ is observed. Now write the joint pmf plugging in everything we know; what is this a function of? Is this function a pmf?

3. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 2.3$.

4. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 1$.

5. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 5$.

6. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 3.5$.

7. Suppose the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ is observed. Which value — 1, 2.3, 3.5, or 5 — would you choose as your estimate for μ? Why?

8. We obviously have more choices for our estimate of μ than just 1, 2.3, 3.5, or 5. Describe in principle the process you would follow to find the estimate of μ based on observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$.

9. Suppose that we observe the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$. Plot the likelihood function; be sure to label your axes. Based on your plot, what would be your estimate of μ?

10. Suppose that we observe the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$. Plot the log of the likelihood function. Compare the plots of the likelihood function and the log-likelihood; what do you notice about where the maximum value occurs?

11. Now consider a general sample of size n. Carefully write the likelihood function, and the log-likelihood function. What are these functions of? How would you use them to determine your estimate of μ given x cars sold in a single day?

- Consider a random sample of n values from a population p_θ with parameter θ: Let $X_1, \ldots, X_n$ be i.i.d. $\sim p_\theta$.

- The **likelihood function** of θ, given $X_1 = x_1, \dots, X_n = x_n$, is

$$L(\theta) \equiv L_{x_1,\dots,x_n}(\theta) = \prod_{i=1}^{n} p_\theta(x_i) = p_\theta(x_1) \times \cdots \times p_\theta(x_n)$$

- That is, the likelihood function is the joint density[2] of the sample $X_1, \dots, X_n$, evaluated at the observed data values $x_1, \dots, x_n$, viewed as *a function of the (unknown) parameter(s)* θ.
- Note that $x_1, \dots, x_n$ represent distinct values (like $x_1 = 3, x_2 = 0, x_3 = 2$ in the example).
- The MLE of θ is the value of θ for which the observed sample data has the highest likelihood of occurring.
- The **maximum likelihood estimator (MLE)** of θ is the RV $\hat{\theta}$ which maximizes the likelihood, i.e. the value $\hat{\theta}$ which satisfies

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta.$$

- Remember that an estimator $\hat{\theta}$ is a function of $X_1, \dots, X_n$.
- The **log-likelihood function** of θ is

$$\begin{aligned} \ell(\theta) \equiv \ell_{x_1,\dots,x_n}(\theta) &= \log L(\theta) \\ &= \sum_{i=1}^{n} \log p_\theta(x_i) = \log p_\theta(x_1) + \cdots + \log p_\theta(x_n) \end{aligned}$$

- Because log is an increasing function it follows that $\hat{\theta}$ is the MLE of θ if and only if

$$\ell(\hat{\theta}) \geq \ell(\theta) \text{ for all } \theta.$$

Example 28.3. Suppose instead of observing the number of cars sold on individual days, you only knew that in a sample of 3 days there were 5 cars sold in total.

1. *Probability question.* If $X_1, \dots, X_n$ are i.i.d. Poisson(μ), what is the distribution of $X_1 + \cdots + X_n$?

2. Write the likelihood function for a sample of 3 days in which a total of 5 cars were sold. How does this compare to the likelihood function based on the sample $(3, 0, 2)$?

[2]Recall: By definition $\text{Var}(Y) = \text{E}[(Y - \text{E}(Y))^2]$ but remember the useful computational formula $\text{Var}(Y) = \text{E}(Y^2) - (\text{E}(Y))^2$. Apply this to $Y = \hat{\theta} - \theta$, and remember that $\hat{\theta}$ is random but θ is not.

3. Find the MLE of μ based on a sample of 3 days in which a total of 5 cars are sold.

- In many situations, the full sample data is not necessary to find the MLE; rather, a few descriptive statistics are often sufficient to evaluate the likelihood and determine the MLE
- Poisson(μ): the MLE of μ is the sample mean $\bar{X}$
- Binomial(n, p), with n known: the MLE of p is the sample proportion $\hat{p} = X/n$, where X is the number of "successes" in the sample
- Exponential(λ): the MLE of λ is the sample rate $1/\bar{X}$; the usual situation is where the X's measure times between "events" and so $\bar{X}$ is the sample mean time between "events".
- Normal(μ, σ), with both μ and σ unknown: the MLEs of μ, σ^2, and σ, are, respectively

$$\hat{\mu} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2$$

$$\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}$$

Example 28.4. Assume a Poisson(μ) model for the number of cars sold in a day at a particular car dealership. In 3 days, 5 cars are sold.

1. Consider $\sqrt{\mu}$. Is $\sqrt{\mu}$ a parameter or a statistic? How would you interpret $\sqrt{\mu}$ in this context?

2. What do you think the MLE of $\sqrt{\mu}$ is?

3. Consider $e^{-\mu}$. Is $e^{-\mu}$ a parameter or a statistic? How would you interpret $e^{-\mu}$ in this context?

4. What do you think the MLE of $e^{-\mu}$ is?

5. Consider $\mu e^{-\mu}$. Is $\mu e^{-\mu}$ a parameter or a statistic? How would you interpret $\mu e^{-\mu}$ in this context?

6. What do you think the MLE of $\mu e^{-\mu}$ is?

7. Consider $\mu^2 e^{-\mu}/2$. Is $\mu^2 e^{-\mu}/2$ a parameter or a statistic? How would you interpret $\mu^2 e^{-\mu}/2$ in this context?

8. What do you think the MLE of $\mu^2 e^{-\mu}/2$ is?

9. We started by assuming a Poisson model, but how do we know that is a reasonable assumption? Suppose we observe the number of cars sold for a random sample of days. Suggest you might use the sample data to investigate whether a Poisson model is appropriate.

- A parameter is any characteristic of a population distribution
- For many populations, there are a few "main" parameters (like μ for Poisson(μ), or (μ, σ) for Normal(μ, σ), but any characteristic of a population is parameter and can be estimated
- **Invariance property of MLEs.** If $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$.
- That is, the "MLE of a function is the function of the MLE"
- The invariance principle says that if you find the MLE of the "main parameters" then you automatically get the MLE of all the parameters that are based on it

Example 28.5. Suppose that within a certain population, credit scores follow a Normal(μ, σ) distribution. In a random sample of 5 individuals from this population, the credit scores are 600, 630, 680, 700, 770.

1. Compute the MLE of μ.

2. Compute the MLE of σ.

3. Explain in words what it means for these parameters to be the MLE.

4. Compute the MLE of the 16th percentile of credit scores for this population

5. Compute the MLE of the 90th percentile of credit scores for this population

6. Compute of the MLE of the population proportion of individuals with a credit score above 750.

- Maximum likelihood estimation procedures are widely applicable and maximum likelihood estimators have many nice theoretical properties.
- However, there are some drawbacks:
  - May require numerical maximization, which can be computationally intensive (especially if many parameters)
  - May not be reliable estimators in small samples. (Many of the nice theoretical properties of MLEs are true for large samples.)

Example 28.6. We wish to estimate the parameter μ for a Poisson(μ) distribution based on a single observed value X.

1. Suppose that $x = 3$. Carefully write the likelihood function.

2. Suppose that $x = 3$. Carefully write the log-likelihood function.

3. Use calculus to find the MLE of μ given $x = 3$. Compare to what we found previously using graphical methods.

4. Now consider a general x. Carefully write the likelihood function.

5. Now consider a general x. Carefully write the log-likelihood function.

6. Use calculus to find the MLE of μ given x. Compare to what we found previously using graphical methods.

Example 28.7. We wish to estimate the parameter μ for a Poisson(μ) distribution based on a random sample of n values $X_1 \ldots, X_n$.

1. Suppose that $n = 3$ and the sample is (3, 0, 2). Carefully write the likelihood function.

2. Suppose that $n = 3$ and the sample is (3, 0, 2). Carefully write the log-likelihood function.

3. Use calculus to find the MLE of μ given $n = 3$ and the sample (3, 0, 2). Compare to what we found previously using graphical methods.

4. Now consider a general sample of size n. Carefully write the likelihood function.

5. Now consider a general sample of size n. Carefully write the log-likelihood function.

6. Use calculus to find the MLE of μ given a sample of size n. Compare to what we found previously using graphical methods.

29 Bias of Estimators

- MLE is one procedure for finding an estimator of a parameter θ, but there any others
- When several estimators of θ are available, how do we decide which is "better"?
- The value of a parameter θ is unknown, so it is impossible to determine if any single estimate of θ is good or bad.
- An estimator is a random variable that returns different estimates of θ for different samples. So even if an estimator produces "good" estimates of θ for some samples, it might produce "bad" estimates of θ for others.
- Therefore, we can never determine for any particular sample if an estimator produces a good estimate of the true unknown value of θ.
- Rather, we evaluate the estimation *procedure*: Over many samples, does an estimator tend to produce reasonable estimates of θ for a range of potential values of θ?
- Remember: a statistic/estimator is a random variable whose sampling distribution describes how values of the estimator vary from sample-to-sample over many (hypothetical) samples.
  - We can estimate the degree of this variability by *simulating many hypothetical samples* from an assumed population distribution and computing the value of the statistic for each sample.
  - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed, and we don't know what the population distribution is.

Example 29.1. Recall that in the car dealership problem we were trying to estimate μ based on a random sample $X_1, \dots, X_n$ from a Poisson(μ) distribution. We considered a few different estimators of μ.

$$\begin{aligned}
\bar{X} &= \frac{1}{n}\sum_{i=1}^{n} X_i \\
S^2 &= \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \\
\hat{\sigma}^2 &= \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \\
\hat{\mu} &= \frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)
\end{aligned}$$

Let's see how these estimators perform when $\mu = 2$ for samples of size $n = 3$.

1. Describe in detail how you could use simulation to approximate the distribution of $\bar{X}$ for samples of size $n = 3$ when $\mu = 2$.

2. Conduct the simulation and approximate $\mathrm{E}(\bar{X})$. Interpret this value.

3. Repeat parts 1 and 2 for S^2.

4. Repeat parts 1 and 2 for $\hat{\sigma}^2$.

5. Repeat parts 1 and 2 for $\hat{\mu}$.

6. Identify if each estimators tend to overestimate μ, underestimate μ, or neither, when $\mu = 2$. But then explain why this information by itself itself helpful.

- $\hat{\theta}$ is an **unbiased** estimator of θ if

$$\mathrm{E}\left(\hat{\theta}\right) = \theta \quad \text{for } \textit{all} \text{ potential values of } \theta$$

- Roughly speaking, an unbiased estimator does not tend to systematically overestimate or underestimate the parameter of interest, regardless of the true value of the parameter.
- The **bias (function)** of an estimator[1] is the difference between its expected value (over many hypothetical samples) and the true (but unknown) value of the parameter

$$\text{bias}(\hat{\theta}) = \mathrm{E}(\hat{\theta} - \theta) = \mathrm{E}(\hat{\theta}) - \theta, \qquad \text{a function of } \theta$$

- If an estimator is unbiased its bias function is 0
- In general, the sample mean $\bar{X}$ is an *unbiased* estimator of the population mean μ.

$$\mathrm{E}(\bar{X}) = \mu$$

- In general, the "$1/(n-1)$" sample variance S^2 is an *unbiased* estimator of the population variance σ^2

$$\mathrm{E}(S^2) = \sigma^2$$

Example 29.2. Consider estimating μ for a Poisson(μ) distribution. Determine which of the following estimators are unbiased estimators of μ based on a random sample from a Poisson(μ) distribution. If the estimator is not unbiased, identify and plot its bias function.

1. $\bar{X}$.

2. S^2

3. $\hat{\sigma}^2$. (Hint: $\hat{\sigma}^2 = \frac{n-1}{n} S^2$.)

[1] We are only considering bias in *estimation*: is there bias in how we use the data to estimate the parameter? There could also be *bias in data collection*, in how the sample is selected and how the variables are measured. In this course we always assume that we have a "perfect" random sample from the population; in practice, that will never be true, but it might be a reasonable assumption based on how the data are collected. In any case, the bias function only quantifies bias in estimation, not bias in data collection.

4. $\hat{\mu} = \frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)$.

Example 29.3. Continuing the Poisson(μ) problem. Let $\theta = e^{-\mu}$. We know $\bar{X}$ is an unbiased estimator of μ. We also know that $\hat{\theta} = e^{-\bar{X}}$ is the MLE of $\theta = e^{-\mu}$. Is $\hat{\theta}$ an unbiased estimator of θ?

1. Describe in detail how you would use simulation to approximate the bias of $\hat{\theta}$ as an estimator of θ when $\mu = 2$.

2. Describe in detail how you would use simulation to approximate the bias *function* of $\hat{\theta}$.

3. Explain the reason why $e^{-\bar{X}}$ is a biased estimator of $e^{-\mu}$ even though $\bar{X}$ is an unbiased estimator of μ.

- In general, if $\hat{\theta}$ is an unbiased estimator of θ, $g(\hat{\theta})$ is usually NOT an unbiased estimator of $g(\theta)$ (unless g is linear).
  - Because in general $\mathrm{E}(g(\hat{\theta})) \neq g(\mathrm{E}(\hat{\theta}))$.
- For example, even though the "$1/(n-1)$" sample variance S^2 is an unbiased estimator of the population variance σ^2, the corresponding sample standard deviation S is a biased estimator of the population standard deviation σ

Example 29.4. Continuing the Poisson(μ) problem. Consider the estimator $\hat{\sigma}^2$. Recall that the bias function is $\mathrm{bias}(\hat{\sigma}^2) = -\frac{\mu}{n}$. What happens to the bias function as n increases. What does this mean?

- An estimator $\hat{\theta}_n$ is an **asymptotically unbiased** estimator of θ if $\mathrm{E}(\hat{\theta}_n) \to \theta$ as $n \to \infty$ for all θ, that is, if $\mathrm{bias}(\hat{\theta}_n) \to 0$ as $n \to \infty$ for all θ.
- For an asymptotically unbiased estimator, if the sample size is large $\mathrm{E}(\hat{\theta}) \approx \theta$, regardless of the true value of θ. That is, if the sample size is large the estimator is "nearly unbiased".
- MLEs are often biased, but in general, MLEs are asymptotically unbiased

30 Variance of Estimators

- The bias function of an estimator measures the estimator's tendency, over many hypothetical samples, to overestimate or underestimate the parameter, as a function of potential values of the parameter.
- But bias is only one consideration. Bias only considers the average value of the estimator over many samples, which is just one feature of the estimator's sample-to-sample distribution.

Example 30.1. Consider again estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n. We have seen that both $\bar{X}$ and S^2 are unbiased estimators of μ. If we want to choose between these two estimators, how do we decide?

1. Assume $n = 3$ and $\mu = 2$. Describe in full detail how you could conduct a simulation to approximate the sample-to-sample distribution of $\bar{X}$ and its and its expected value and standard deviation. Then conduct the simulation and record the results. What does the standard deviation measure?

2. Repeat part 1 for S^2.

3. Compare the simulation results for $\bar{X}$ and S^2 when $n = 3$ and $\mu = 2$. Based on the simulation results, which estimator of μ is preferred when $\mu = 2$: $\bar{X}$ or S^2? Why? But then explain why this information by itself isn't very helpful.

- A statistic (or estimator) is a characteristic of the *sample* which can be computed from the data. More precisely, a statistic is a function of $X_1, \dots, X_n$, but not of θ.
- Because the sample is random, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
- Statistics exhibit **sample-to-sample variability**: the value of a statistic varies from sample to sample.
  - For many statistics (e.g., means and proportions)— but not all— statistics from larger random samples vary less, from sample to sample, than statistics from smaller random samples.
  - For example, $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$.
- When choosing between two *unbiased* estimators, the one with smaller variance is generally preferred.
- Remember that the variance of an estimator will be a function of the unknown parameter θ, so we need to consider the variance *function* for various potential values of θ.

Example 30.2. Consider again estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n. We have seen that both $\bar{X}$ and S^2 are unbiased estimators of μ. If we want to choose between these two estimators, how do we decide?

1. Identify the variance function of $\bar{X}$.

2. Describe in full detail how you could use simulation to approximate the variance function of S^2.

3. It can be shown that

$$\text{Var}(S^2) = \frac{2\mu^2}{n-1} + \frac{\mu}{n}, \qquad \text{when the population distribution is Poisson}(\mu)$$

   Sketch a plot of the variance functions of both $\bar{X}$ and S^2.

4. Which of these two unbiased estimators of μ, $\bar{X}$ or S^2, is preferred? Why?

5. Suppose $n = 3$ and the sample is $(3, 0, 2)$. For this sample $\bar{x} = 1.67$ and $s^2 = 2.33$. Which number, 1.67 or 2.33, is a better estimate of μ? Explain.

6. Suppose $n = 3$ and the sample is $(3, 0, 2)$. For this sample $\bar{x} = 1.67$ and $s^2 = 2.33$. Which number, 1.67 or 2.33, would you choose as the estimate of μ based on this sample? Why?

31 Mean Square Error of an Estimator

- MLE is one procedure for finding an estimator of a parameter θ, but there any others
- When several estimators of θ are available, how do we decide which is "better"?
- The value of a parameter θ is unknown, so it is impossible to determine if any single estimate of θ is good or bad.
- An estimator is a random variable that returns different estimates of θ for different samples. So even if an estimator produces "good" estimates of θ for some samples, it might produce "bad" estimates of θ for others.
- Therefore, we can never determine for any particular sample if an estimator produces a good estimate of the true unknown value of θ.
- Rather, we evaluate the estimation *procedure*: Over many samples, does an estimator tend to produce reasonable estimates of θ for a range of potential values of θ?
- Remember: a statistic/estimator is a random variable whose sampling distribution describes how values of the estimator vary from sample-to-sample over many (hypothetical) samples.
  - We can estimate the degree of this variability by *simulating many hypothetical samples* from an assumed population distribution and computing the value of the statistic for each sample.
  - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed, and we don't know what the population distribution is.
- Roughly, there are two sources of differences between a statistic and a corresponding parameter
  - systematic errors due to estimation bias
  - "chance errors" due to natural sample-to-sample variability

Example 31.1. Continuing the situation of estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \ldots, X_n$ of size n. Consider the estimators

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 = \left(\frac{n-1}{n}\right)S^2$$

We have seen that S^2 is an unbiased estimator of μ but $\hat{\sigma}^2$ is not. Does that mean we should prefer S^2 over $\hat{\sigma}^2$?

1. Assume $n = 3$ and $\mu = 2$. Use simulation to approximate the distribution of $\hat{\sigma}^2$ and its expected value and variance.

2. Compare the simulation results to those for S^2 (from a previous handout.) For which estimator is the bias smaller? For which estimator is the variance smaller? Is there a clear preference between the two estimators when $\mu = 2$?

- Estimators can be evaluated based on both bias and variability.
- Mean square error is a combined measure of both an estimator's bias and its variability.
- The **mean square error (MSE)** of an estimator $\hat{\theta}$ of a parameter θ is

$$\text{MSE}(\hat{\theta}) = \text{E}\left(\left(\hat{\theta} - \theta\right)^2\right), \qquad \text{a function of } \theta$$

- Mean square error measures, on average, how far the estimator deviates from the parameter it is estimating.
- The MSE of an estimator is a function of the parameter θ. It can be shown that[1]

$$\begin{aligned} \text{MSE}(\hat{\theta}) &= \text{Var}(\hat{\theta}) + \left(\text{E}(\hat{\theta}) - \theta\right)^2 \\ &= \text{Var}(\hat{\theta}) + \left(\text{bias}(\hat{\theta})\right)^2 \end{aligned}$$

- Remember that bias is measured in the measurement units of the variable, while variance is measured in squared units. So MSE basically puts bias and variability on a common unit scale and sums to get a total measure.
- There can be many "reasonable" estimators of a parameter. Given two estimators, it's often the situation that neither one has smaller MSE for all potential values of θ. Choosing an estimator often involves a tradeoff between bias and variability.

3. Use the simulation results to compute the MSE of S^2 and $\hat{\sigma}^2$ when $\mu = 2$. Determine which estimator has smaller MSE when $\mu = 2$, but then explain why this information by itself is not very useful.

[1] Recall: By definition $\text{Var}(Y) = \text{E}[(Y - \text{E}(Y))^2]$ but remember the useful computational formula $\text{Var}(Y) = \text{E}(Y^2) - (\text{E}(Y))^2$. Apply this to $Y = \hat{\theta} - \theta$, and remember that $\hat{\theta}$ is random but θ is not.

4. Recall

$$\mathrm{Var}(S^2) = \frac{2\mu^2}{n-1} + \frac{\mu}{n}, \qquad \text{when the population distribution is Poisson}(\mu)$$

   Find and plot the MSE functions of S^2 and $\hat{\sigma}^2$. Which estimator is preferred? Discuss.

Example 31.2. Continuing estimating μ for a Poisson(μ) distribution. Consider $\bar{X}$ and the constant estimator 4.2.

1. Esplain what it means to use 4.2 as an estimation procedure.

2. Find the MSE of the constant estimator 4.2.

3. Find the MSE of $\bar{X}$.

4. Suppose $n = 3$. Plot the MSE functions of the two estimators. Does either estimator have a better MSE? Explain.

5. Donny Don't says: "neither estimator has smaller MSE for all values of μ. So we'll use the constant estimator 4.2 when μ is near 4.2 (between 3.17 and 5.56 if $n = 3$) and we'll use $\bar{X}$ for other values of μ". Do you agree with Donny's strategy? Do you see any problems with it?

6. What happens to the MSEs as n increases? In particular, what happens to the range of values of μ for which the constant estimator 4.2 has smaller MSE than $\bar{X}$?

- Given a parameter θ, it is never possible to find a single estimator that has the smallest MSE for all potential values of θ.
- Choosing an estimator often involves a tradeoff between bias and variability.
- MLEs generally have good properties. For many situations, roughly,
  - The MLE is asymptotically unbiased; that is, the bias is small when the sample size is large
  - The variance of an MLE is about as small as it can be for an unbiased estimator
- But there are many reasonable estimation procedures besides MLE (e.g., Bayes estimators).
- Bias is estimation is not necessarily bad. Being willing to accept a little bit of bias can often lead to a beneficial reduction in variability.
  - On the other hand, try to minimize bias in *data collection*: how the sample is selected, how the variables are measured, etc
  - A lot of the theory assumes we have a perfect random sample, which is never true in practice

32 Central Limit Theorem

- Many statistical applications involve *random sampling* from a population.
- And one of the most basic operations in statistics is to average the values in a sample.
- If a sample is selected at random, then the sample mean is a random variable that has a distribution. How does this distribution behave?

Example 32.1. A random sample of n customers at a snack bar is selected, independently. Let T_n by the total dollar amount spent by the n customers in the sample, and let $\bar{X}_n = T_n/n$ represent the sample mean dollar amount spent by the n customers in the sample.

Assume that

- 40% of customers spend 5 dollars
- 40% of customers spend 6 dollars
- 20% of customers spend 7 dollars

Let X denote the amount spent by a single randomly selected customer.

1. Find $\mathrm{E}(X)$. We'll call this value the population mean μ; explain what this means.

2. Find $\mathrm{SD}(X)$. We'll call this value the population standard deviation σ; explain what this means.

3. Randomly select two customers, independently, and let X_1 and X_2 denote the amounts spent by the two people selected. Make a table of all possible (X_1, X_2) pairs and their probabilities, and use the table to find the distribution of $\bar{X}_2$. Interpret in words in context what this distribution represent.

4. Compute and interpret $\mathrm{P}(\bar{X}_2 > 6)$.

5. Compute $\mathrm{E}(\bar{X}_2)$. How does it relate to μ?

6. There are 3 "means" in the previous part. What do they all mean?

7. Compute $\mathrm{Var}(\bar{X}_2)$ and $\mathrm{SD}(\bar{X}_2)$. How do these values relate to the population variance σ^2 and population standard deviation σ?

8. Describe in words in context what $\mathrm{SD}(\bar{X}_2)$ measures variability of.

- The **population distribution** describes the distribution of values of a variable over all *individuals* in the population.
  - The **population mean**, denoted μ, is the average of the values of the variable over all individuals in the population.
  - The **population standard deviation**, denoted σ, is the standard deviation of all the individual values in the population.

- A **(simple) random sample** of size n is a collection of random variables $X_1, \dots, X_n$ that are *independent* and *identically distributed* (*i.i.d.*)
- The **sample mean** is

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n} = \frac{T_n}{n},$$

- Because the sample is randomly selected, the sample mean $\bar{X}_n$ is a random variable that has a distribution. This distribution describes how sample means vary from sample-to-sample over many random samples of size n.
- Over many random samples, sample means do not systematically overestimate or underestimate the population mean.

$$\mathrm{E}(\bar{X}_n) = \mu$$

- Variability of sample means depends on the variability of individual values of the variable; the more the values of the variable vary from individual-to-individual in the population, the more the sample means will vary from sample-to-sample. However, sample means are less variable than individual values of the variable. Furthermore, the sample-to-sample variability of sample means decreases as the sample size increases

$$\mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$
$$\mathrm{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

Example 32.2. Continuing Example 32.1. Now suppose we take a random sample of 30 customers; consider $n = 30$ and $\bar{X}_{30}$.

1. Find $\mathrm{E}(\bar{X}_{30})$, $\mathrm{Var}(\bar{X}_{30})$, and $\mathrm{SD}(\bar{X}_{30})$.

2. Use simulation to determine the approximate shape of the distribution of $\bar{X}_{30}$.

3. Simulation shows that $\bar{X}_{30}$ has an approximate Normal distribution. Use this Normal distribution to approximate $\mathrm{P}(\bar{X}_{30} > 6)$, and interpret the probability.

- The **(Standard) Central Limit Theorem**. Let $X_1, X_2, ...$ be independent and identically distributed (i.i.d.) random variables with finite mean μ and finite standard deviation σ. Then

$$\bar{X}_n \text{ has an approximate } N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \text{ distribution, if } n \text{ is large enough.}$$

$$T_n \text{ has an approximate } N(n\mu, \sigma\sqrt{n}) \text{ distribution, if } n \text{ is large enough}$$

- The CLT says that if the sample size is large enough, the sample-to-sample distribution of sample means (or sums) is approximately Normal, *regardless of the shape of the population distribution.*

Example 32.3. Continuing Example 32.1. Now suppose that each customer spends 5 dollars with probability 0.4, 6 with probability 0.4, 7 with probability 0.19, and 30 with probability 0.01 (maybe they treat a few friends).

1. Use simulation to approximate the distribution of $\bar{X}_{30}$; is it approximately Normal?

2. Use simulation to approximate the distribution of $\bar{X}_{100}$; is it approximately Normal?

3. Use simulation to approximate the distribution of $\bar{X}_{300}$; is it approximately Normal?

- The CLT says that if the sample size is large enough, the sample-to-sample distribution of sample means is approximately Normal, *regardless of the shape of the population distribution.*
- However, the shape of the population distribution — which describes the distribution of individual values of the variable — does matter in determining *how large* is large enough.

  - If the population distribution is Normal, then the sample-to-sample distribution of sample means is Normal for any sample size.
  - If the population distribution is "close to Normal" — e.g., symmetric, light tails, single peak — then smaller samples sizes are sufficient for the sample-to-sample distribution of sample means to be approximately Normal.
  - If the population distribution is "far from Normal" — e.g., severe skewness, heavy tails, extreme outliers, multiple peaks — then larger sample sizes are required for the sample-to-sample distribution of sample means to be approximately Normal.

- You should certainly be aware that there is no magic number and that some population distributions require a large sample size for the CLT to kick in. However, in many situations the sample-to-sample distributions of sample means (or other statistics) is approximately Normal even for small or moderate sample sizes.

Example 32.4. The number of free throws that a particular basketball player attempts in a game has a Poisson distribution with parameter 2.7, independently from game to game. Approximate the probability that the player attempts more than 60 free throws in the next 20 games.

Example 32.5. Ten independent resistors are connected in series. Each has a resistance Uniformly distributed between 215 and 225 ohms, independently. Compute the probability that the mean resistance of this series system is between 219 and 221 ohms.

Example 32.6. A fair six-sided die is rolled until the total sum of all rolls exceeds 300. Approximate the probability that at least 80 rolls are necessary.

Example 32.7. Suppose that average annual income for U.S. households is about \$100,000, and that the standard deviation of income is about \$230,000.

1. What can you say about the probability that a single randomly selected household has an income above \$120,000?

2. Donny Don't says: "Since a sample size of one million is extremely large, the CLT says that the incomes in a sample of one million households should follow a Normal distribution". Do you agree? If not, explain to Donny what the CLT does say.

3. Use the CLT to approximate the probability that in a sample of 1000 households the sample mean income is above \$120,000.

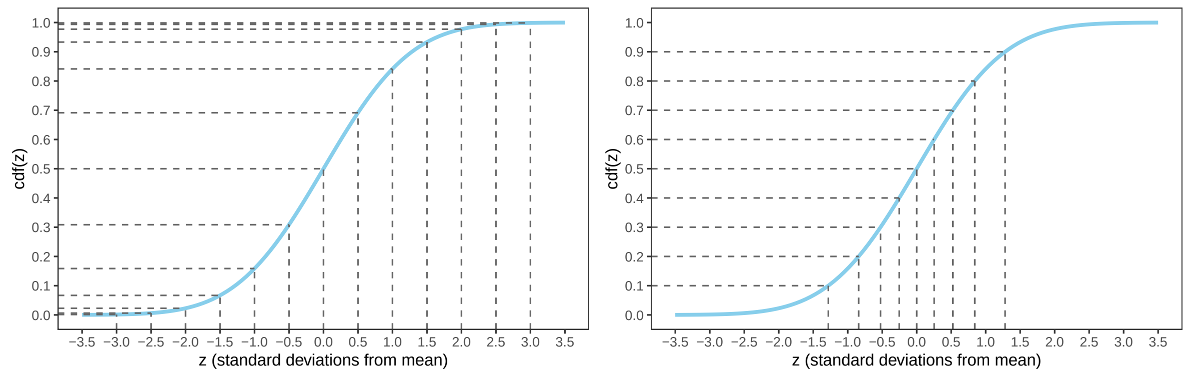
4. Donny says: "Wait, the distribution of household income is highly skewed to the right. Wouldn't it be more appropriate to use the sample median than the sample mean"? Do you agree? Explain.

References

Summary of Some Common Distributions

Name and parameters	Possible values	pmf/pdf	cdf	Mean	SD
Discrete					
Poisson(μ)	$x = 0, 1, 2, \ldots$	$e^{-\mu}\frac{\mu^x}{x!}$	No closed form	μ	$\sqrt{\mu}$
Binomial(n, p)	$x = 0, 1, 2, \ldots, n$	$\binom{n}{x}p^x(1-p)^{n-x}$	No closed form	np	$\sqrt{np(1-p)}$
Geometric(p)	$x = 1, 2, 3, \ldots$	$p(1-p)^{x-1}$	$1 - (1-p)^x$	$\frac{1}{p}$	$\sqrt{\frac{1-p}{p^2}}$
Negative Binomial(r, p)	$x = r, r+1, r+2, \ldots$	$\binom{x-1}{x-r}p^r(1-p)^{x-r}$	No closed form	$\frac{r}{p}$	$\sqrt{\frac{r(1-p)}{p^2}}$
Hypergeometric(n, N_1, N_0); $N = N_1 + N_0, p = \frac{N_1}{N}$	$x = 0, 1, \ldots, n$ (usually)	$\frac{\binom{N_1}{x}\binom{N_0}{n-x}}{\binom{N_1+N_0}{n}}$	No closed form	np	$\sqrt{np(1-p)\left(\frac{N-n}{N-1}\right)}$
Continuous					
Uniform(a, b)	$a < x < b$	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{b-a}{\sqrt{12}}$
Exponential(λ)	$x > 0$	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda}$
Normal(μ, σ)	$-\infty < x < \infty$	$\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$	No closed form	μ	σ

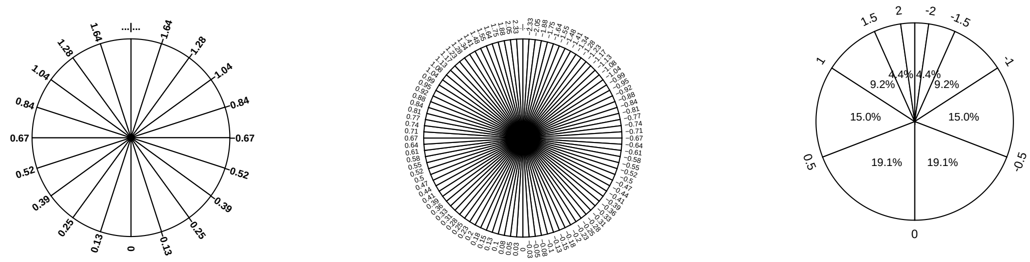
Percentiles for Normal Distributions (Empirical Rule)



(a) Percentiles highlighted in 0.5 increments of standard deviations from the mean

(b) Deciles highligted in terms of standard deviations from the mean

Figure 1: The Normal(0, 1) cdf. Provides the cdf for any Normal distribution if the variable is measured in *standard deviations from the mean*. It's the same cdf in both figures just with different values highlighted.



(a) Each sector represents an interval with 5% probability

(b) Each sector represents an interval with 1% probability

(c) Values labeled in increments of 0.5 standard deviations from the mean

Figure 2: A continuous *Normal(0, 1)* spinner. The same spinner is displayed in all figures, with different features highlighted. This spinner can be used to simulate values from any Normal distribution by simulating values in terms of standard deviations from the mean.

Table 2: Select percentiles for a Normal distribution in terms of standard deviations from the mean

(a) In 0.5 increments of standard deviations from the mean

Percentile	Standard deviations from the mean
0.02%	-3.5
0.1%	-3.0
0.6%	-2.5
2.3%	-2.0
6.7%	-1.5
15.9%	-1.0
30.9%	-0.5
50%	0.0
69.1%	0.5
84.1%	1.0
93.3%	1.5
97.7%	2.0
99.4%	2.5
99.9%	3.0
99.98%	3.5

(b) Deciles and quantiles as standard deviations from the mean

Percentile	Standard deviations from the mean
1%	-2.33
5%	-1.64
10%	-1.28
20%	-0.84
25%	-0.67
30%	-0.52
40%	-0.25
50%	0.00
60%	0.25
70%	0.52
75%	0.67
80%	0.84
90%	1.28
95%	1.64
99%	2.33

The empirical rule is often described in terms of “within [blank] standard deviations of the mean” as in the following:

- 38% of values are within 0.5 standard deviations of the mean
- 50% of values are withing 0.67 standard deviations of the mean
- 68% of values are within 1 standard deviation of the mean
- 87% of values are within 1.5 standard deviations of the mean
- 95% of values are within 2 standard deviations of the mean
- 99% of values are within 2.6 standard deviations of the mean
- 99.7% of values are within 3 standard deviations of the mean
- 99.99% of values are within 4 standard deviations of the mean